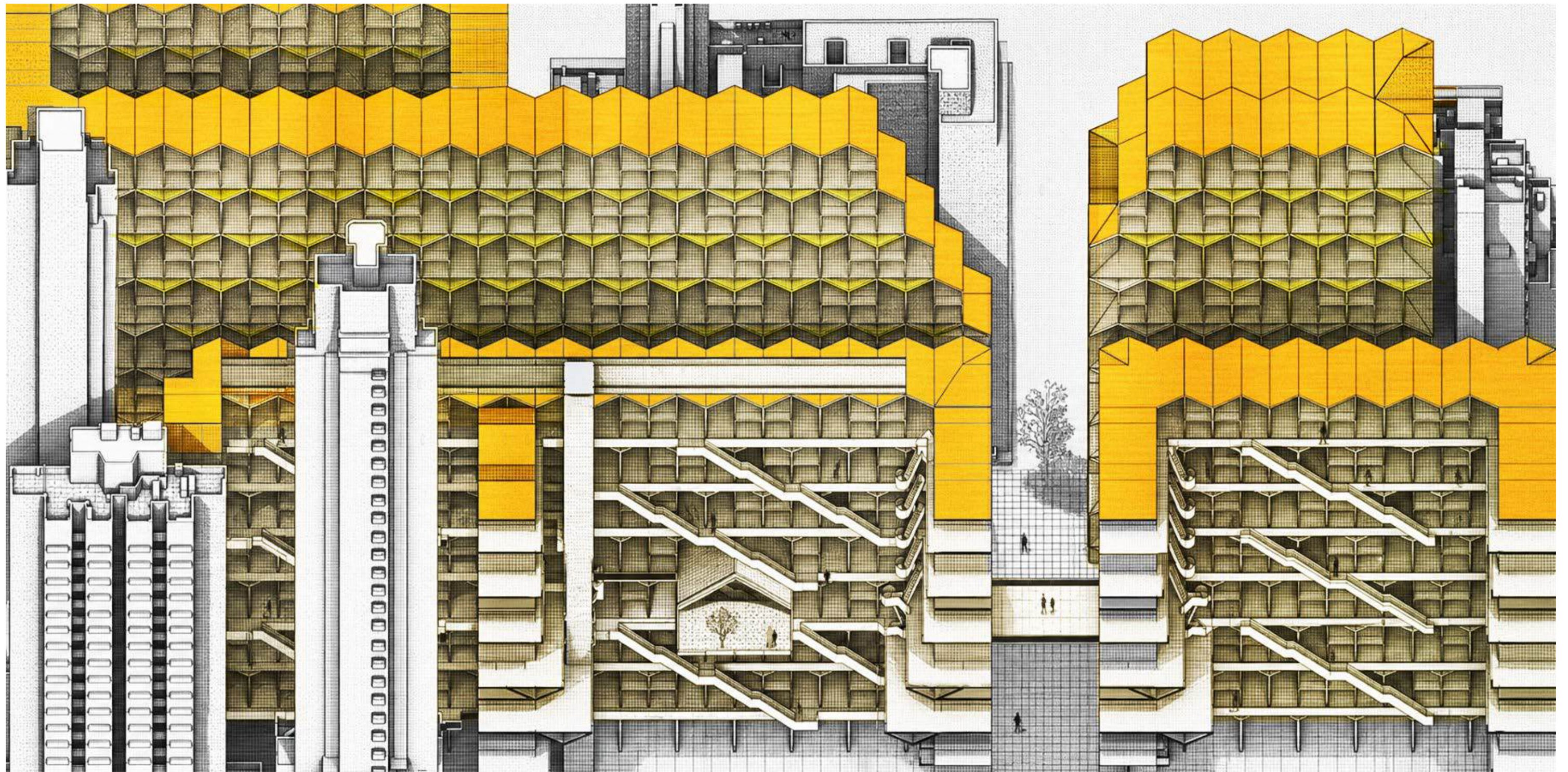




CHARLES TO
ARCHITECTURE PORTFOLIO

TO WAI LAM, CHARLES		
(852) 5161 6760 charleswrtsy@gmail.com GitHub: https:// github.com/charleswrtsy		
EDUCATION		
The University of Hong Kong		CURRENT
Master of Architecture		
University of South Carolina		2022
International Business and Studio Art Photography (Exchange)		
The Chinese University of Hong Kong		2018-2022
Bachelor of Social Science (Architectural Studies)		
SKILLS		
Software	AutoCAD Rhinoceros Adobe Photoshop, Illustrators, InDesign Basic Python	
Languages	Native in Cantonese Fluent in English and Mandarin Basic Japanese	
Interest	soccer, photography, travelling, exploring Web3	
WORK EXPERIENCE		
Teaching Assistant at The University of Hong Kong		Jan–July 2026
- Delivered tutorials to undergraduate students on foundational coding, design methodologies, and integrating generative AI tools into creative and analytical workflows. - Facilitated course delivery focused on utilizing Large Language Models (LLMs), generative solutions, and agentic scenarios to solve complex global challenges.		
Wong & Ouyang (HK) Ltd		April 2023–July 2024
Architectural Assistant		
- Worked on schematic and detail design stages for University Laboratory Campuses and a large-scale Hospital. - Meeting with clients and coordinate with project teams. - Involved in Tender Bids for a Government Community Health Centre.		
Condition_Lab		July-August 2021
- Developed a prototypal tracing machine to convert natural conditions into drawings. - longlisted in DEZEEN AWARD and in TRINITY BUOY WHARF DRAWING PRICE.		
COMPETITION		
HK University Web3 Quant Trading Hackathon (Sponsored by Flow Traders and Jane Street)		Nov - Dec 2025
- First Price, designed and back tested multi-factor trading strategies, incorporating cross-validation, signal delay simulation, transaction cost analysis, and volatility-adjusted position sizing. - Developed a real-time Python execution engine for signal aggregation, comprehensive risk management, and dynamic order monitoring.		
Kaira Looro Architecture Competition, Top 30		June 2020
- Participated in an international architecture competition - Designed an children health care centre in Africa, work published on their yearbook.		
HKIP YPG Transitional Housing Planning Competition, Silver Price		2022
GIS-driven research to inform and propose a transitional housing solution tailored to urban needs.		
PROJECTS		
Quant Backtest Engine		
- Developed a systematic CTA model including momentum, mean-reversion, and sentiment-driven strategies. - Built a non-price factor research engine to identify alpha signals across derivatives and blockchain-based data, implemented with walk-forward testing, parameters optimization, and performance analysis.		

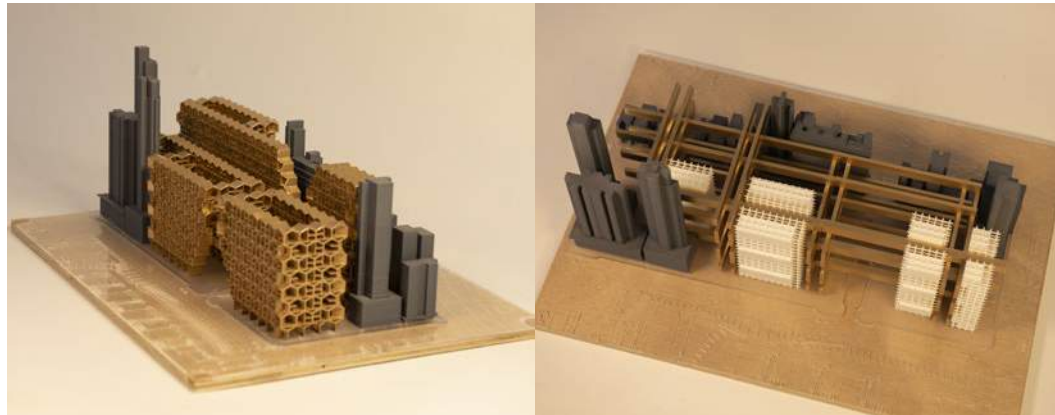
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SUPER LIQUID HOUSING:

Redefining Spatial Politics & Fractional Ownership
through Financial Model

Thesis Project, HKU
Instructed by Dr. Holger Kehne
January- May 2026



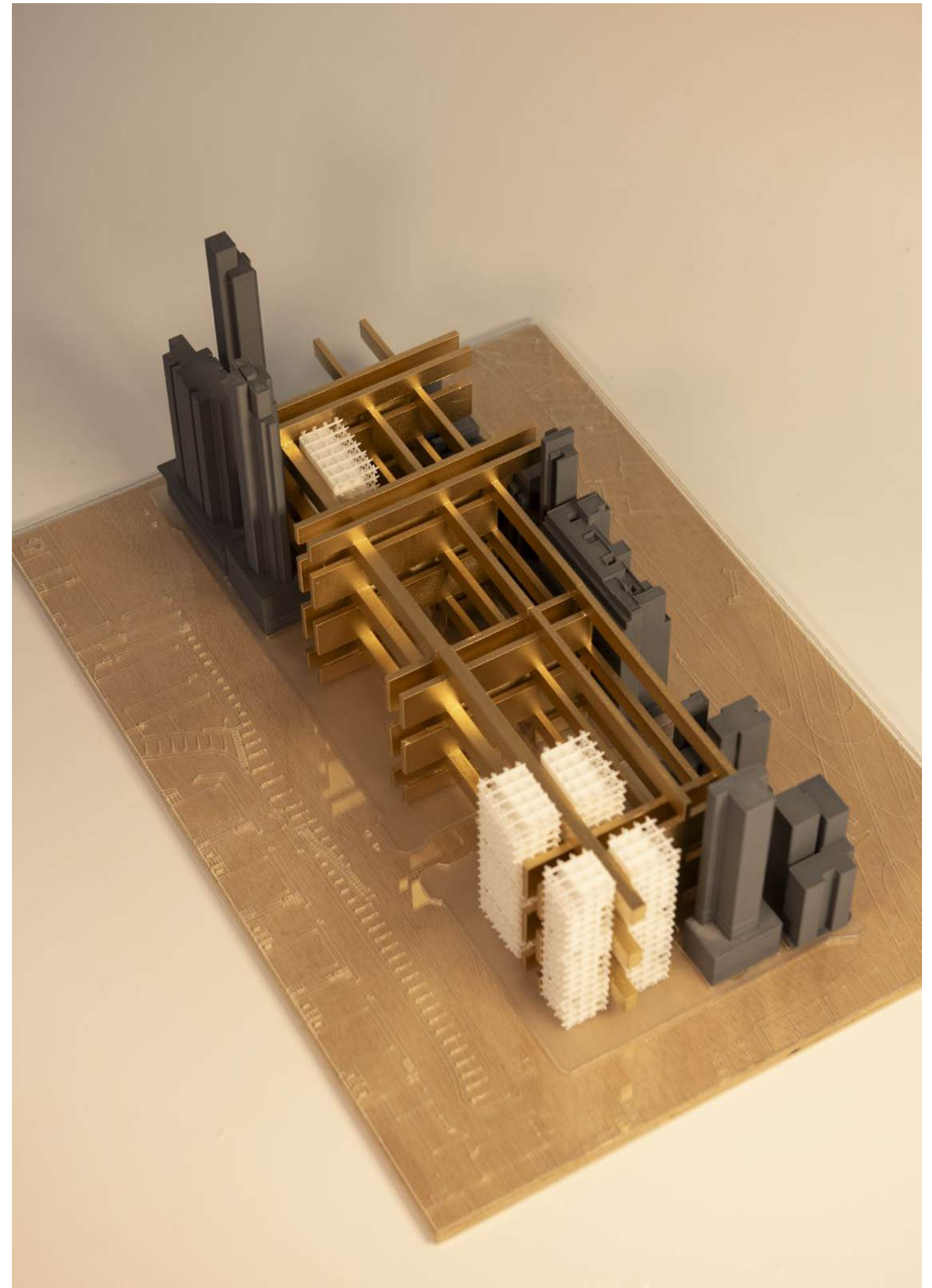
Thesis Abstract

This thesis introduces an architectural and financial model designed to dismantle the housing crisis as asset trap facing Hong Kong's marginalized "sandwich class". By leveraging the concept of fractional ownership and tokenization in housing model, the project aims to decouple the housing of property's assetization with spatial ownership and autonomy.

A system of hyper-durable infrastructural frames was explored with two prototypes that rules interconnectivity, degree of expansion, light and air compliance, and formal expression. The permanent infrastructure "Vault" not only acts as value storage, hedged against a gold-backed reserve to secure long-term liquid value, but also serves as framework for expandable modular living units deployed within system.

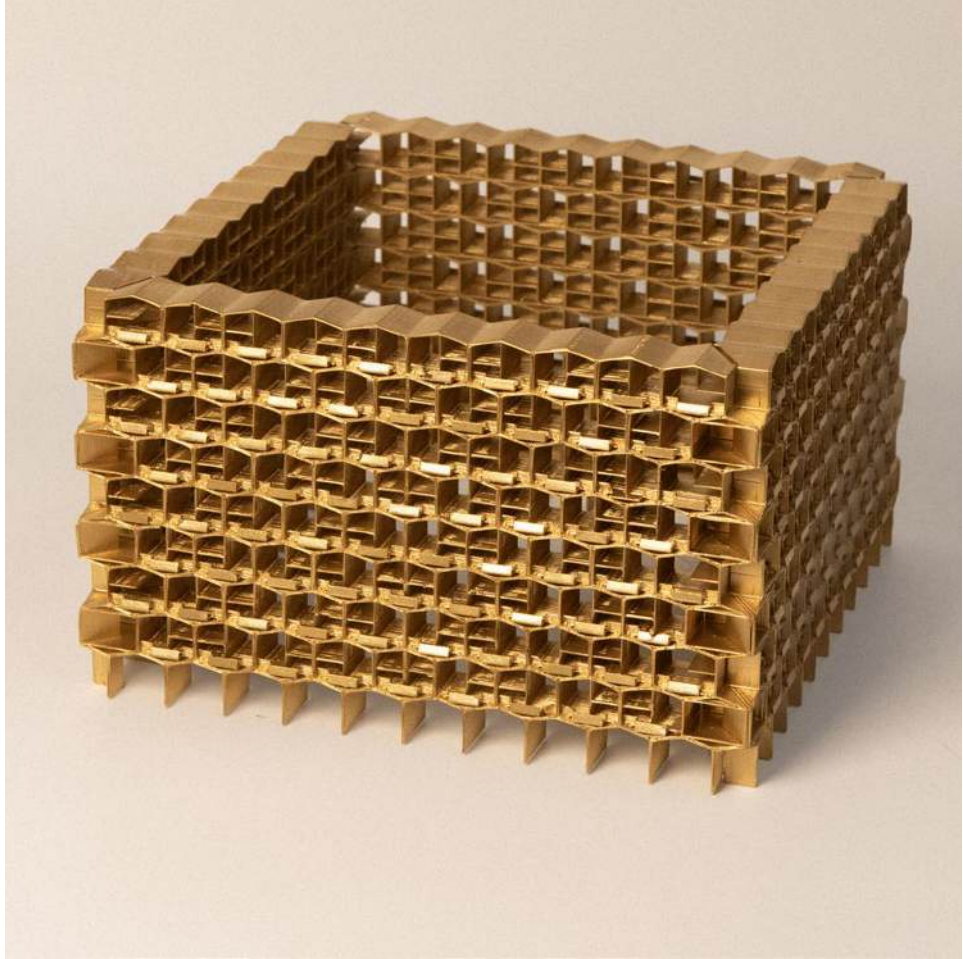
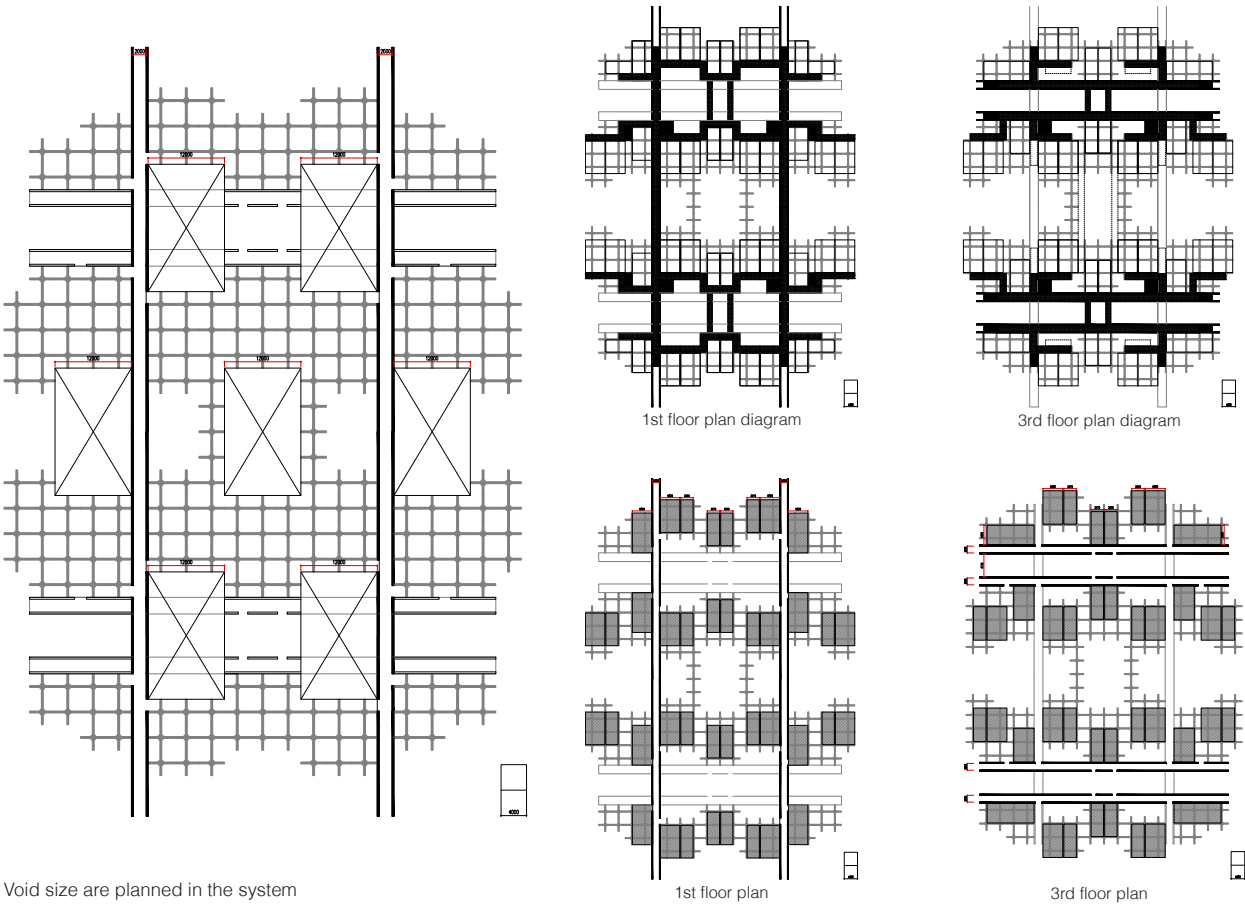
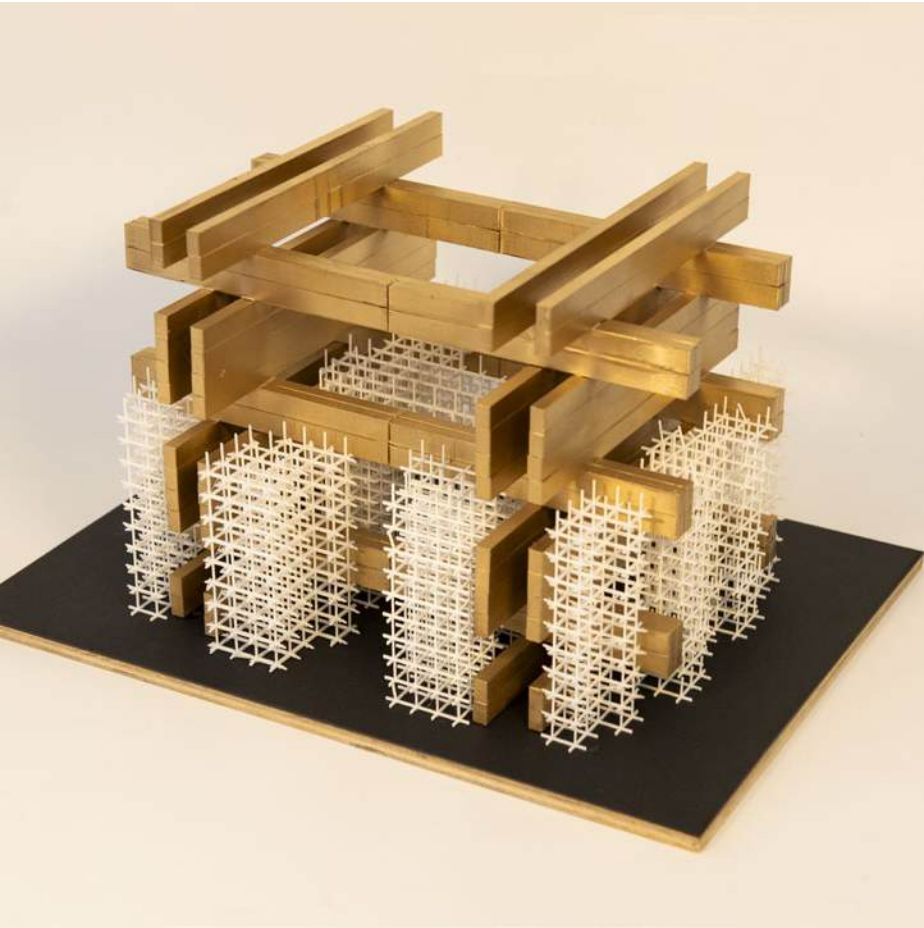
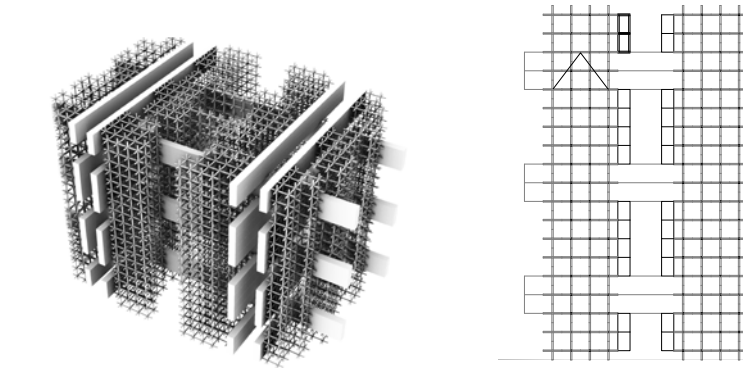
Moving beyond rigid property titles, the ecosystem introduces a 100-year spatial bond tied strictly to spatial volume rather than individual occupants, compressing monthly costs and unlocking unprecedented user autonomy. A foundational set up guarantees that half of the volumetric voids remain communal, allowing private spaces to dynamically expand or contract across human lifecycles. With the investment of non-occupant token owners, the community gained more mobile chances to expand in urban scale.

Tested against the paralyzed Civil Servants' Co-operative Building Society (CBS) schemes in To Kwa Wan, Super Liquid Housing establishes a highly scalable, democratic urban paradigm where housing functions flawlessly as a home, not a financial commodity.



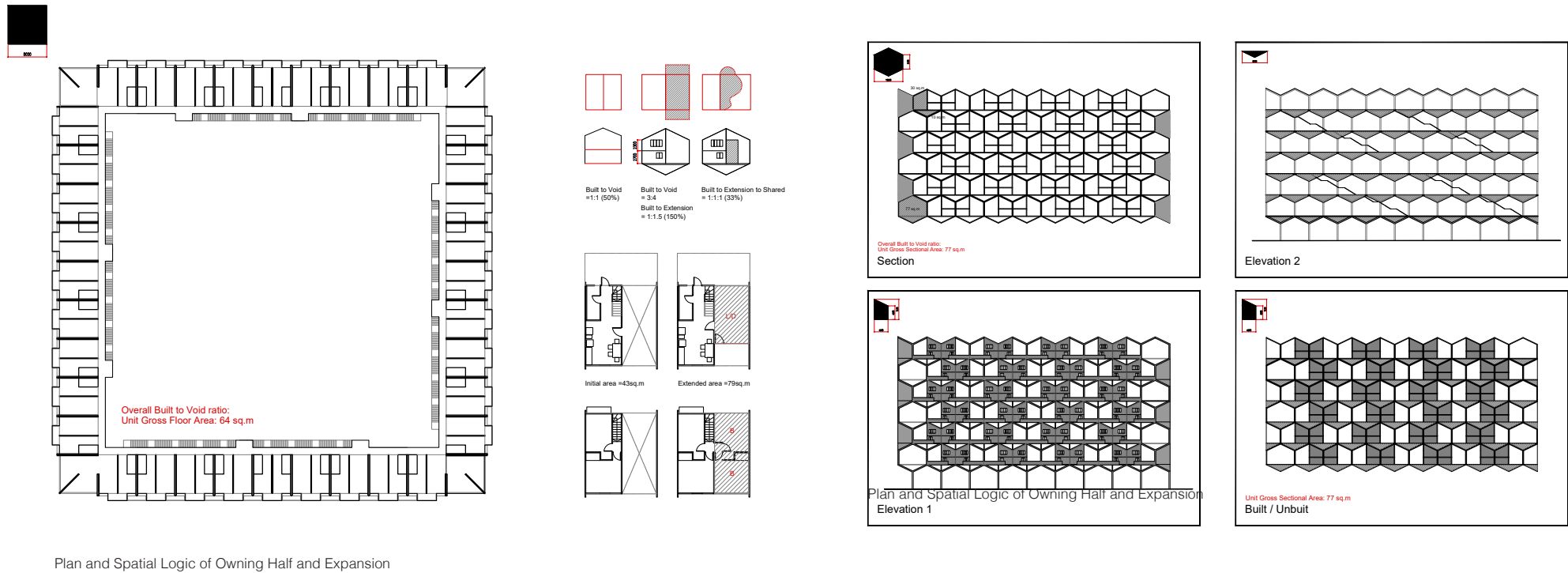
Prototype 1 The Ingot Infrastructure

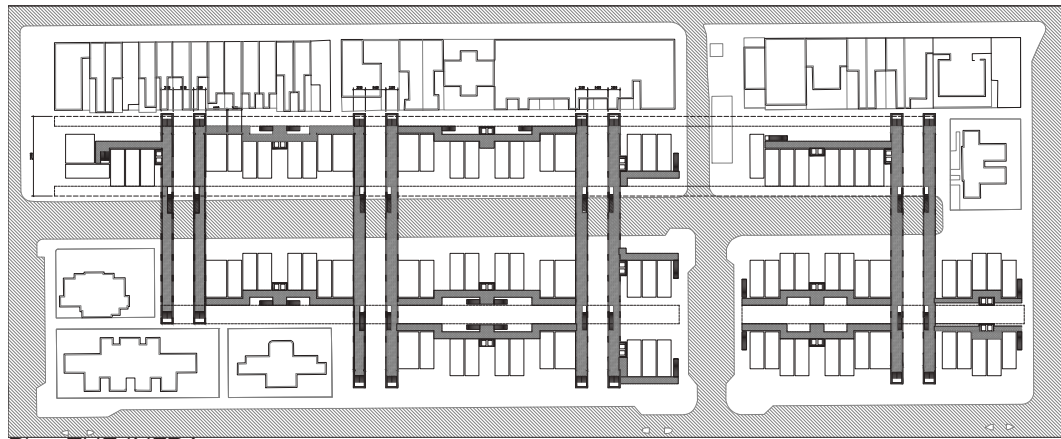
The key idea of this prototype is a durable mega infrasture that provide horizontal connectivity across vertical residential blocks. The value of the building go to the infrastructure and the whole developemnt is fractionalzed and packged into a comodity token for trading. Quality spaces are then returned to resiudents for modular expansion. The infrastructure at the same time act as spatial organizer that lmits the extent of expansion and size of voids and air gap. The infrastruer itself is wide enough to provide communal spaces that form clusters every 4 floors.



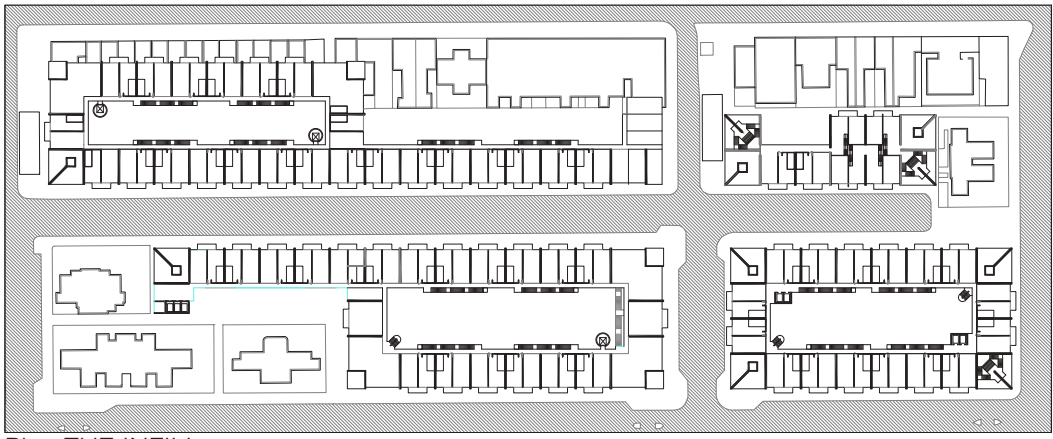
Prototype 2 The Ingot Infill

When the pricing of a space shift from the calculation of Gross Floor Area to Cross Sectional Area, volumetric spaces beome a potential driver of module size. This prototype designed a hexagonal frame that is durable for 100 years. It first day provide void space is halfy built, borrowing idea of Half House by The Elemental. Voids act importastnly as ventilator and communal space before expansion.

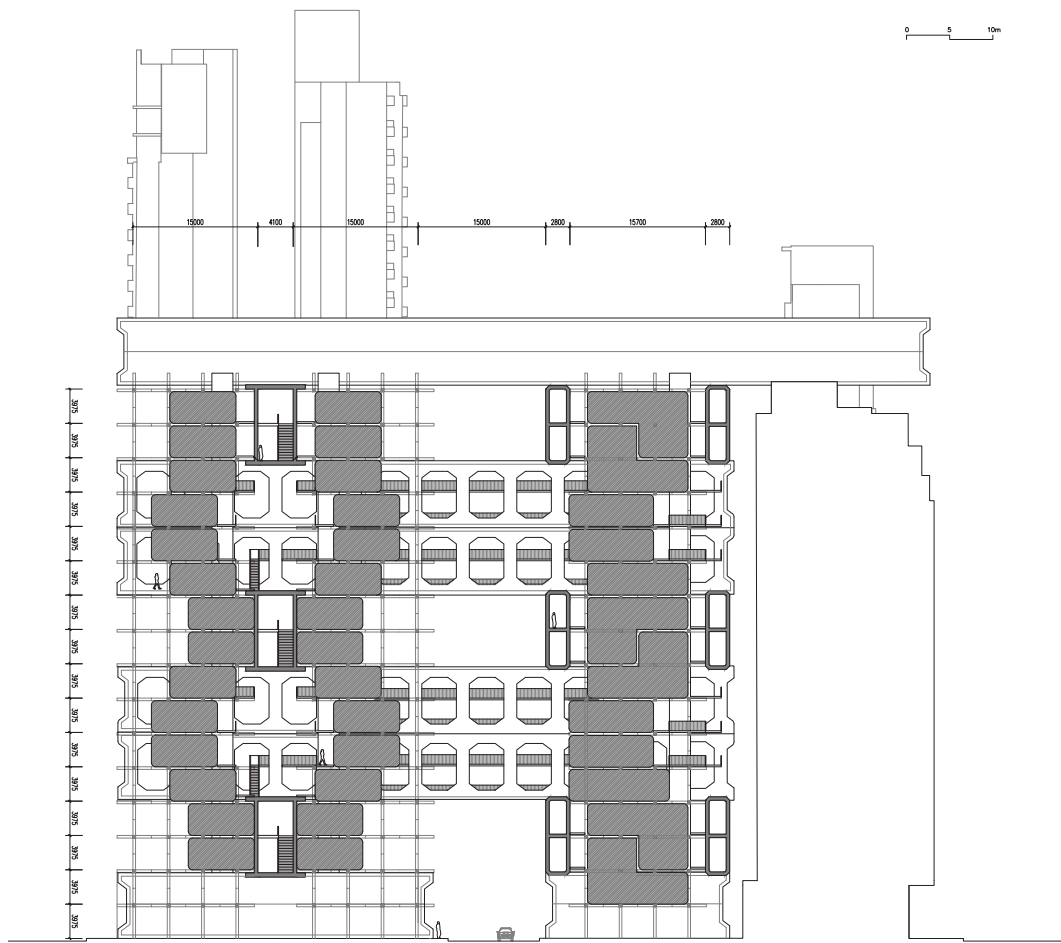




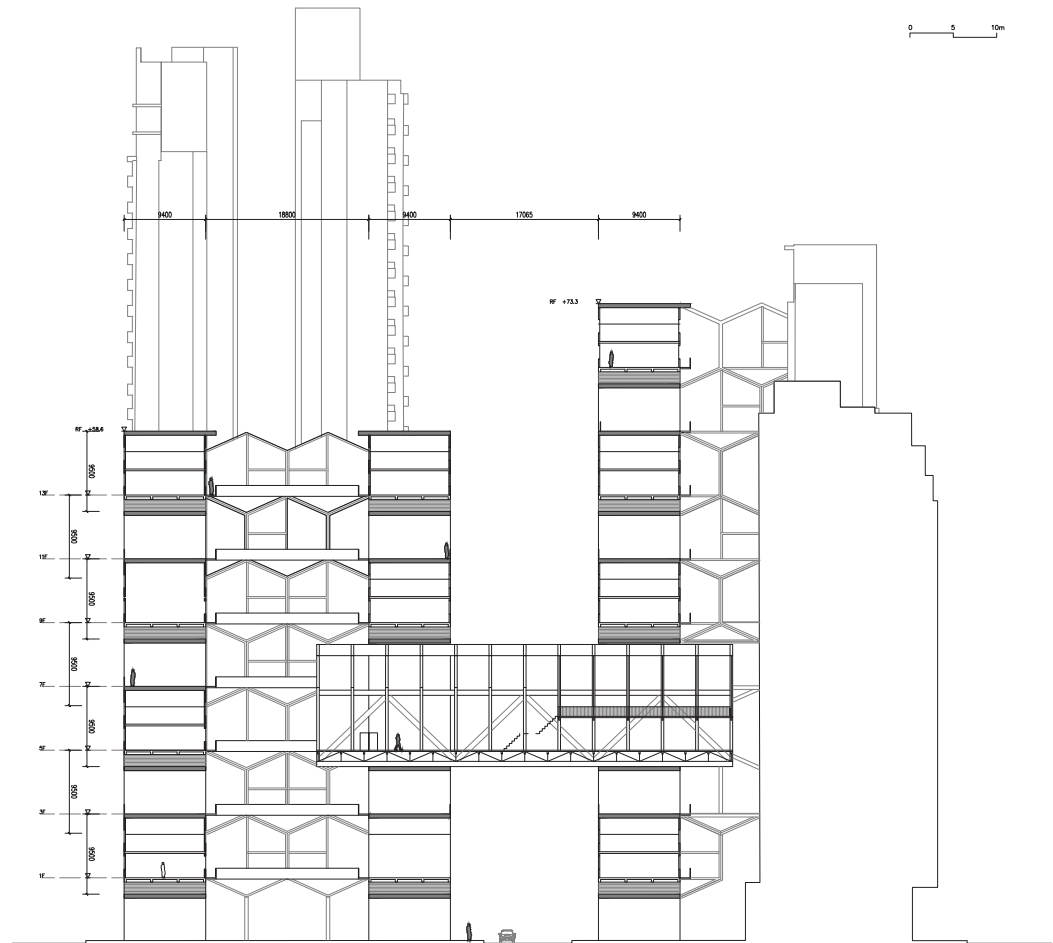
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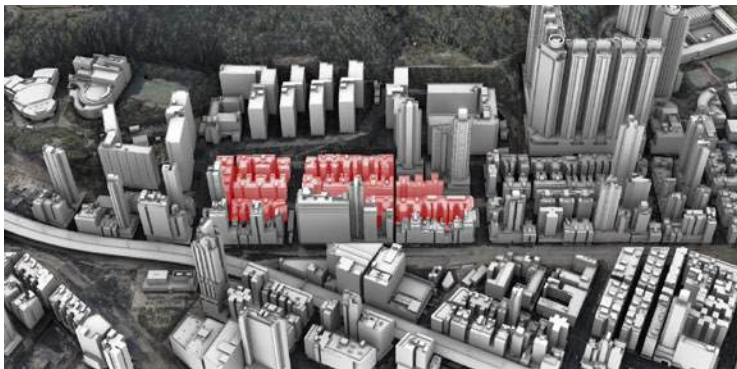
Plan THE INFILL



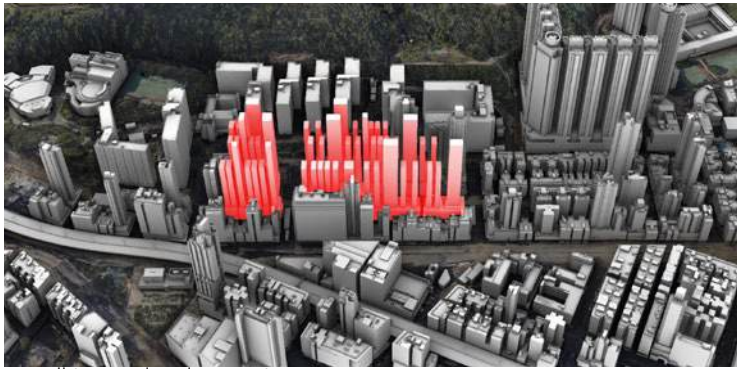
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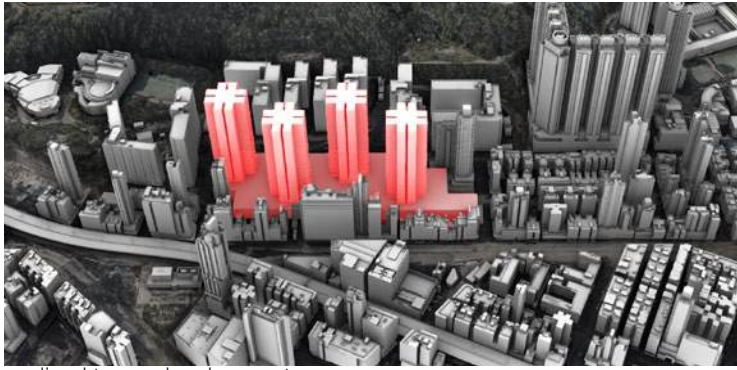
Section THE INFILL



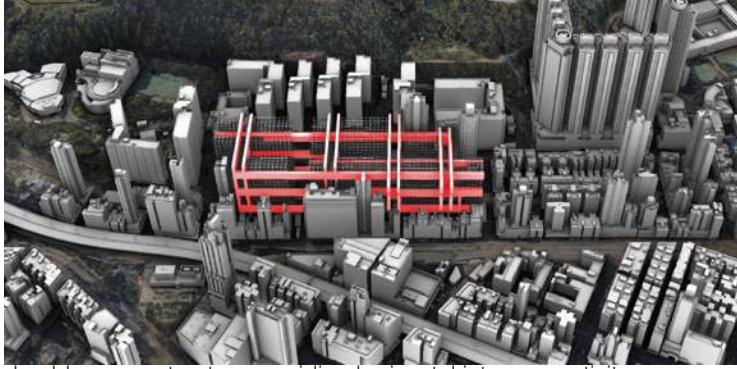
existing buildings



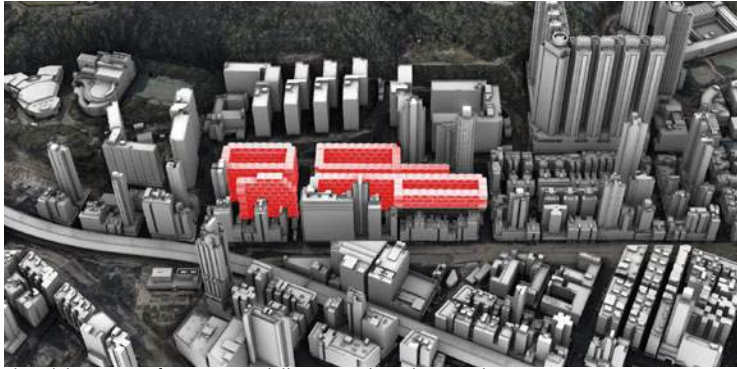
pencil tower development



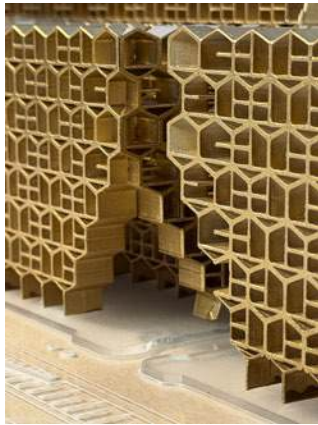
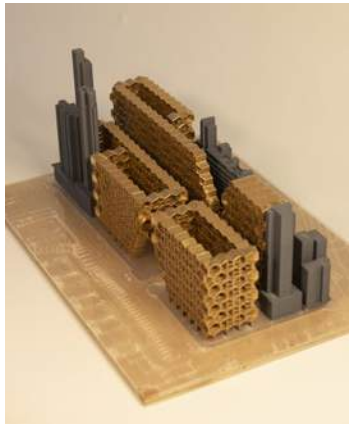
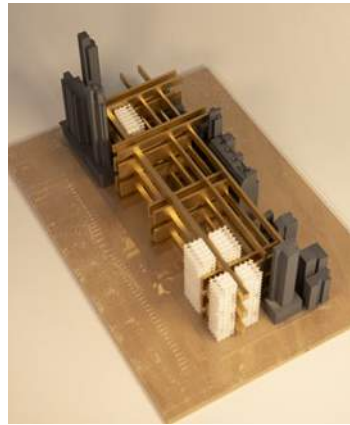
podium tower development

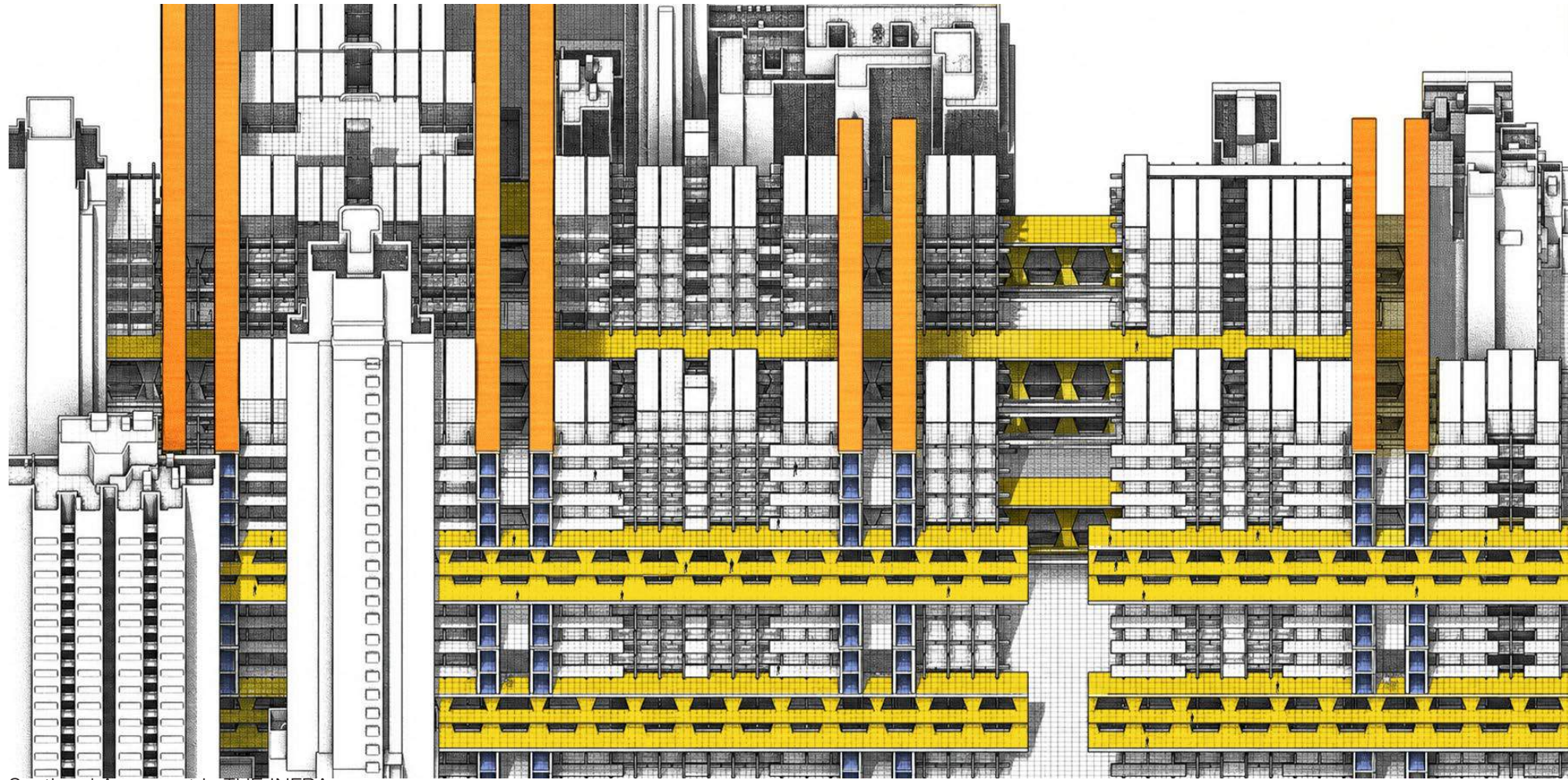


durable mega-structure providing horizontal interconnectivity

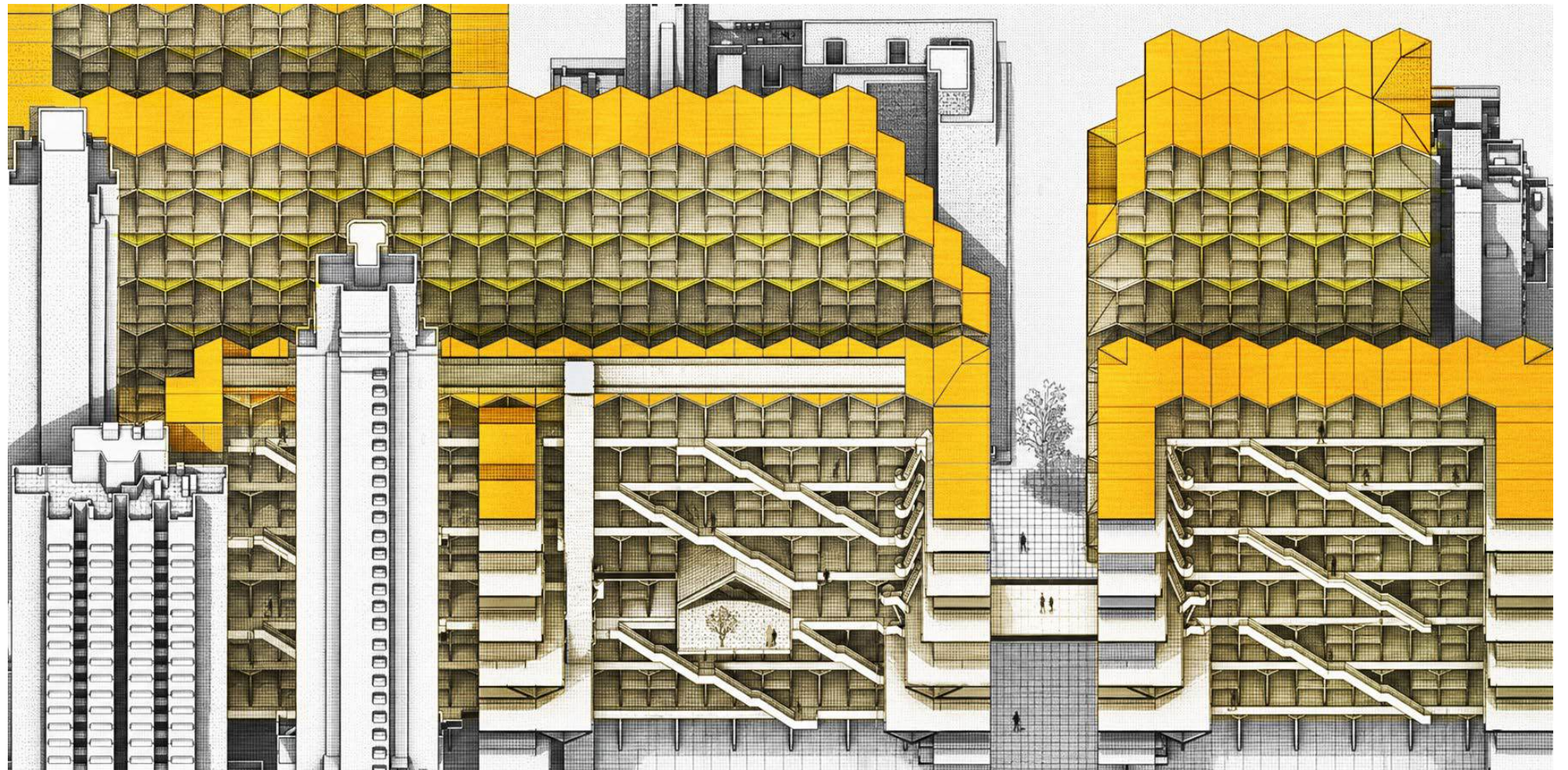
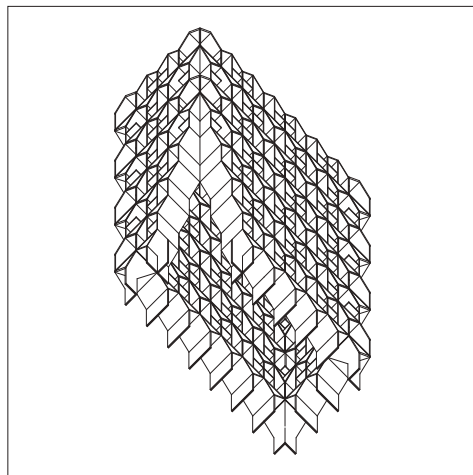
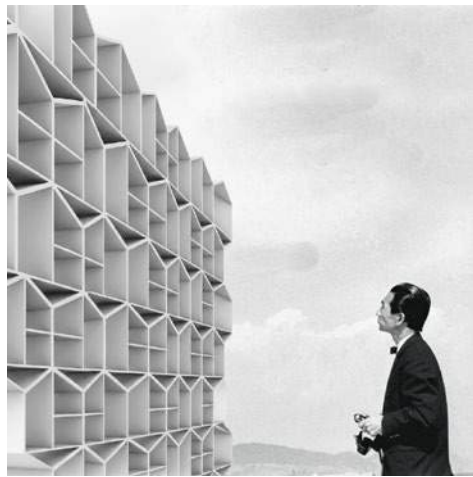
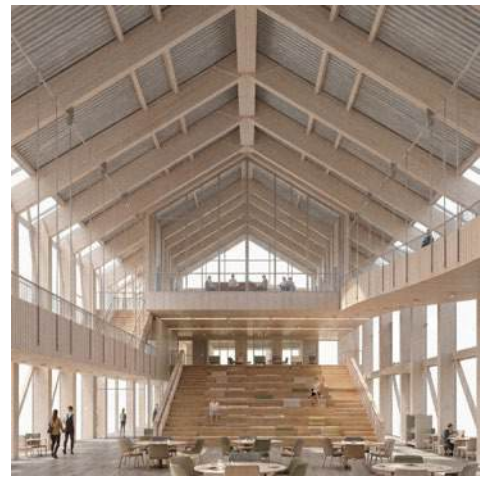
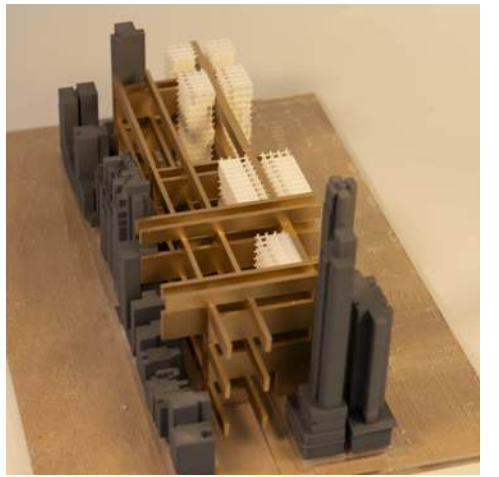
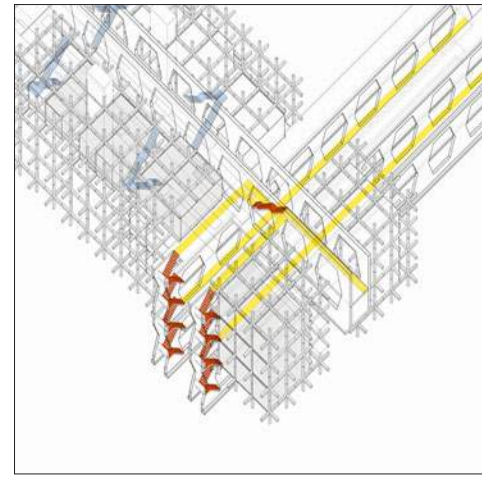
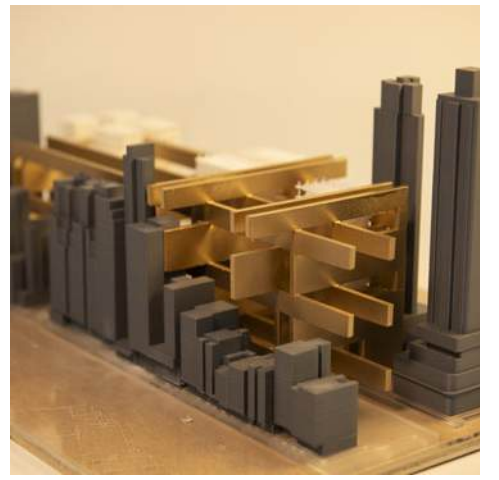
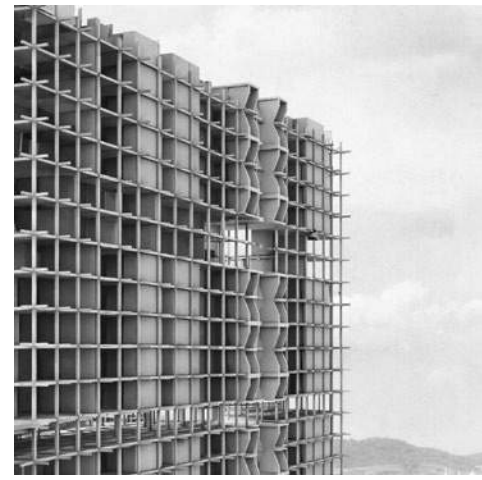


durable mega-frame providing sectional porosity





Sectional Axonometric THE INFRA-



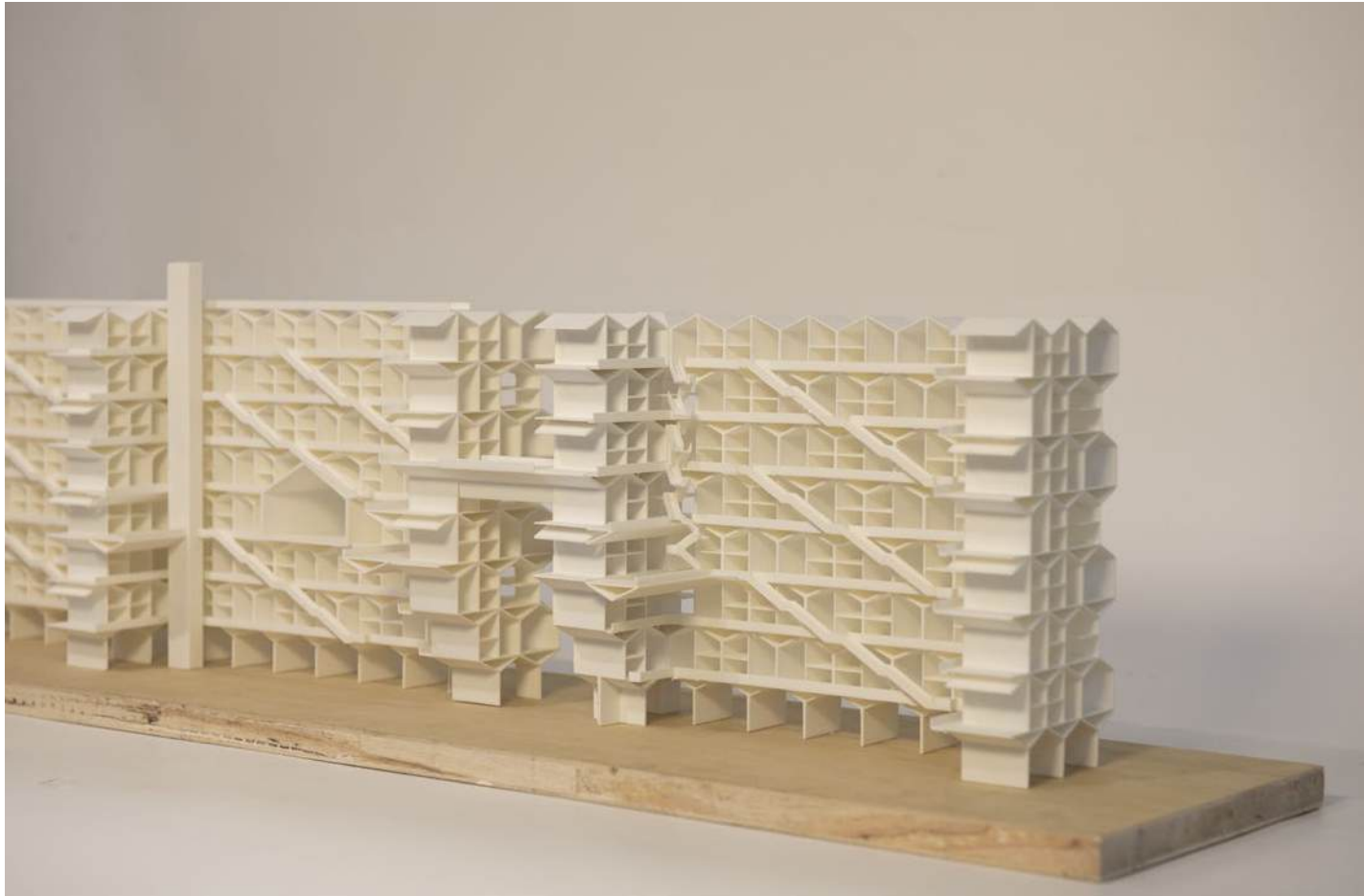
Sectional Axonometric THE INFILL



Public Space - THE INFILL



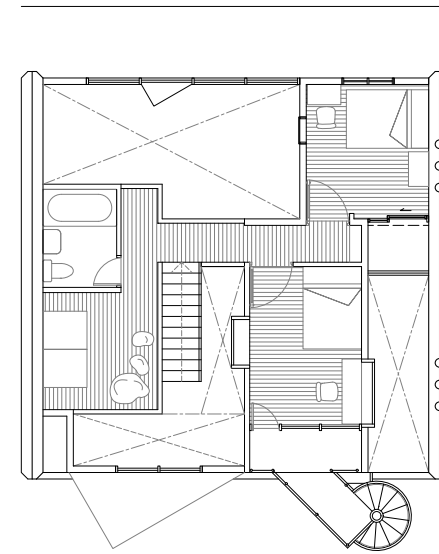
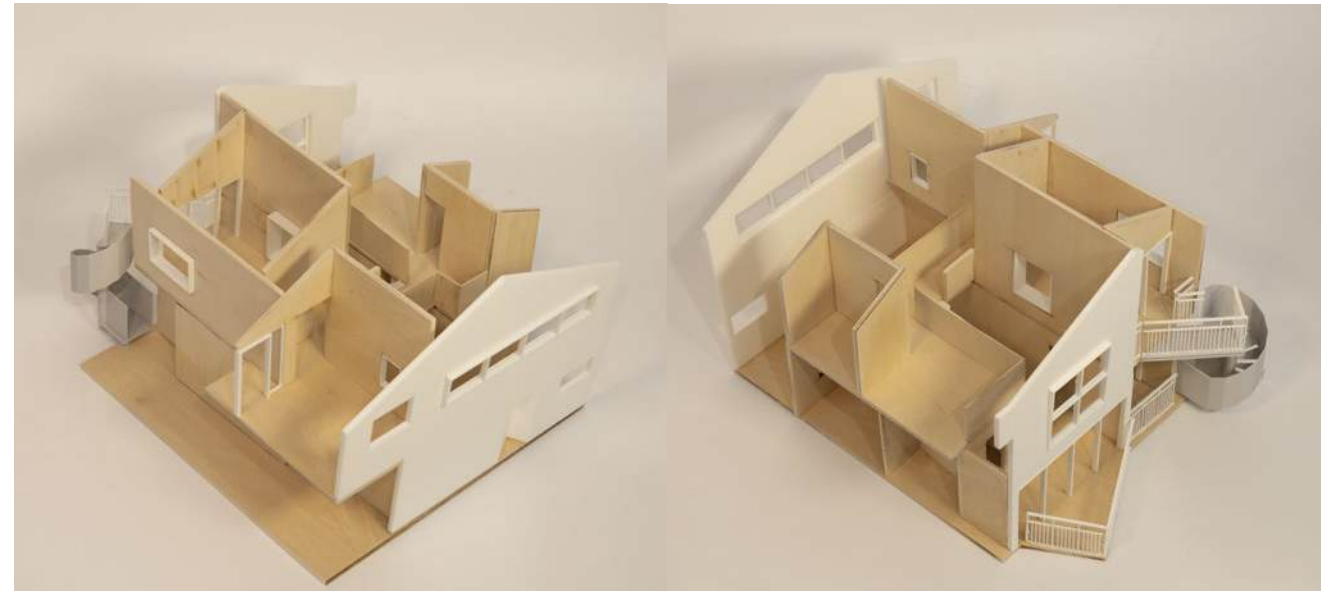
Corridor - THE INFRA



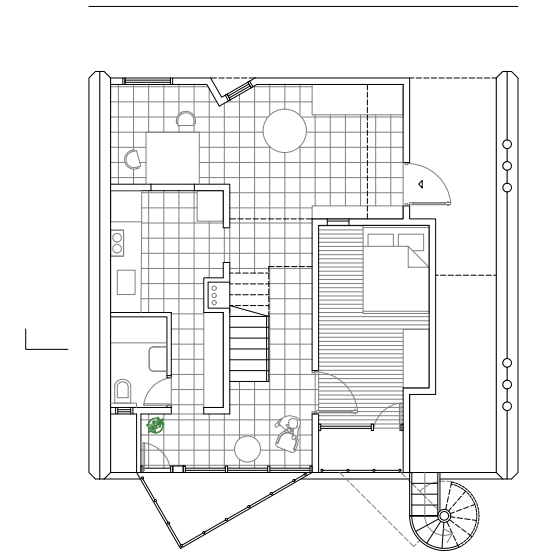
Communal Corculation - THE INFILL



Facade Diversity - THE INFILL



1:50 1F Plan



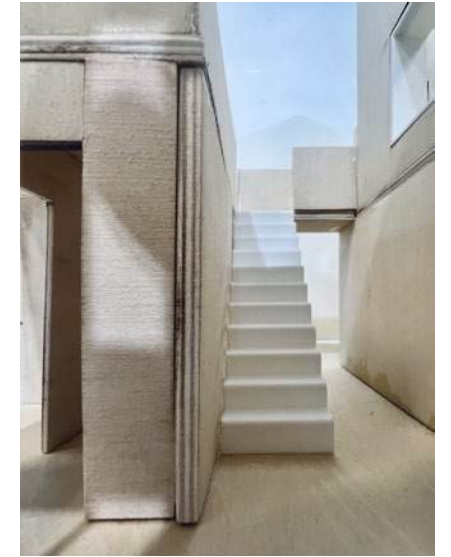
1:50 GF Plan



Modular Void



9m x 9m House Deisign





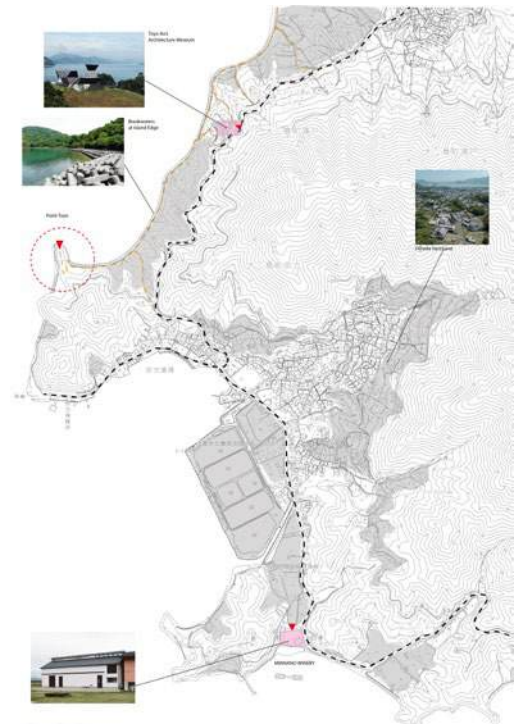
DESIGNING OMISHIMA

Situated on the picturesque island of Omishima in the Seto Inland Sea, this project was developed in collaboration with Toyo Ito, and Kyoto University. Driven by Ito's ongoing initiative to revitalize the aging island, the architectural proposal introduces a comprehensive spatial strategy designed to attract international visitors, youth, and nature enthusiasts. Following immersive on-site research, the intervention unfolds across three interconnected programmatic zones: a dynamic guest house and exhibition space for cyclists, the ecological restoration of abandoned farmland into vibrant mikan (mandarin) landscapes, and a residential community for collective farming. A central conceptual driver is the integration of "observatory" moments throughout the site. The architecture acts as a framing device, curating spatial journeys that visually and tactilely connect visitors to the surrounding trees and the expansive Seto Inland Sea, even allowing them to directly touch the water.

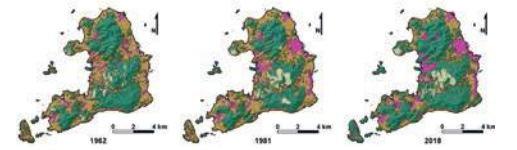
Conceived as a programmatic extension of the Toyo Ito Museum of Architecture, the project aims to foster a renewed community spirit. Ultimately, the proposal culminated in an exhibition at the Toyo Ito Museum—which I co-curated alongside Tokyo University—sparking meaningful dialogue and garnering highly positive responses from Omishima's local residents.

Instructed by Lily Zhang
Omishima, Japan
Sept - Dec 2025

Omishima Island, Japan, within archipelago Geiyo Islands
Seto Naikai or Inland Sea
Longitude: 133°00'00E Latitude: 34°14'34N Area: 64 sq.km



Abandoned Farmland



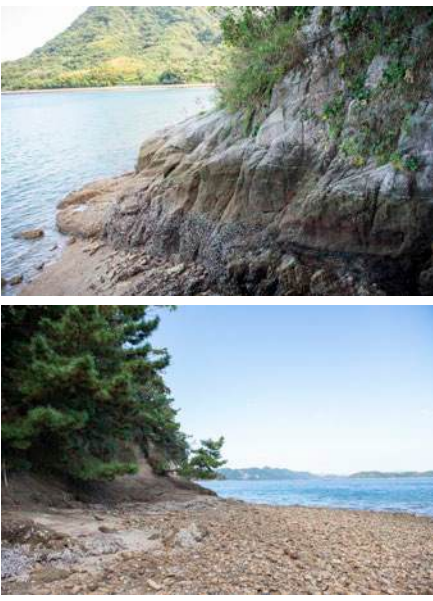
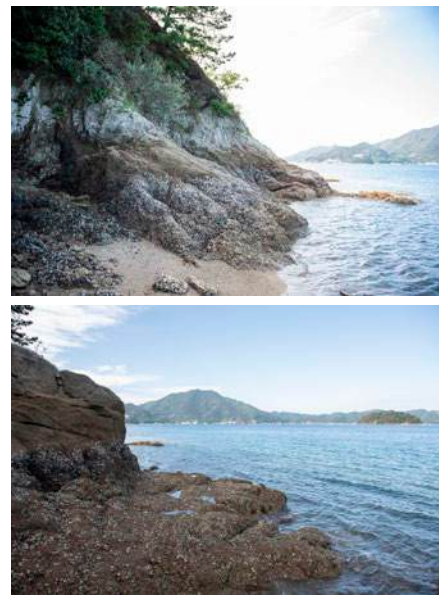
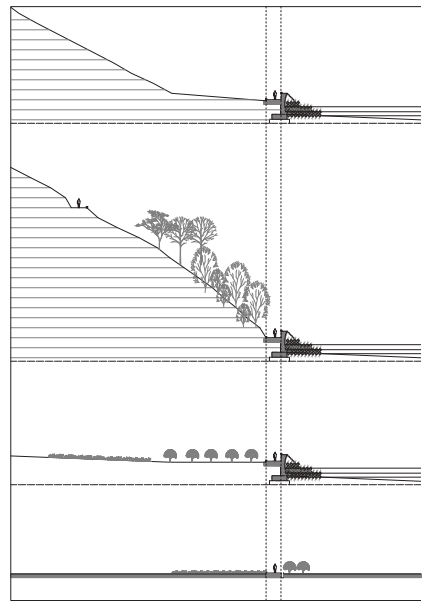
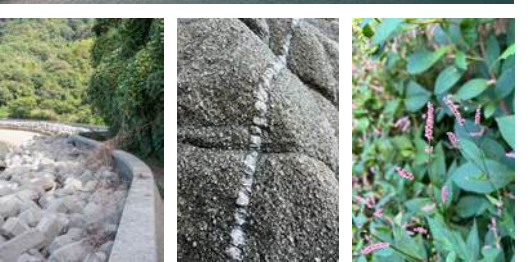
The farmland to developed land ratio reduce due to development of the island. However, due to aging population, abandoned farmland appears more. Japan classifies abandoned farmland into "Type 1 Abandoned Farmland" (recoverable for cultivation) and "Type 2 Abandoned Farmland" (difficult to recover for cultivation due to soil degradation and lack of infrastructure). Renewed buildings should prioritize "Type 2 Abandoned Farmland", as the conversion approval for such land is more lenient (e.g., no review by the Agricultural Committee is required.)



Type 1 Subdivided Flat Land



Type 2 Fragmented Terrain Farm (inaccessible)

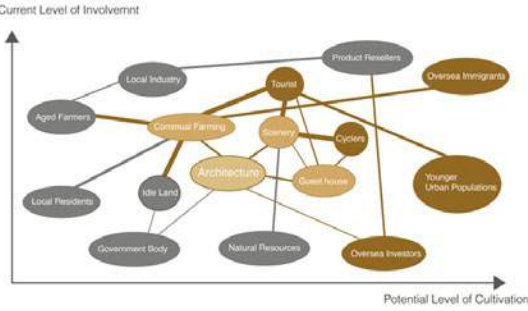


Agriculture + Cultural Tourism: Lightweight Experiential Architecture

Japan's "Regional Revitalization" policy encourages rural areas to develop cultural tourism and attract urban populations for "short-term return to rural areas". For areas with coastal or natural landscapes like Oomishima, renovating "low-impact" experiential buildings on the edge of abandoned farmland can activate land value while avoiding large-scale development that damages the ecology.

Stakeholders & Significance of Renovation

The core of architectural renovation for abandoned farmland in Japan is not to "build new buildings", but to use architecture as a "medium" to connect the land, local residents, and external resources.

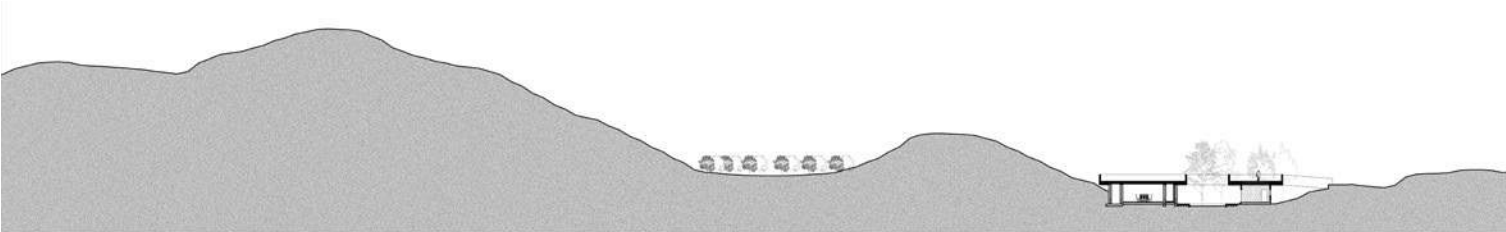


Design Concept & Programms

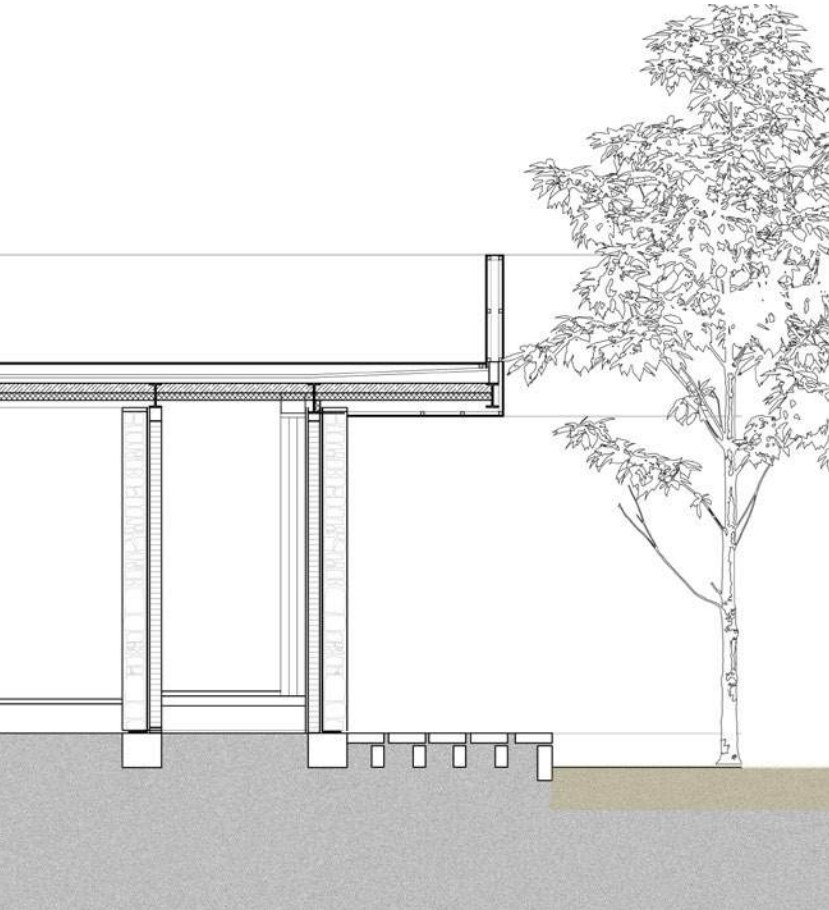
Communal Farming
Farmland transfer is Experienced farmers and overseas natural farming practitioner collaborate to manage a communal garden, providing light physical work opportunities (such as vegetable garden management) to sign therapeutic buildings and reception services in cultural tourism buildings, increasing income while preserving their "sense of village belonging". With the aid of outcoming farmers, crops yield retained for product exporting externally attract overseas investment to the place.

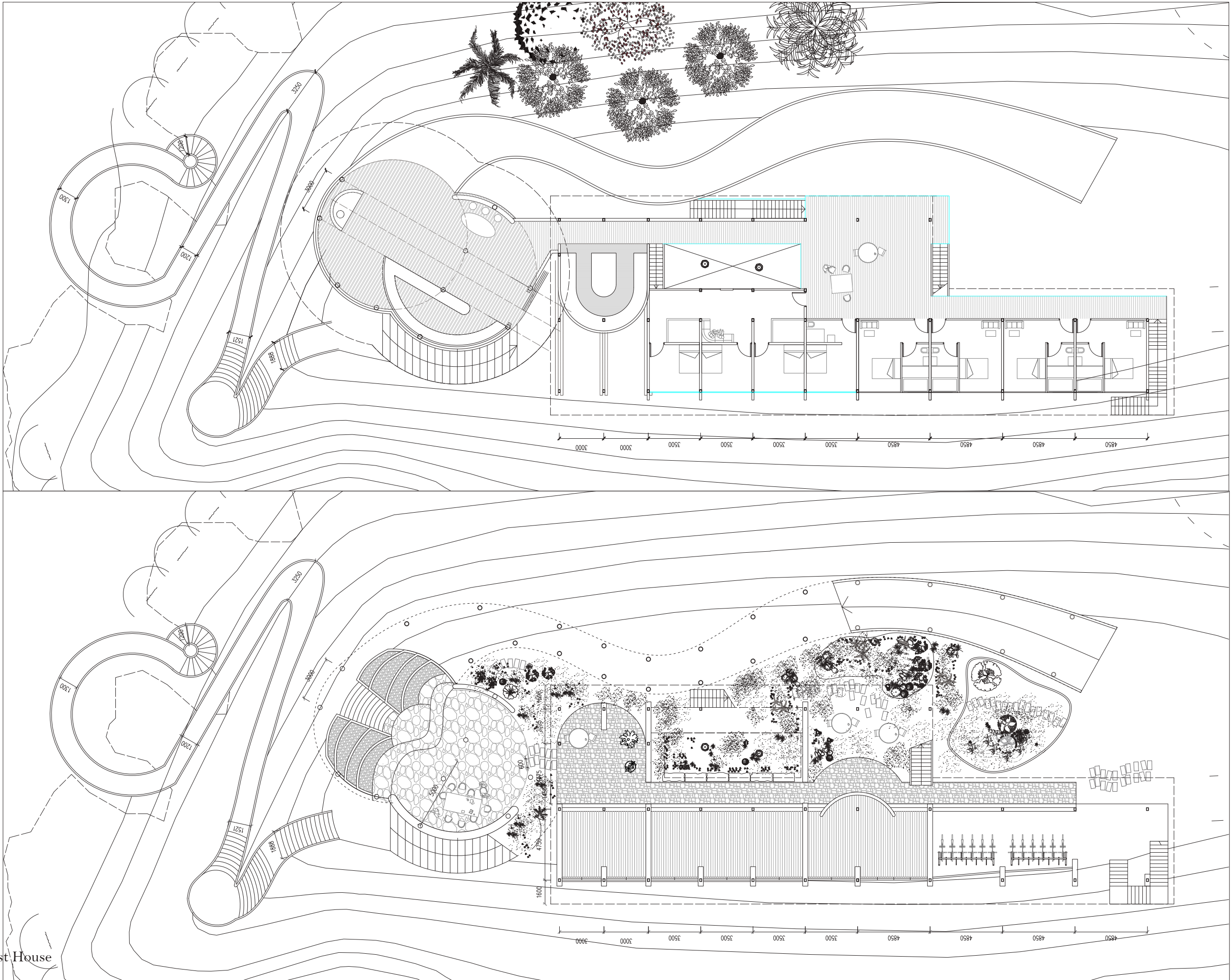
Guest House
To urban population, architecture pieces become entrances to rural areas. Through experiential functions, urban people can rediscover the value of rural areas for scenery, indirectly promoting farmland transfer (e.g., urban entrepreneurs learn about local farmland conditions through buildings and then rent land to develop organic agriculture).

Scenery
Unique island scenery around Seto Island Sea provided opportunities to focus especially for cyclist around the world. For Policy Implementation: Architectural renovation is a "visual center" of Japan's policies such as "effective farmland utilization" and "regional revitalization". It can serve as a demonstration project to attract more resources to rural areas, forming a virtuous cycle of "renovation - profit - re-investment".

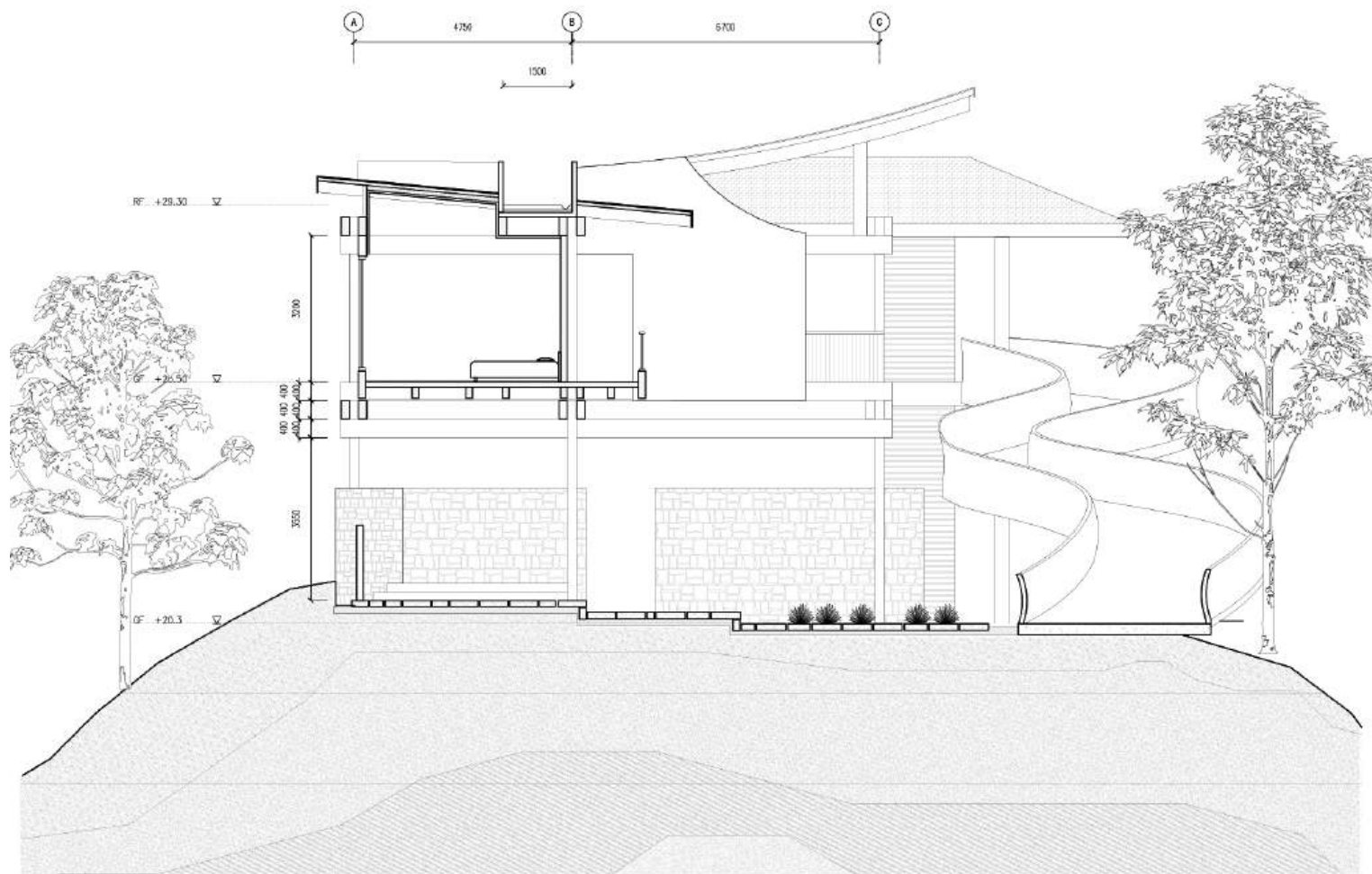


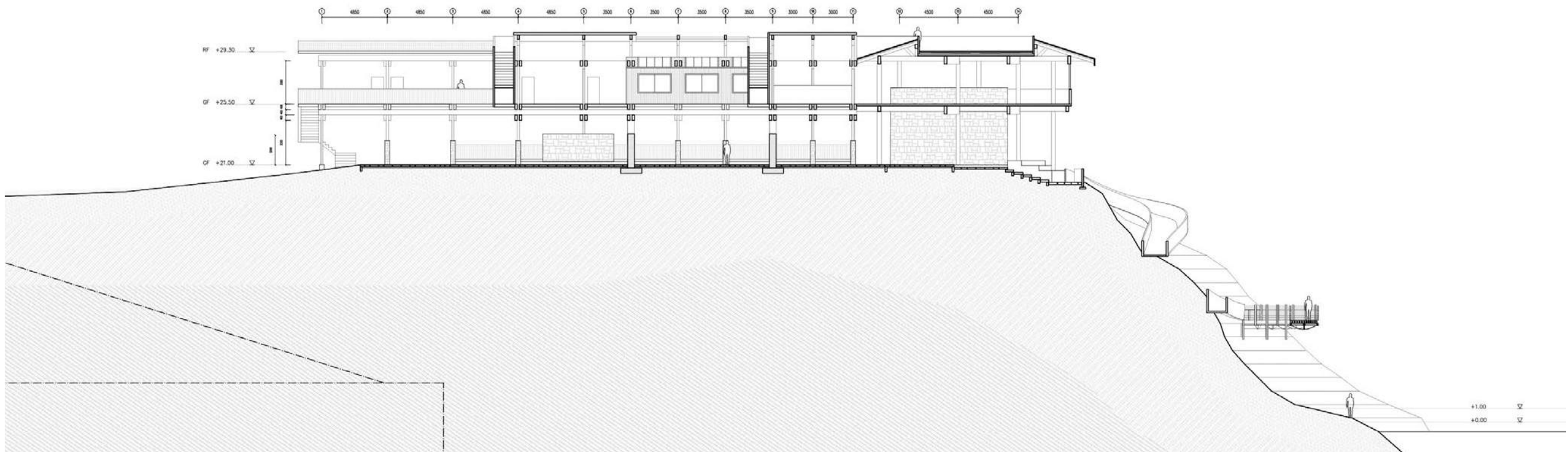
The Co-farming House





The Biker Guest House



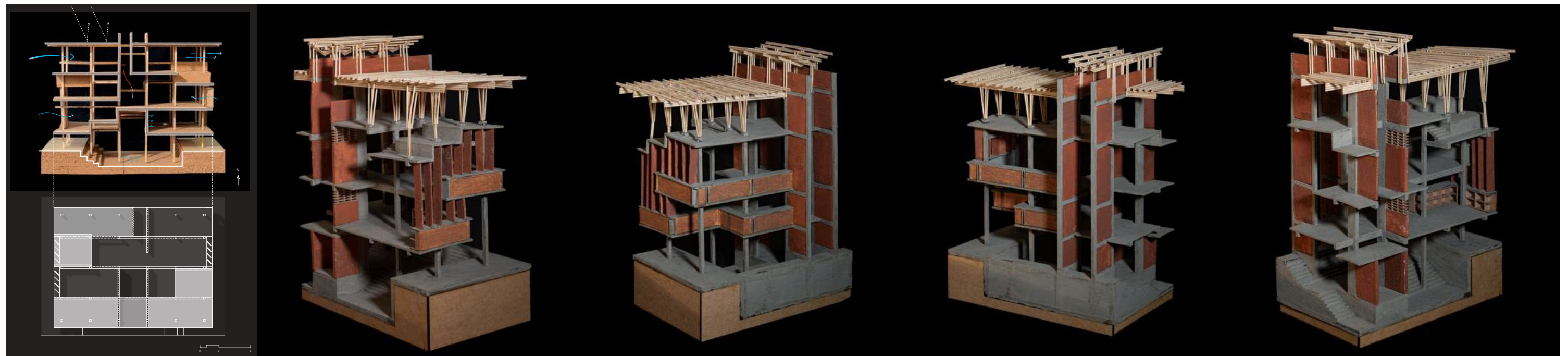
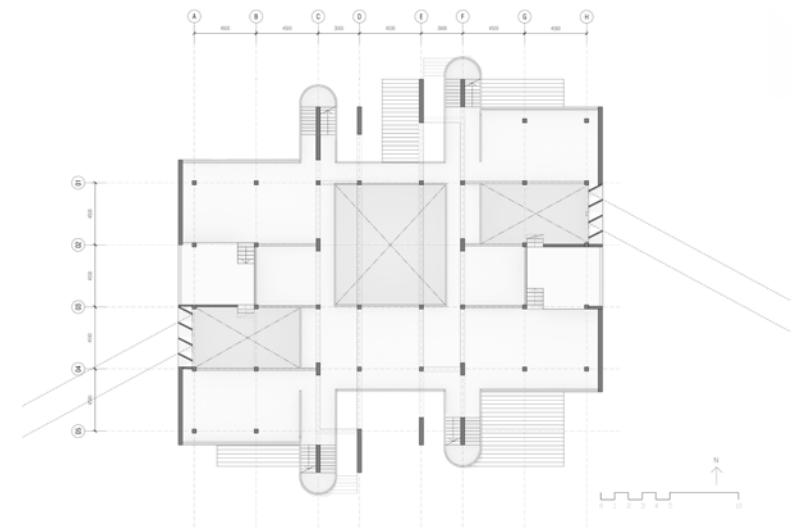


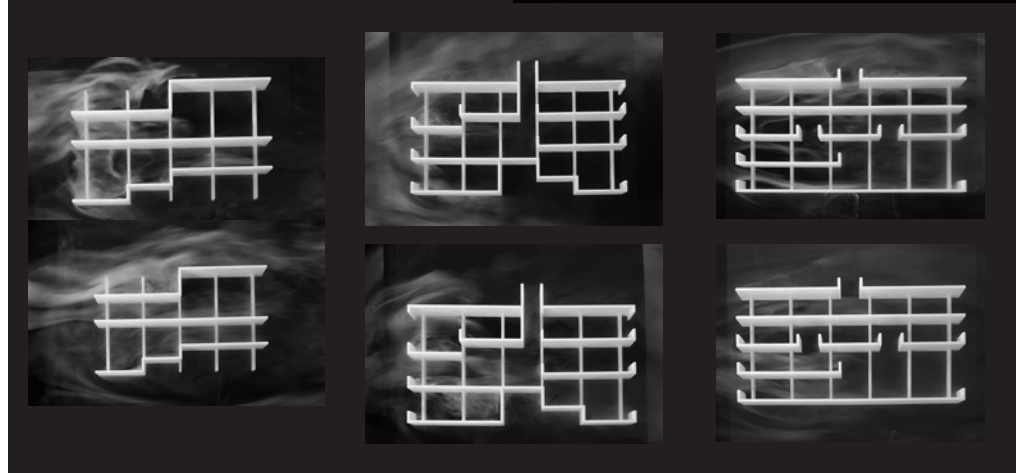
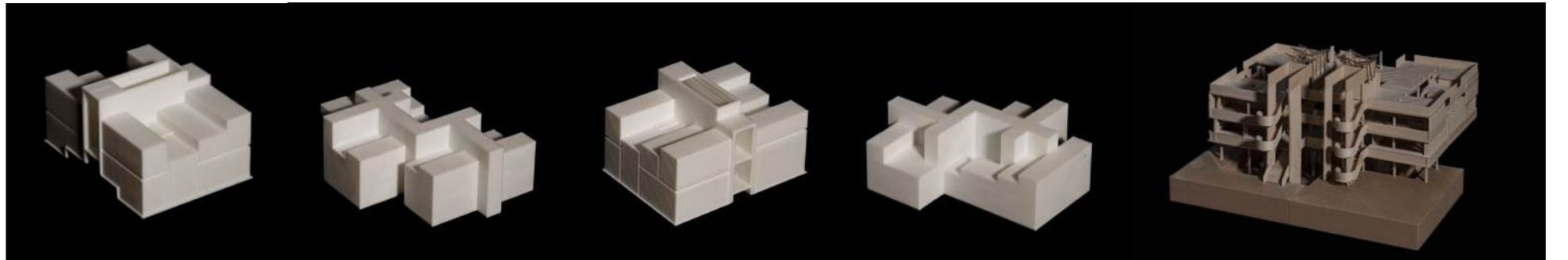
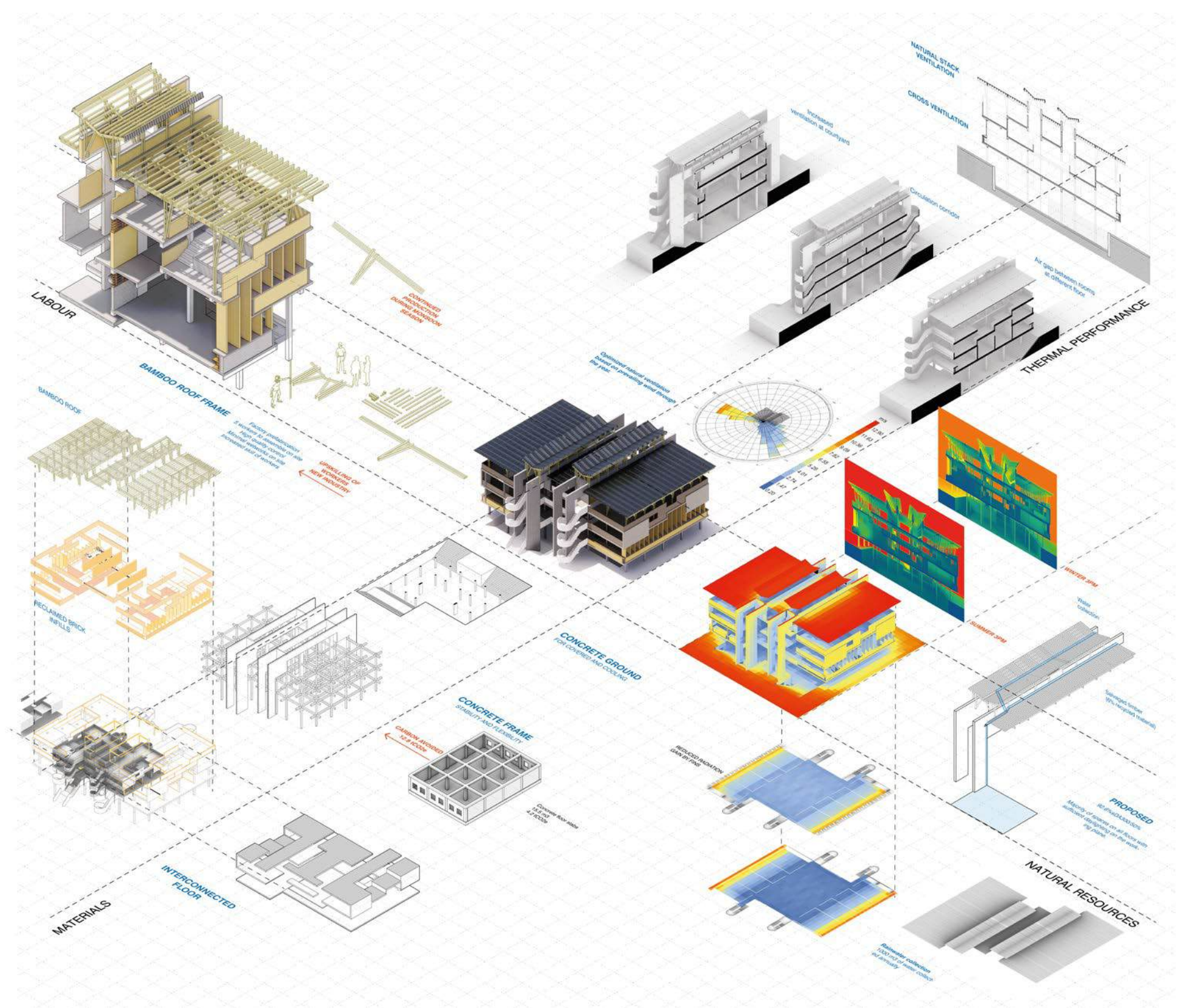
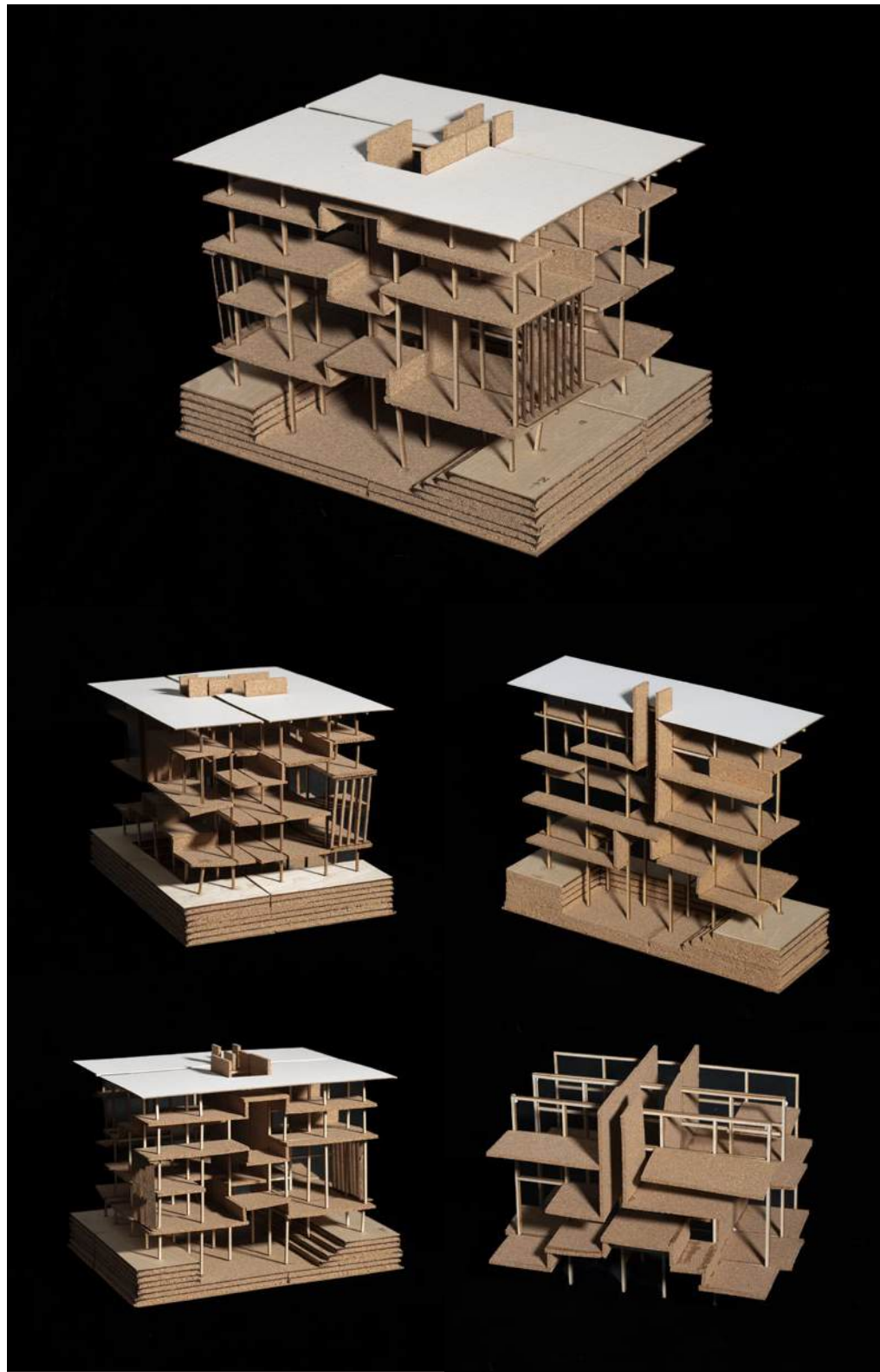


Instructed by Jersey Poon
Hydrabad, India
Jan - Apr 2025

PASSIVE AGGRESSIVE

The Key Concept of the prototype is to boost the porosity of building through shifting interior level of space and providing wind corridor, taken cross ventilation and stack chimney as main passive cooling strategy in hot and semi-arid area Hyderabad. The main building mass is lift up so as to allow air flow through the bottom (underground plaza) and breeze between the bamboo roof and top level of the building, in response to strong and constant prevailing wind both from the east and the west. Differential pressure is created by change in void size. This allow a boost of cross ventilation. In addition, stack chimney is driven by buoyancy improved with reflective metal sheet double roof.





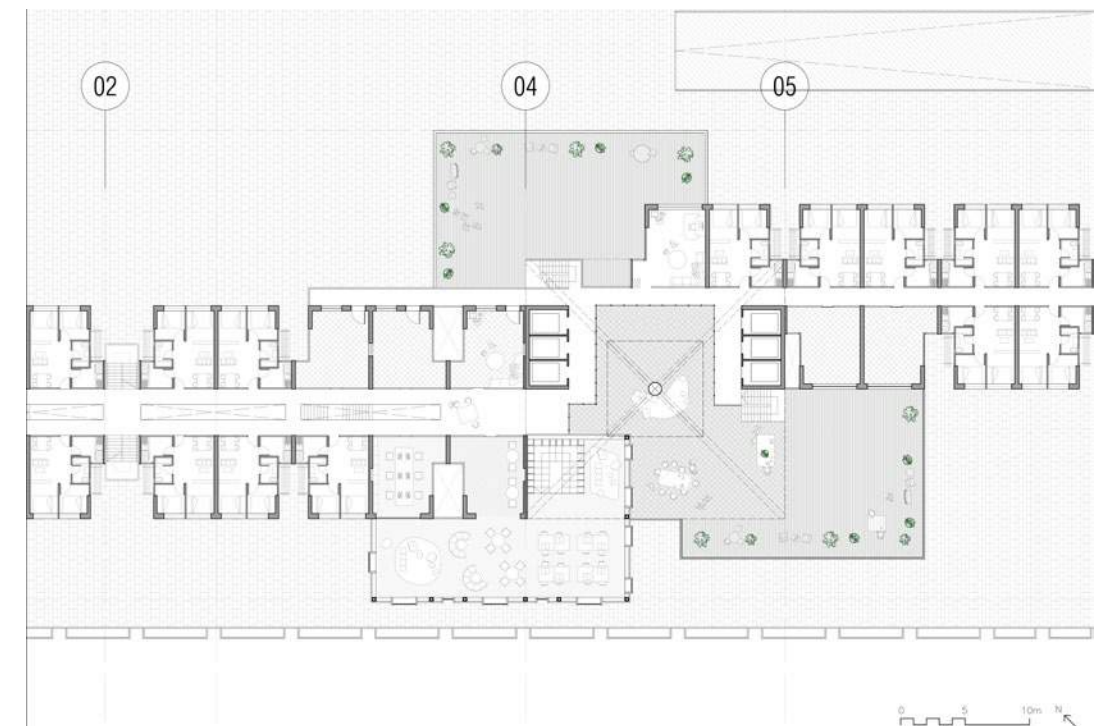


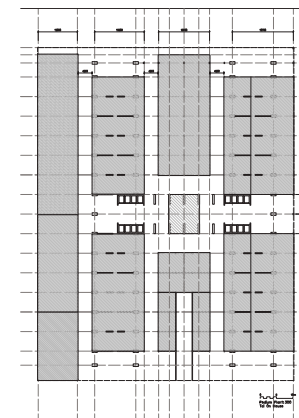
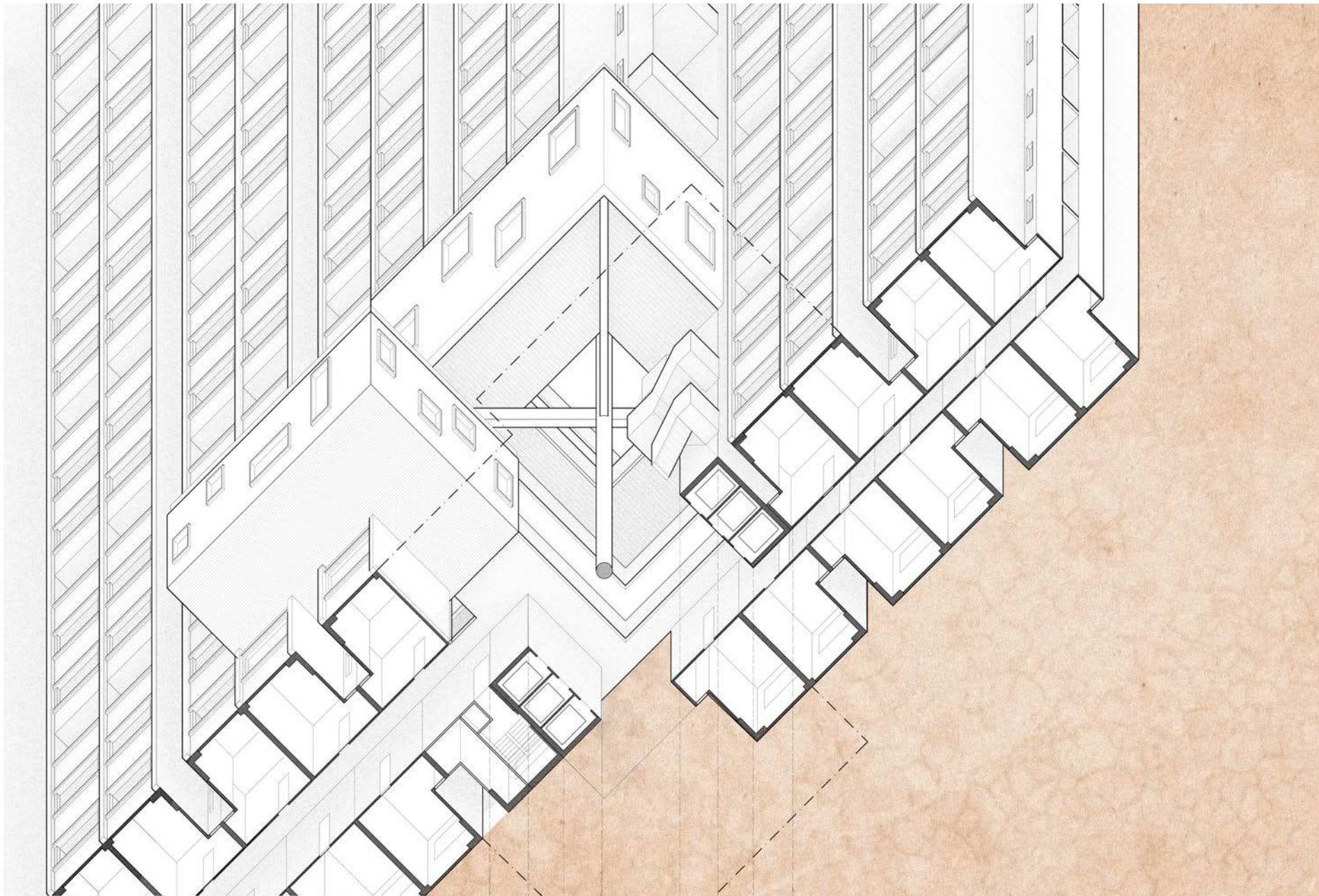
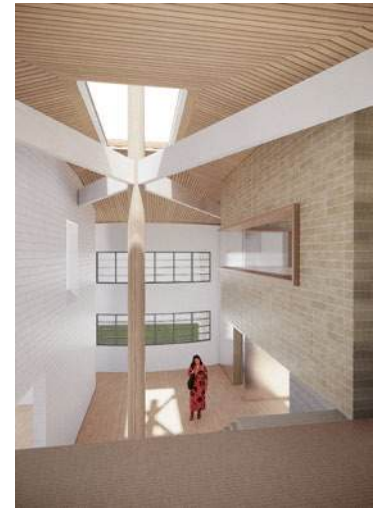
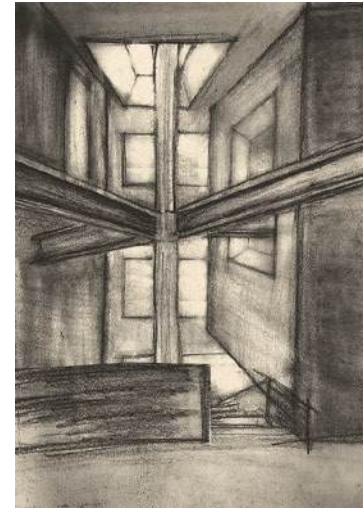
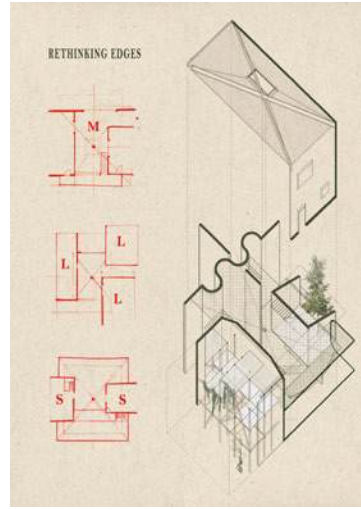
Instructed by Clover Lee
 Tai On Building, Hong Kong
 Sept - Dec 2024

DEFINING MEDIUM SCALE IN HOUSING THROUGH IMAGES

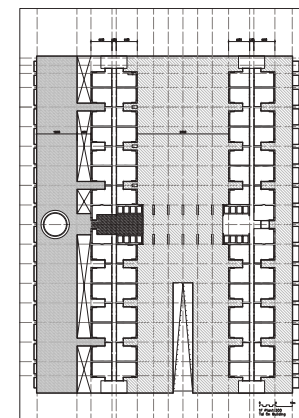


This project is located in Sau Kei Wan. The existing Tai On Building is taken as a reference prototype of a super high density housing project in Hong Kong. The original housing complex is designed in an H-shape form, accommodating 68 units per floor, and over 2100 units in the whole building. The circulation and layouts are designed in the most efficient way. In terms of scale, this building is huge. Through introducing a medium voids with a localised central column and roof, the living spaces of different characteristics are gathered and unified. For example, the hallway in the longer wing is expanded for multifloored interaction. For the people who live in the shorter wing, they have more privacy. Within the building mass, different degrees of privacy and interaction are provided. The medium space brought together people of different generations, both active and less active residents. The design of this medium space is generated from exploration of a medium scale space in different contextual scale. A medium scale space would acts as a cover like these trees to unify the discrete building blocks. Taking inspiration from a fragmented third space in outdoor area, the edges of each zone has the richest interactions that different material meets. The medium space has a variety in materials. It gives an expression that the central column, which is supporting the roof, acts as an envelope to wrap two rooms together. Instead of considering the rooms as compartments of the house, each room has its own quality, but the roof is used to unify them.

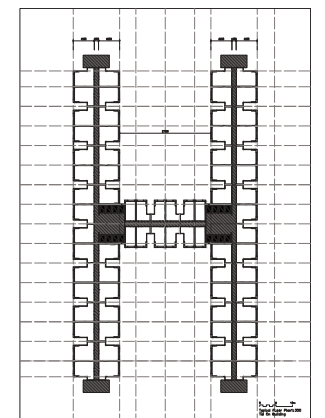




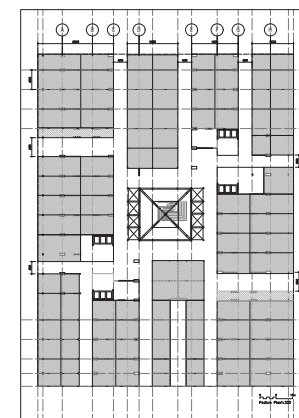
Tai On Building Ground Floor Plan



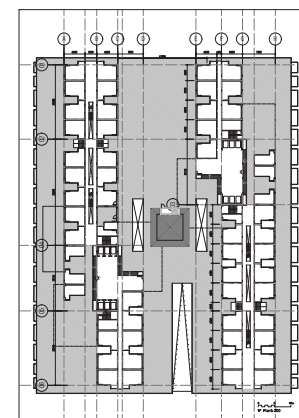
Tai On Building First Floor Plan



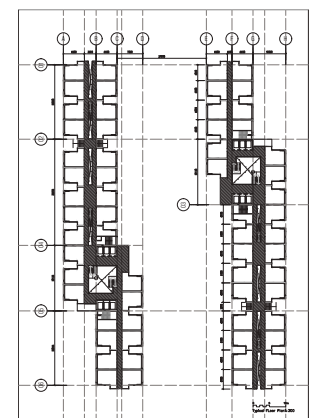
Tai On Building Typical Plan



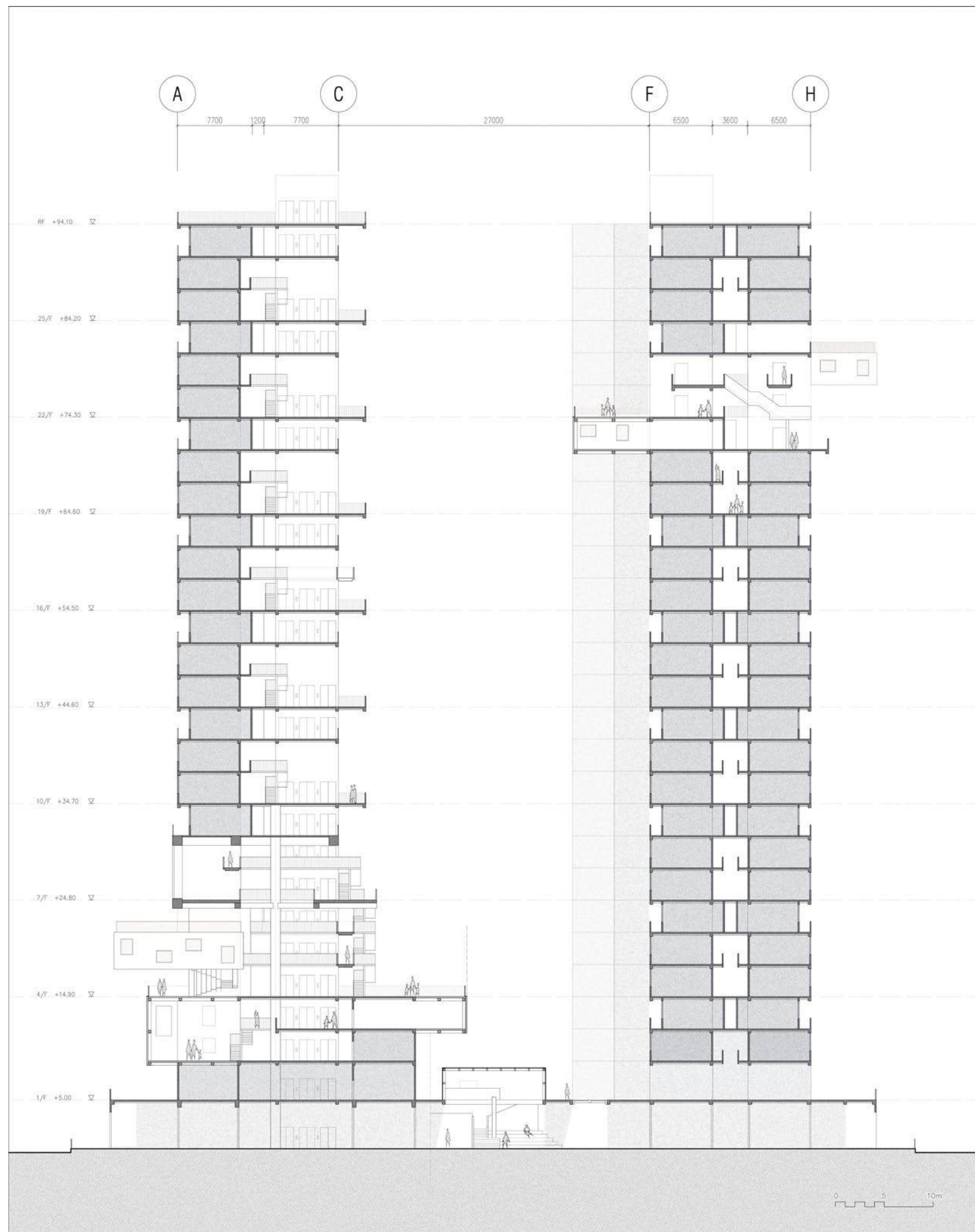
Proposed Ground Floor Plan

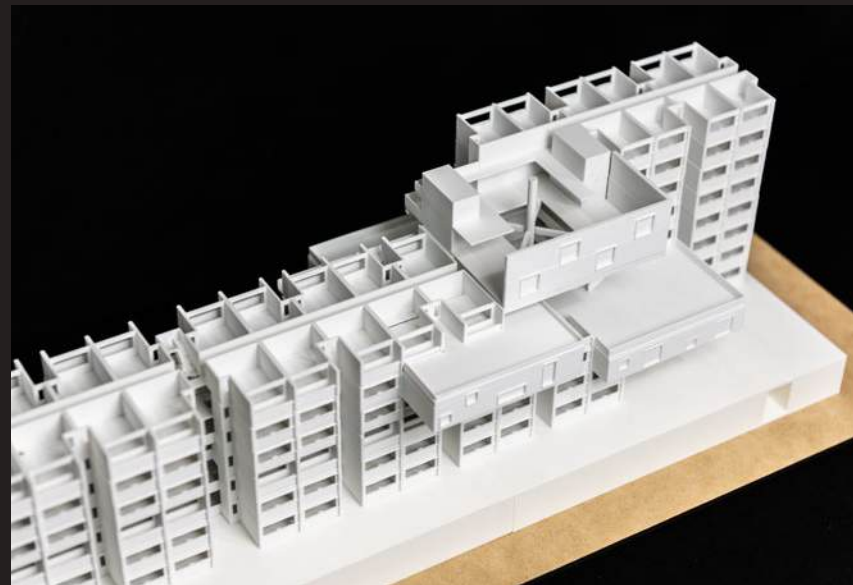
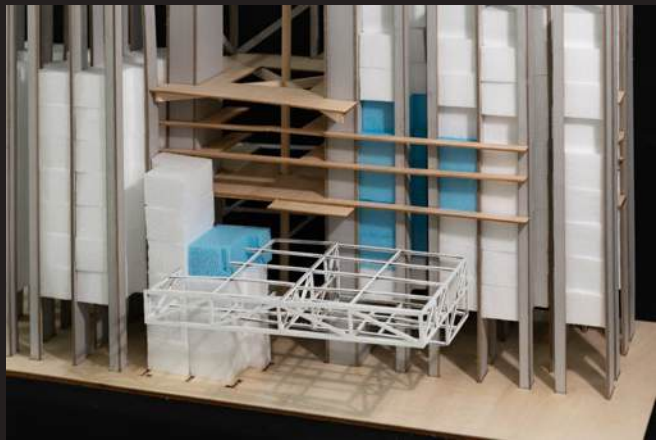


Proposed First Floor Plan



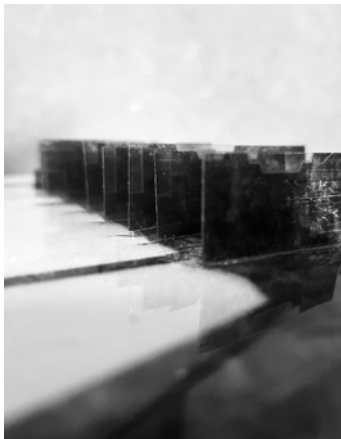
Proposed Typical Plan





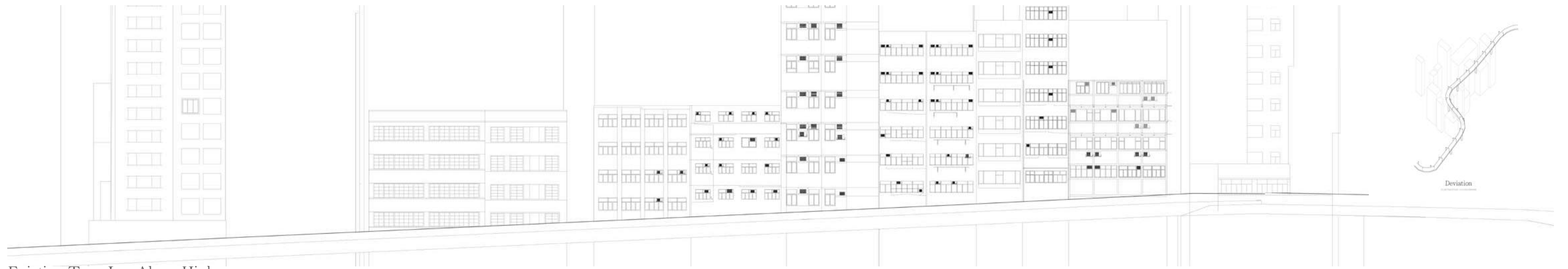
OFFICE OF DEVIATION

A Horizontal Mixed-used Office against High Density District
Instructed by Prof. COMER, Fergus

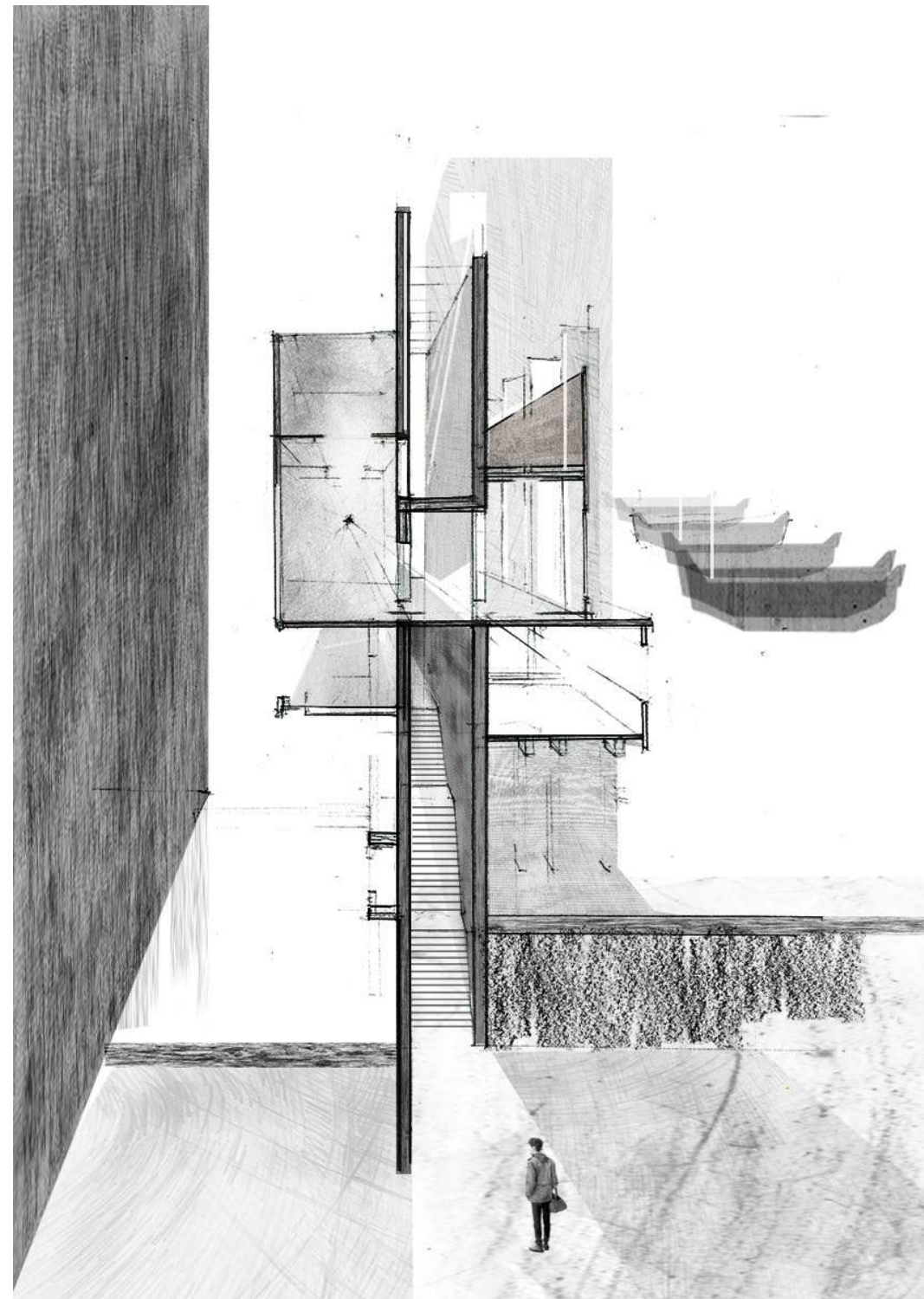


Tin Hau, Hong Kong
Janunary 2022- April 2022





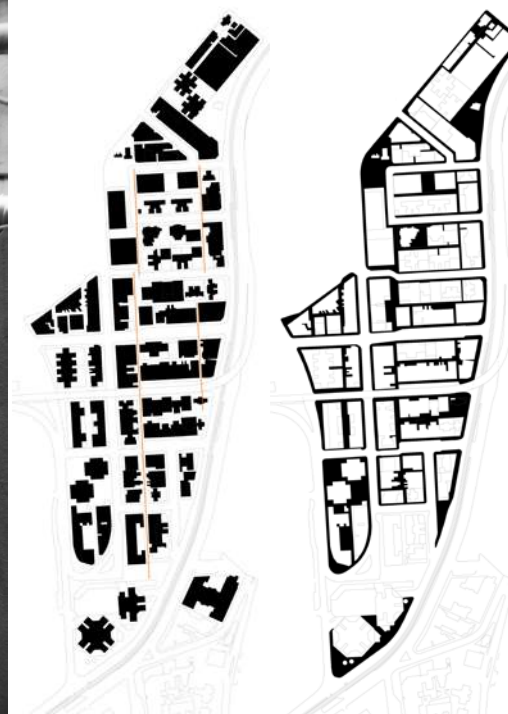
Existing Tong Lau Along Highway



Concept Sketch



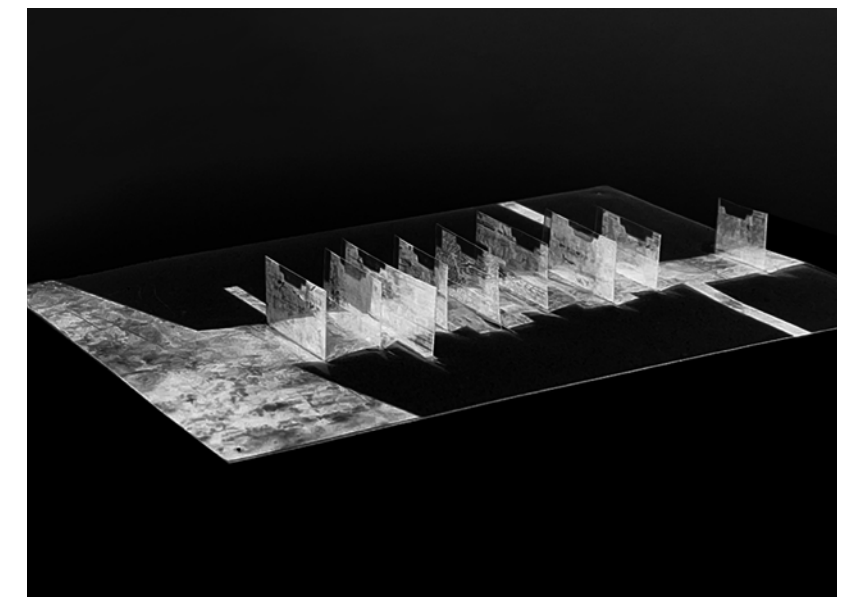
Alley and Density Studies



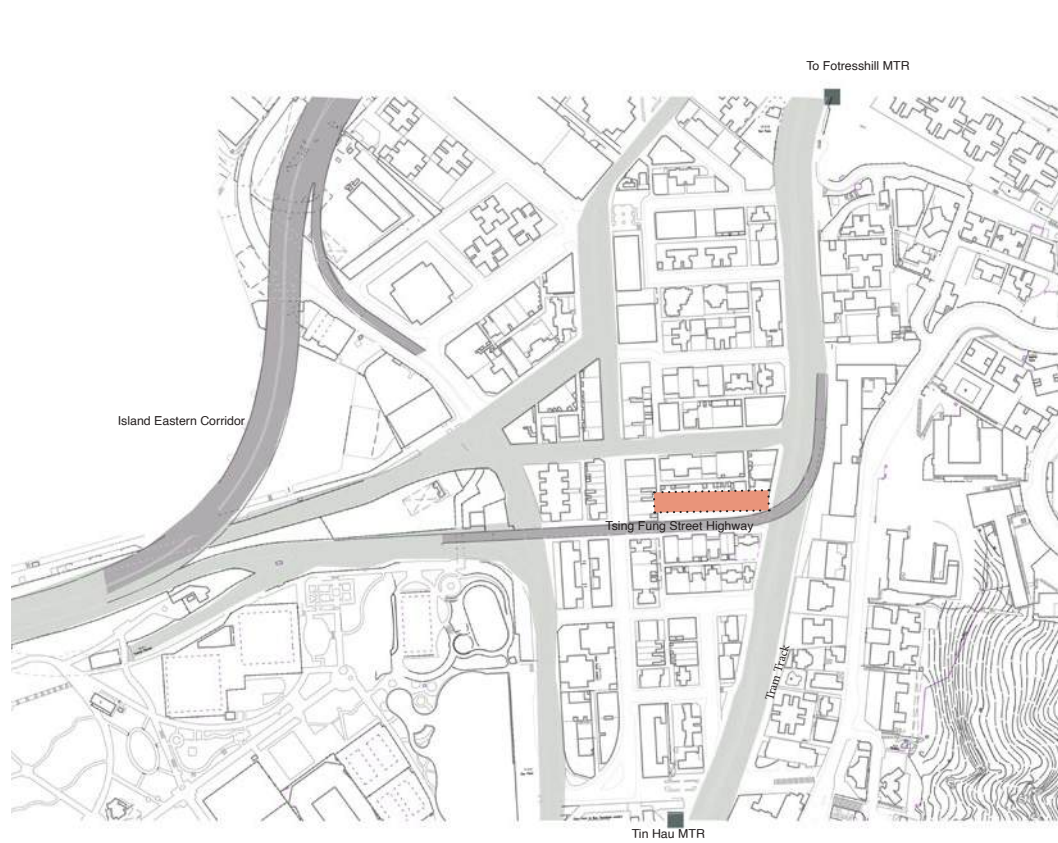
Design Concept

The design intension is to translate a urban gesture of deviation from a highway typology into building scale, in which the spaces are arranged along a curve datum wall, and views wrapped by the deviating highway.

The office space are lifted up while the ground floor is free up for pedetrain and civic activities. The building is oriented close to the highway while leaving wider spaces for the activation of back alley. Programmatically, the building aims to reconnect the public to the history of press printing. As a result, an existing Tong Lau which has the closest distance to the highway were preserved as archive. Co-working space is also designed by resuing and breaking the non-structural wall of existing tong lau to form a linear work space in response to the dynamic with the highway. Communication and interaction are stimulated to generate idea and creativity.



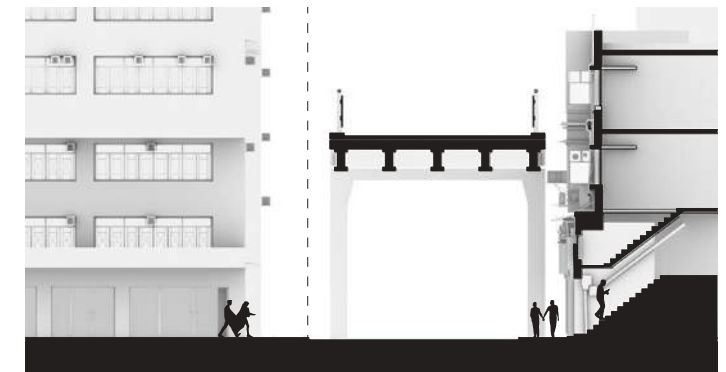
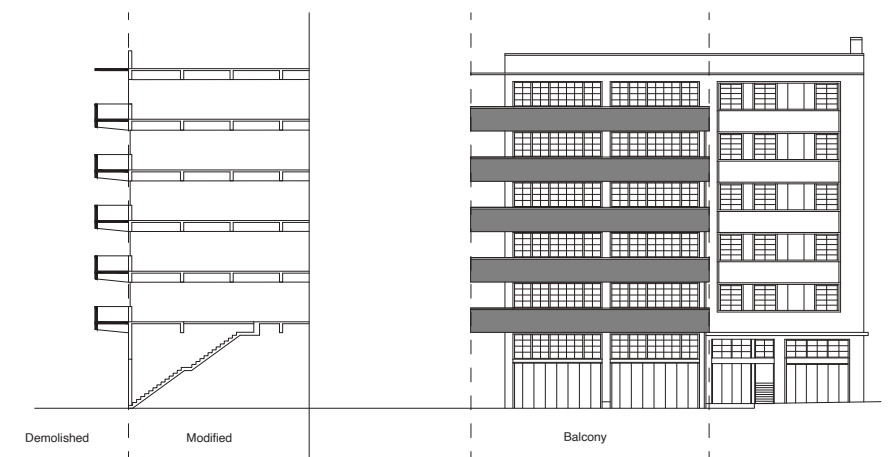
Under Highway Voids



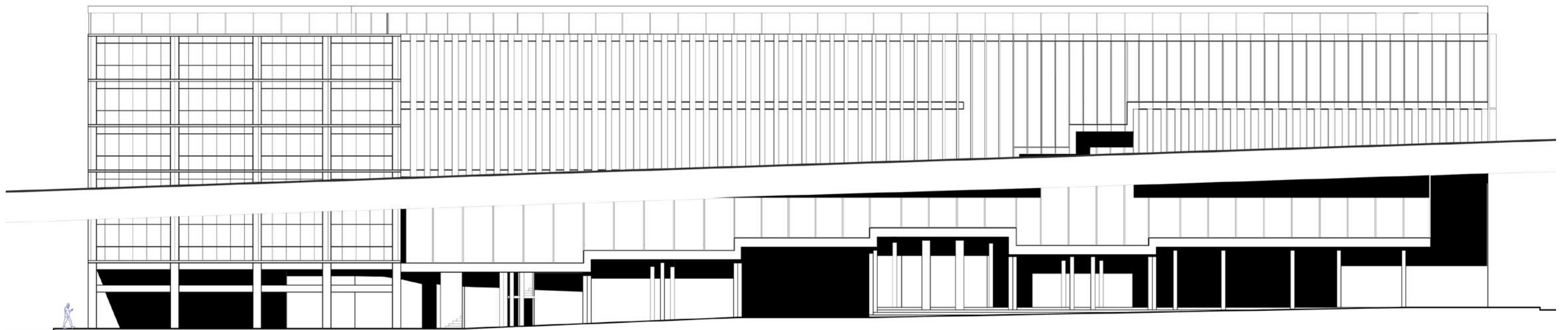
Site Map



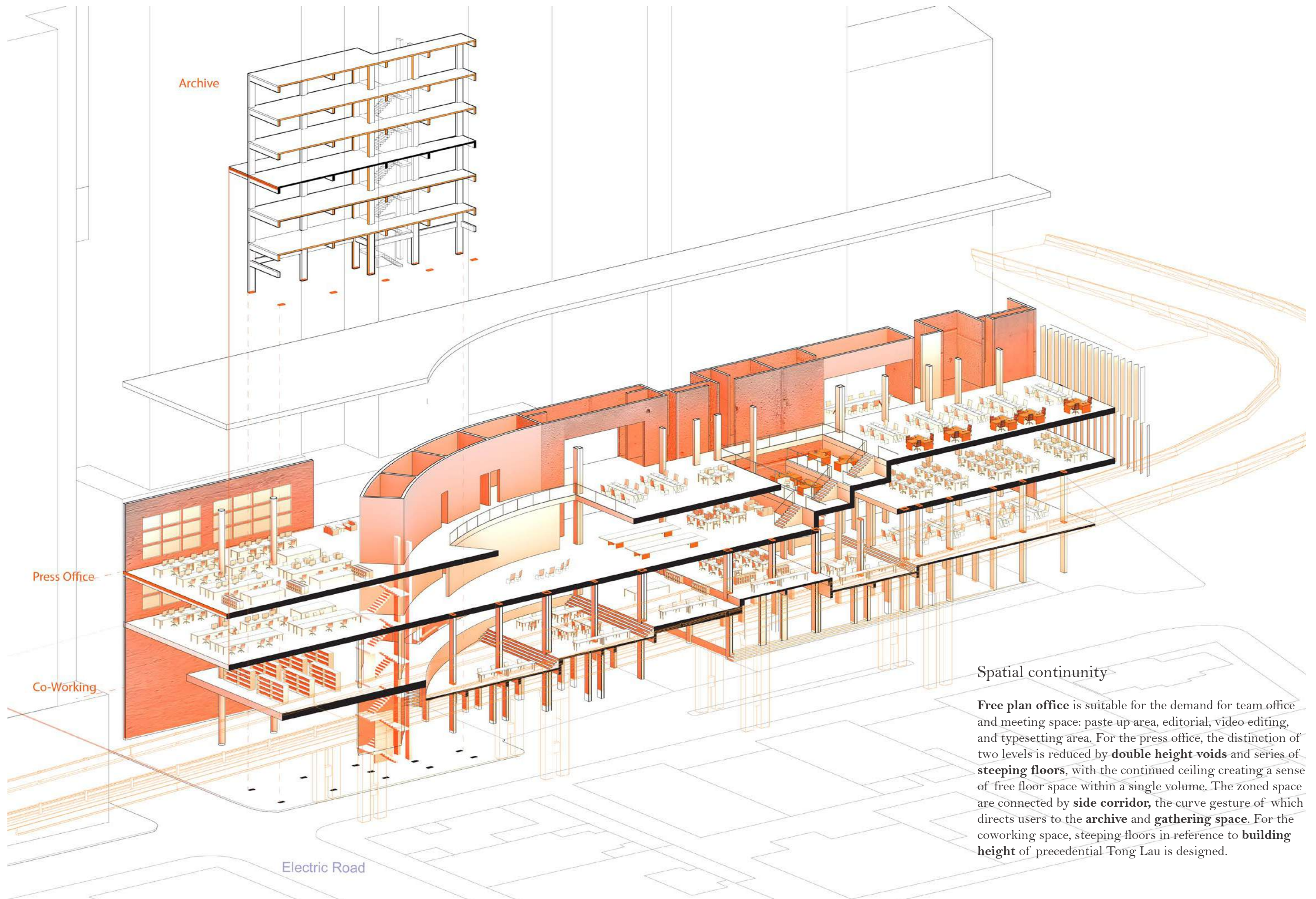
History of Highway and Press Industry



Tong Lau and Highway Conditions

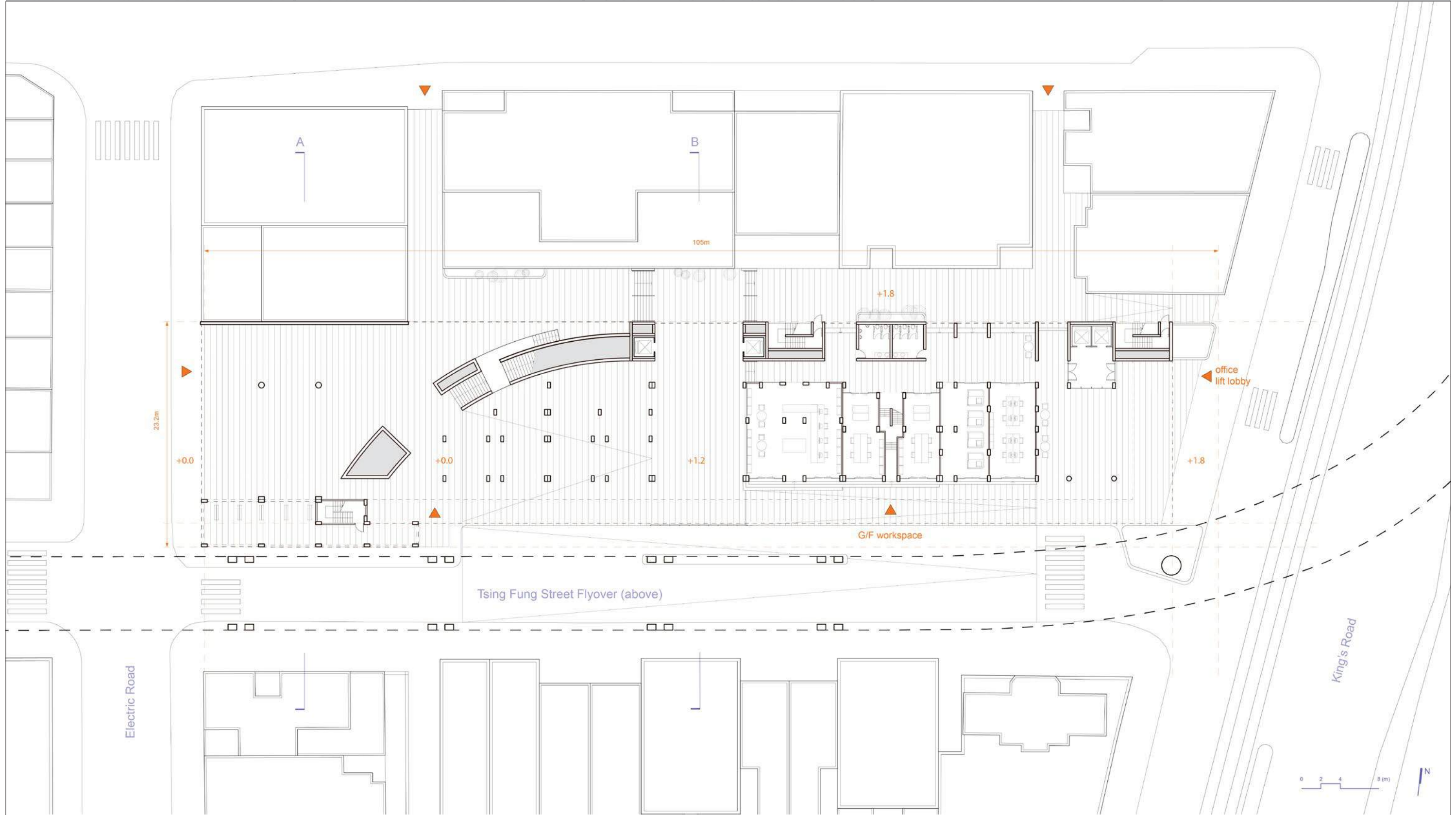
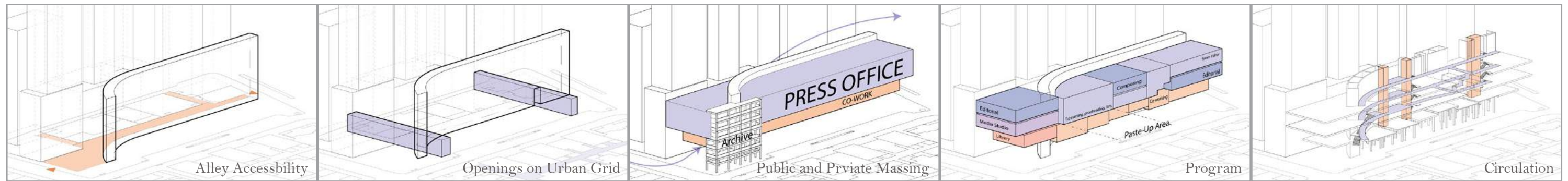


Front Elevation



Spatial continuity

Free plan office is suitable for the demand for team office and meeting space: paste up area, editorial, video editing, and typesetting area. For the press office, the distinction of two levels is reduced by **double height voids** and series of **steeping floors**, with the continued ceiling creating a sense of free floor space within a single volume. The zoned space are connected by **side corridor**, the curve gesture of which directs users to the **archive** and **gathering space**. For the coworking space, steeping floors in reference to **building height** of precedential Tong Lau is designed.



Ground Floor Plan



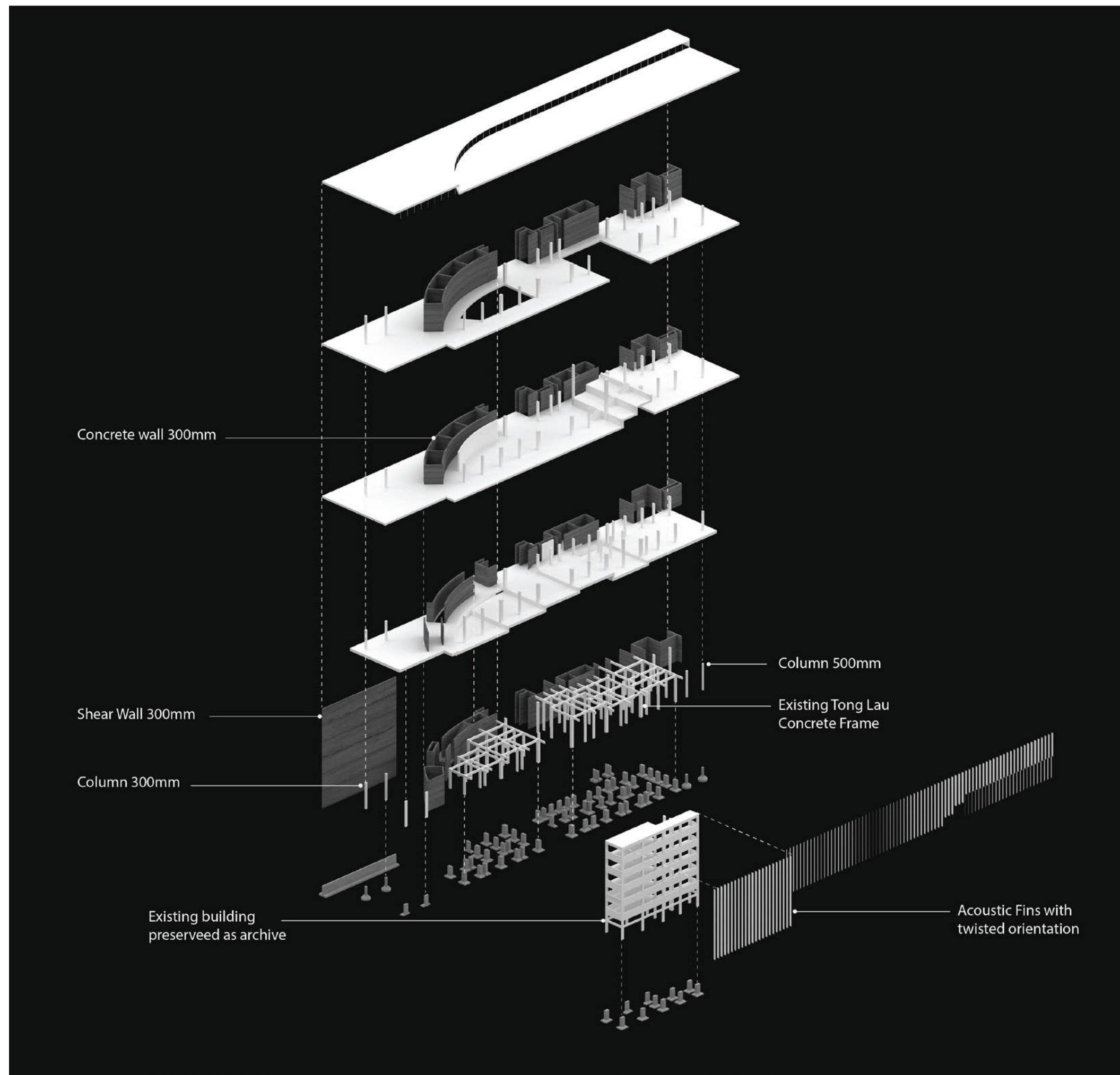
Public Square and Ground Floor Connectivity



West Entrance Street View



Revitalised Open Alley

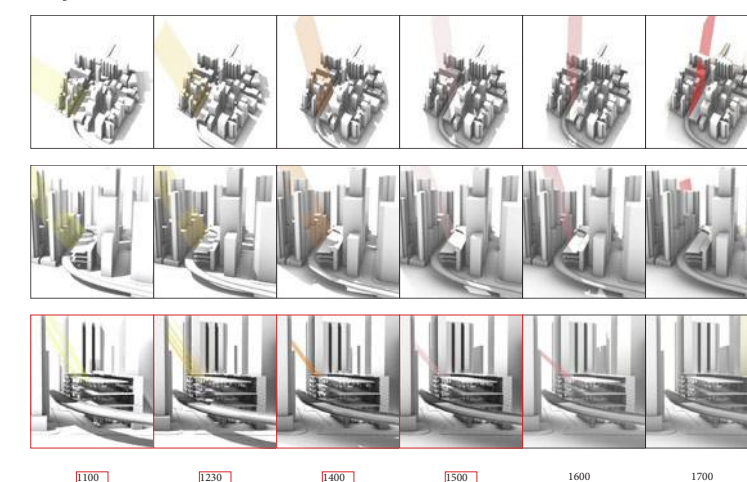


Structure and Facade Design

A Tong Lau at the corner of the urban block is preserved as **an archive** as it demarkcate the history of press industry and how the residential building reacts to the construction highway. The new horizontal building mass intervent old vertical Tong Lau by breaking the non-structural wall while keeping ad reinforcing the **concrete frame of the existing's**.

The building facade was designed according to sunlight studying regarding nearby urban blocks. A gradient vertical fins is deisgned in response to **shading efficiency** and **vision of viecles** on the highway.

Shading Condition on 21 October



Shading in October only



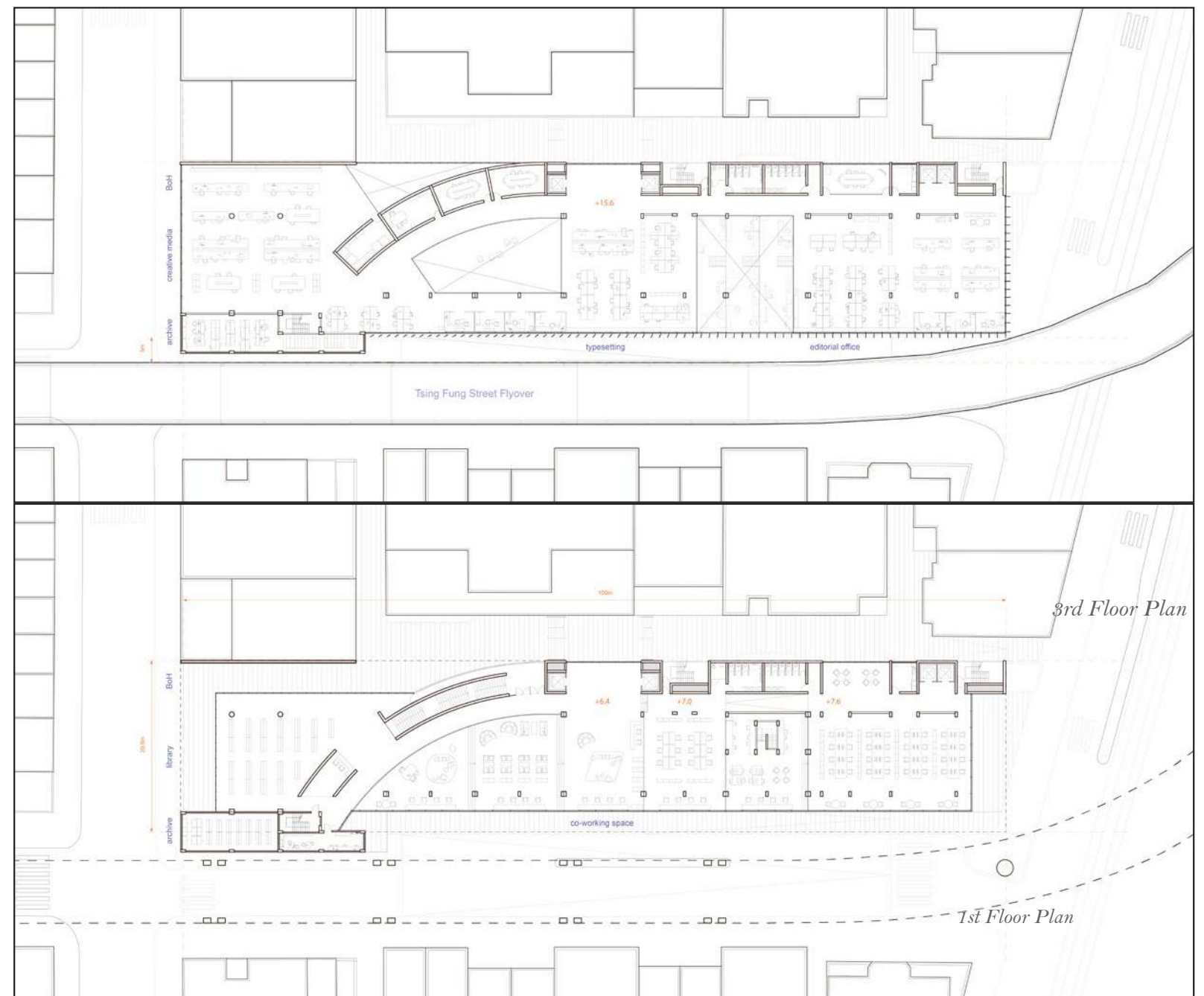
Jun, Sep, Oct Noon

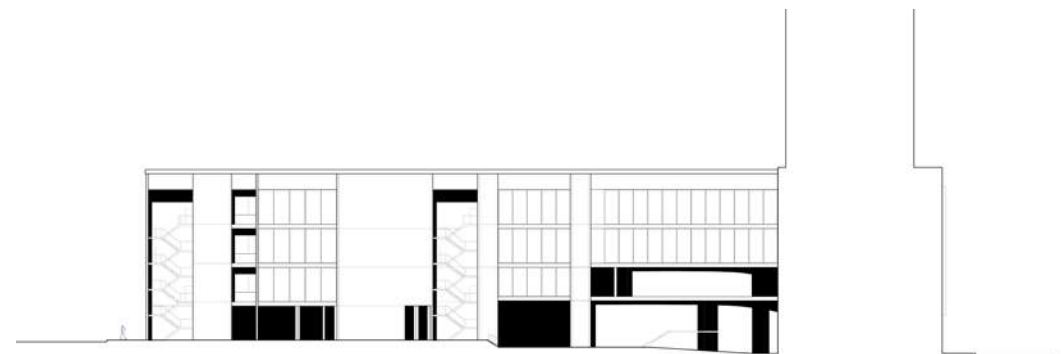
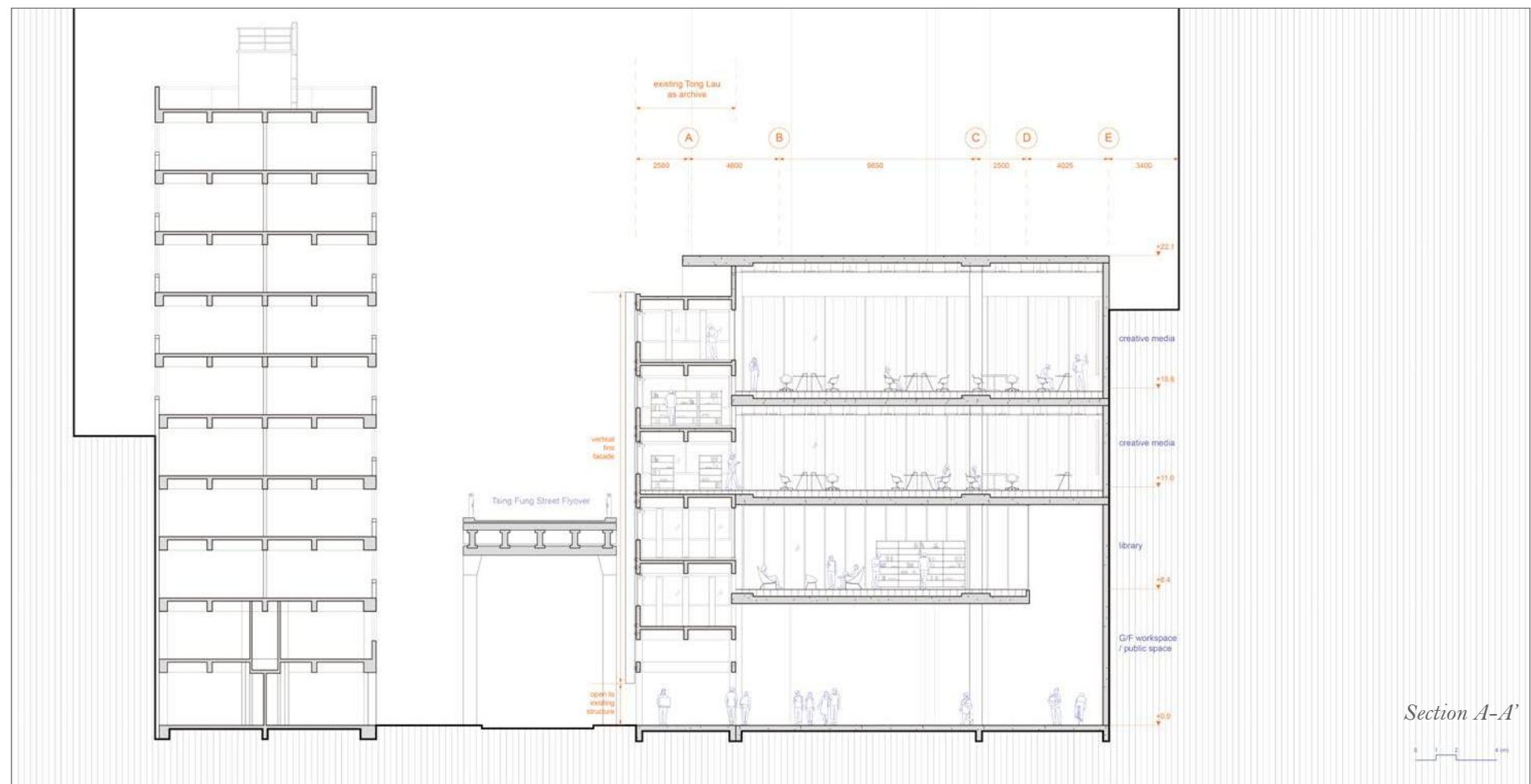


Jun, Sep, Oct Afternoon

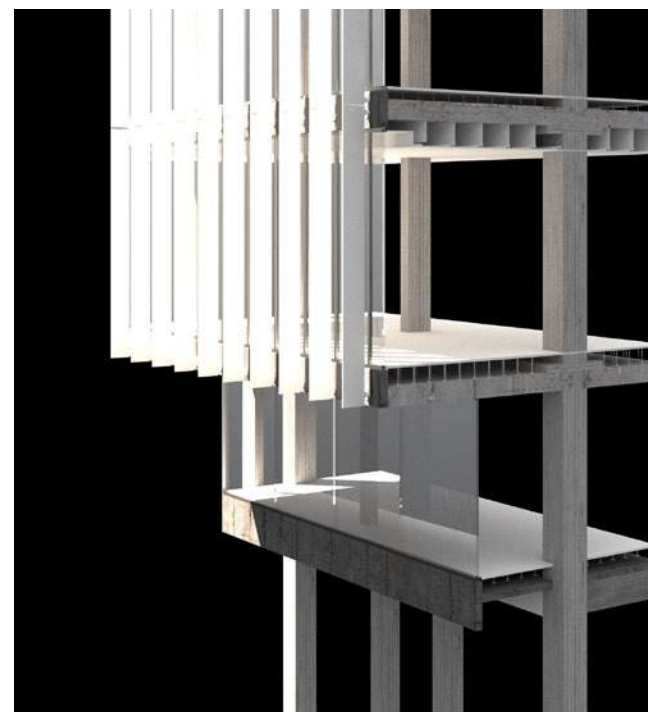
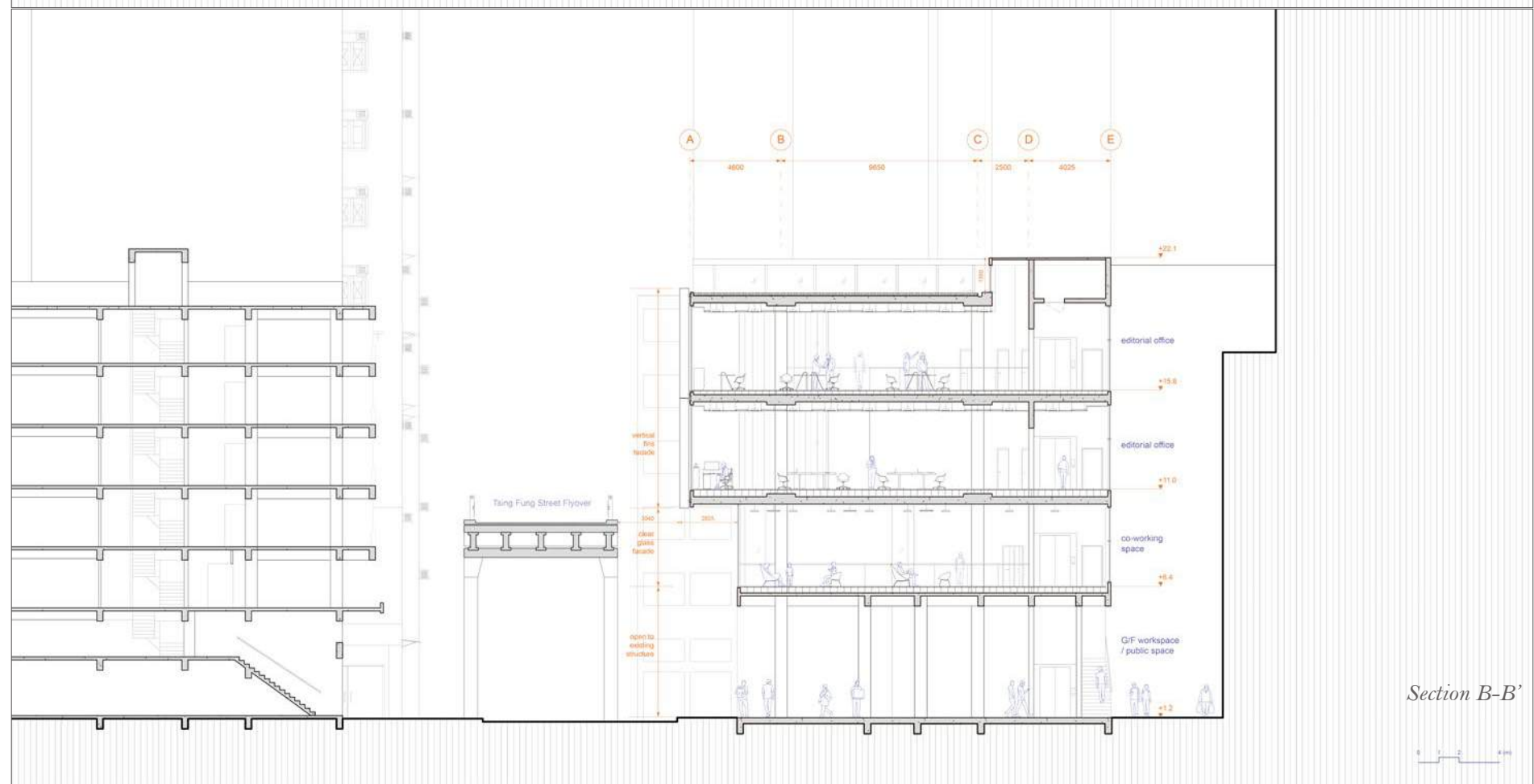
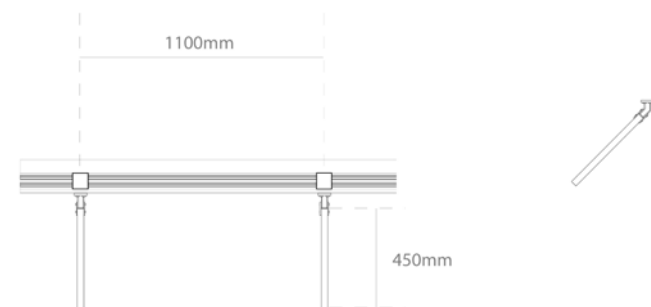


Sunpath and Daylight Studies on Facade

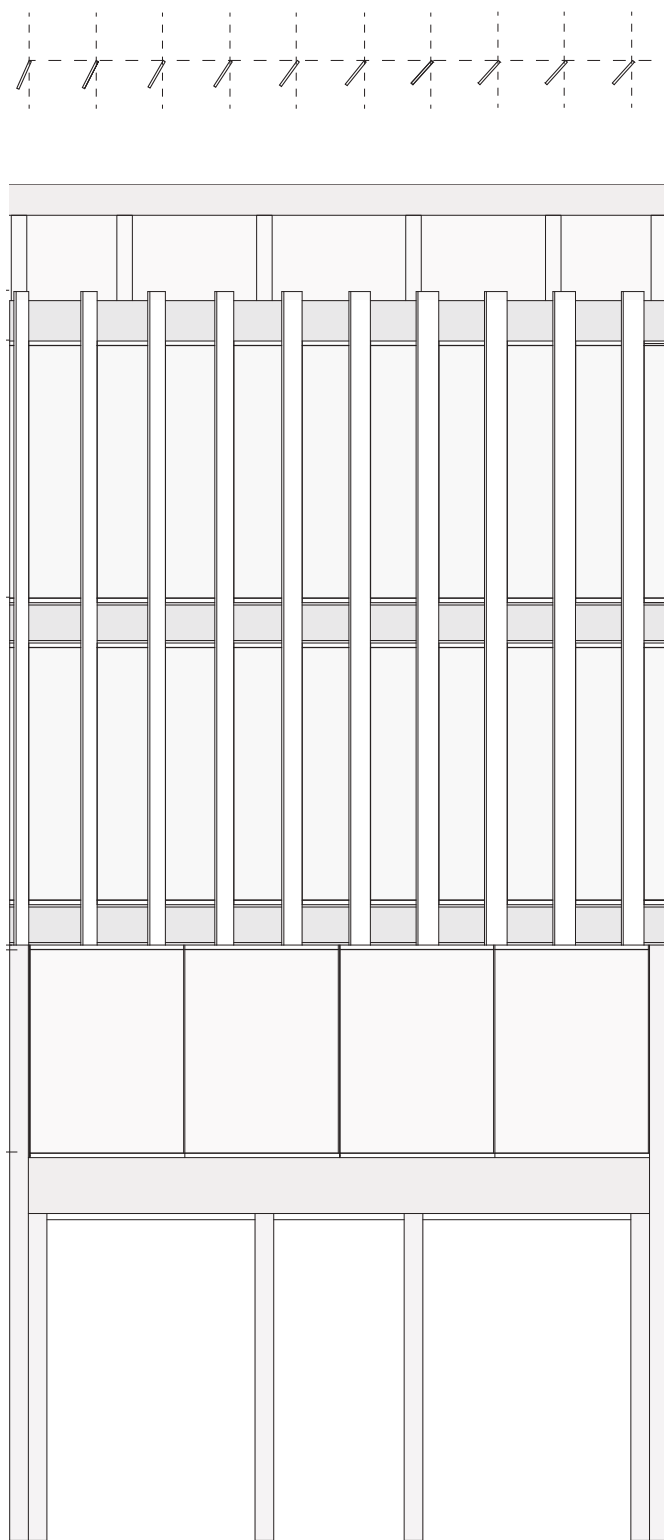




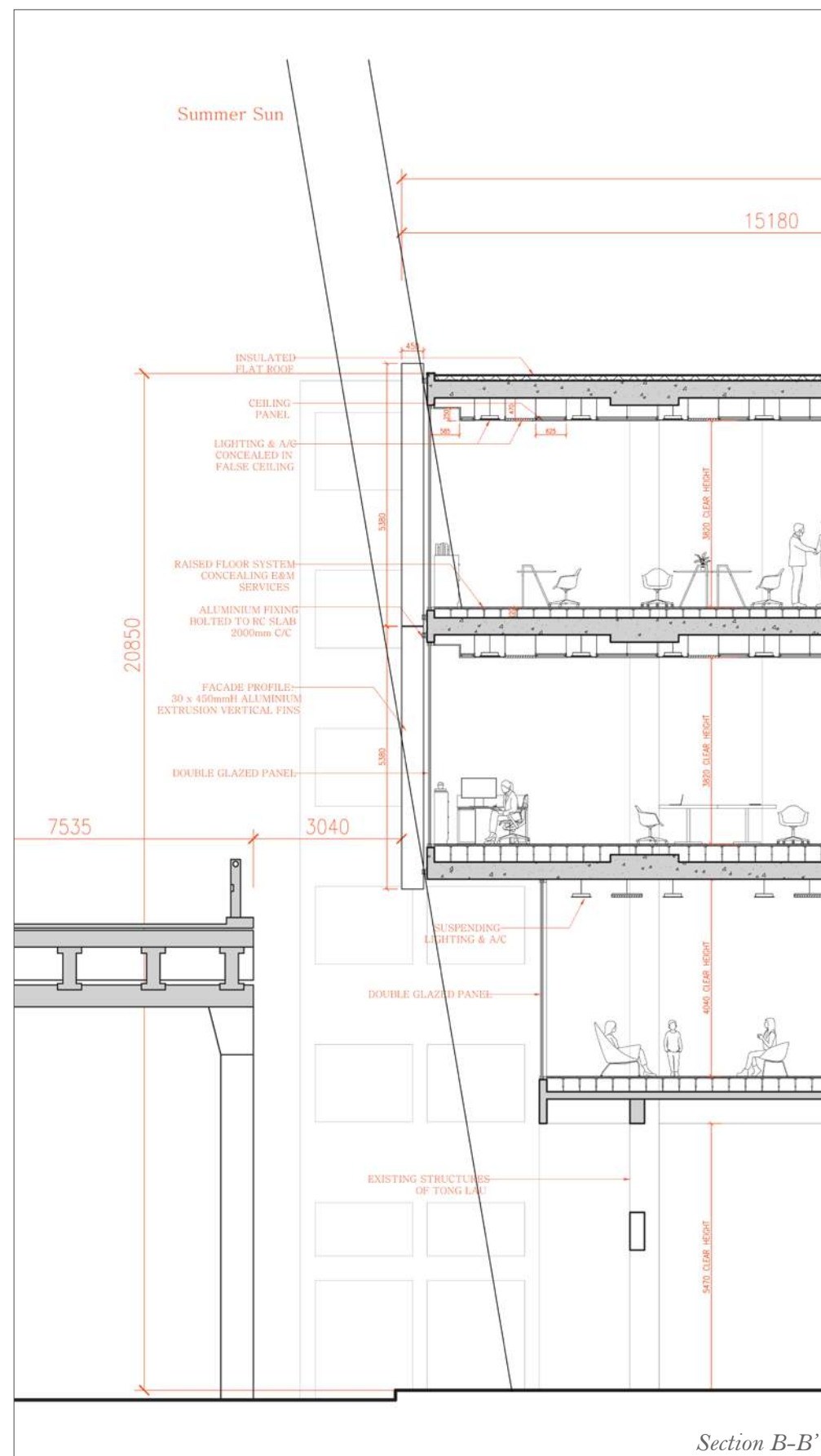
Back Alley Elevation



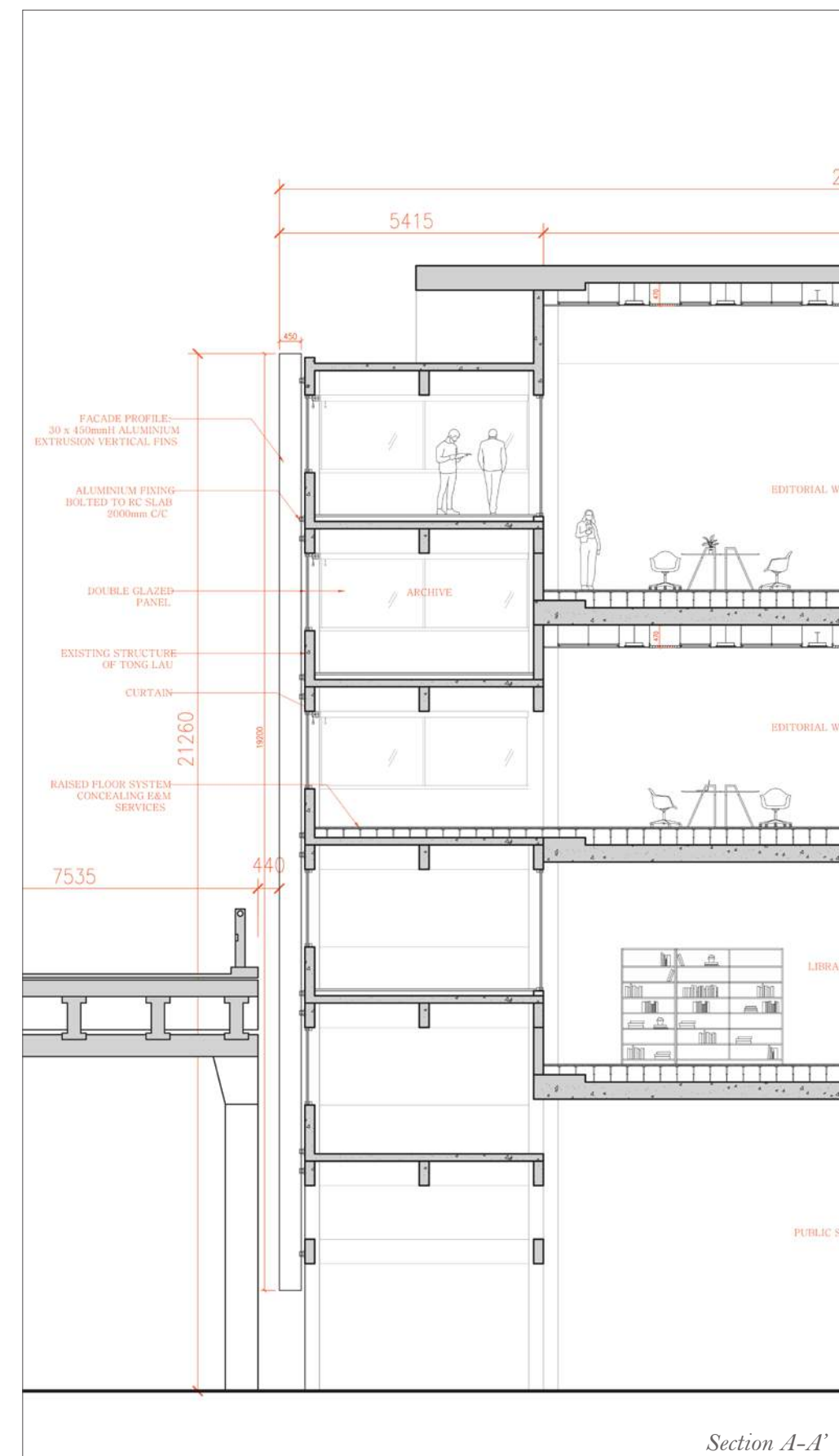
Envelope Details



Partial South Elevation



Section B-B'



Section A-A'

SLOW LIVING ON MULTIGENERATION

A Public Housing Designed for Multigenerational Living
Instructed by Dr. Mo Kar Him
Completed with Taka Tong, Andi Cheung



Shau Kei Wan, Hong Kong
September 2021- December 2021

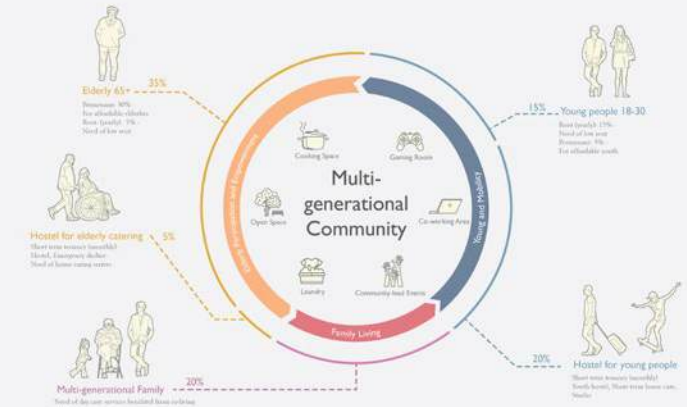




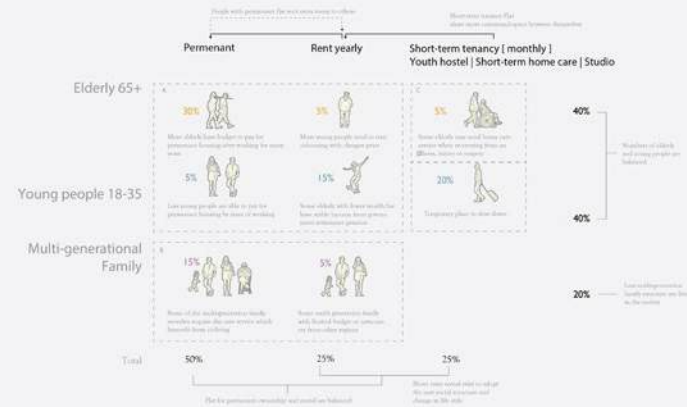
SLOW IN THE FAST PACE

Introduction of Third Place between First and Second Place

The city has been going faster and faster today. People are immersed in the culture of speed and efficiency that we almost fail to notice the toll it takes on every aspect of our lives - especially on interpersonal relationships.

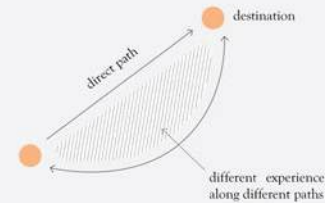
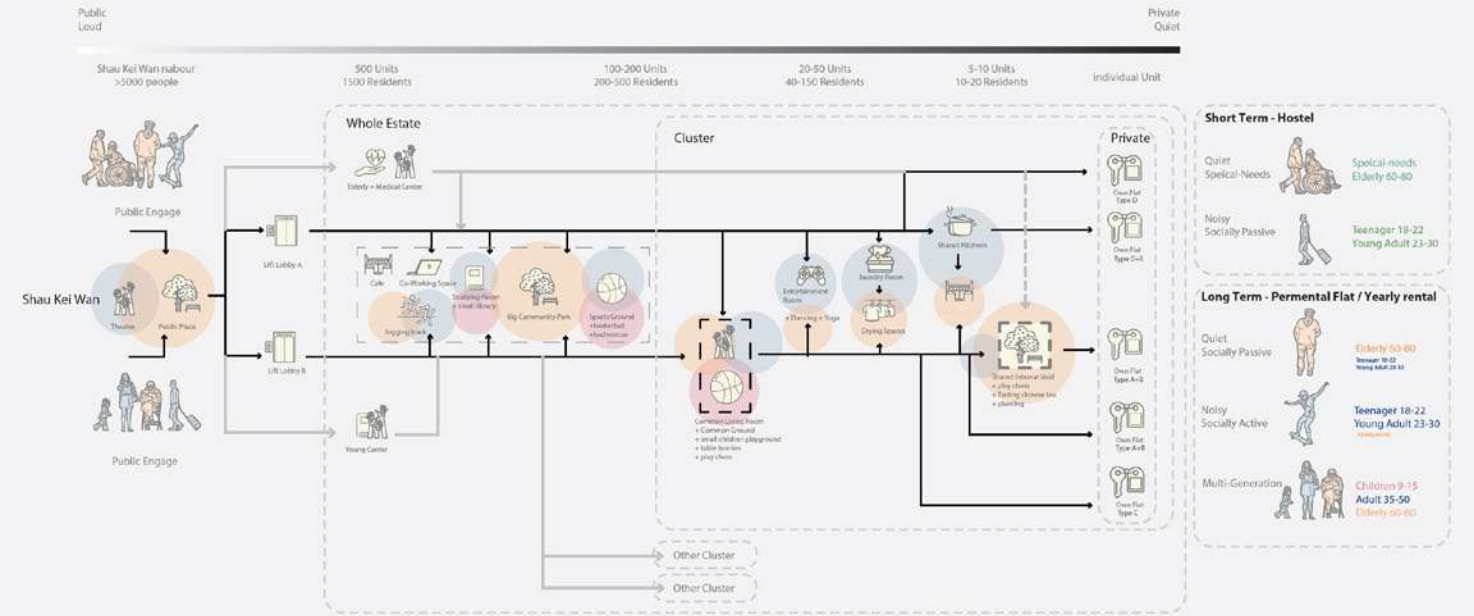


Slow-living Lifestyle



Target Group

Size of Population



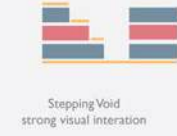
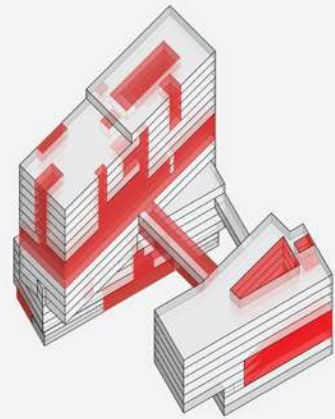
CHOICE MAKING

MULTI-FLOOR VOID

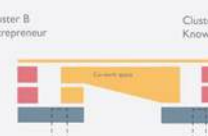
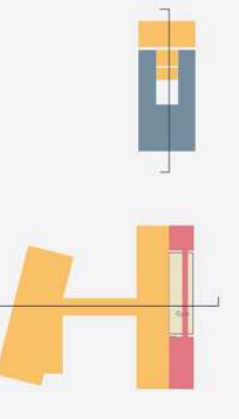


Permanent living place with a comfortable environment and regular community activities in slow living style.
A living place with more energy from the younger who temporarily stay

Temporary place to slow down from fast and stressed life in the busy city.
Communicate and exchange ideas with and learn from the elderly's experience



Void for Physical Connection to gym and garden



Type of Voids



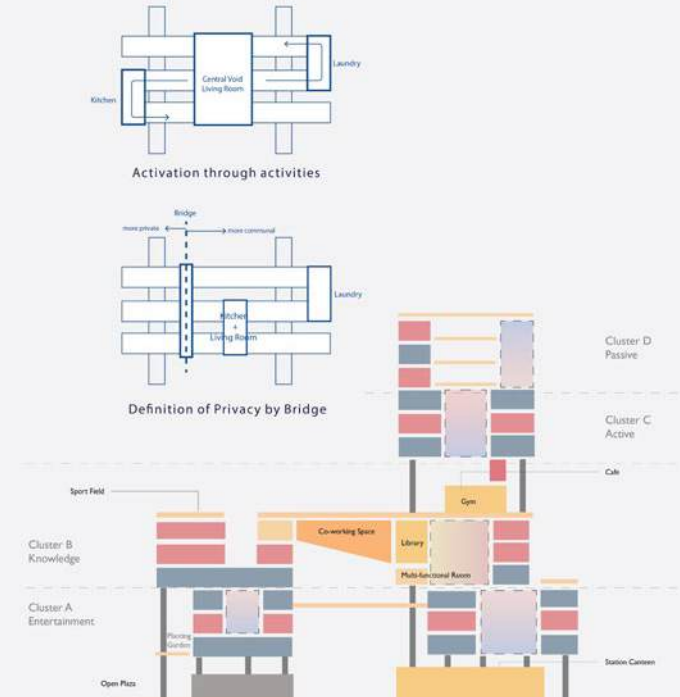
Void for Gardening and Communal Kitchen



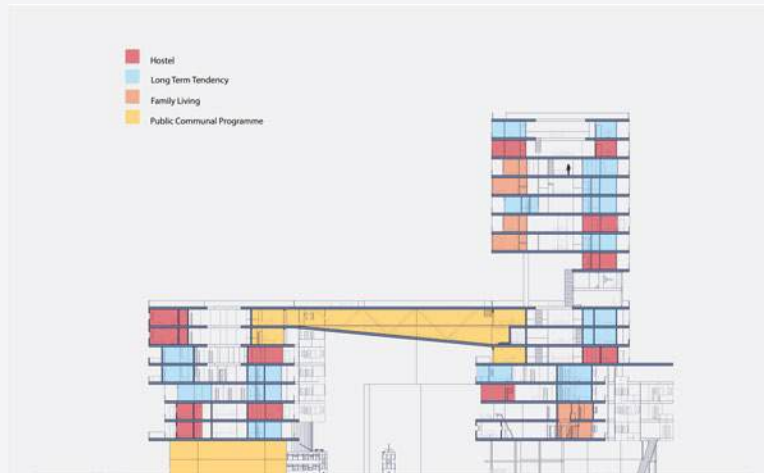
Type of Voids



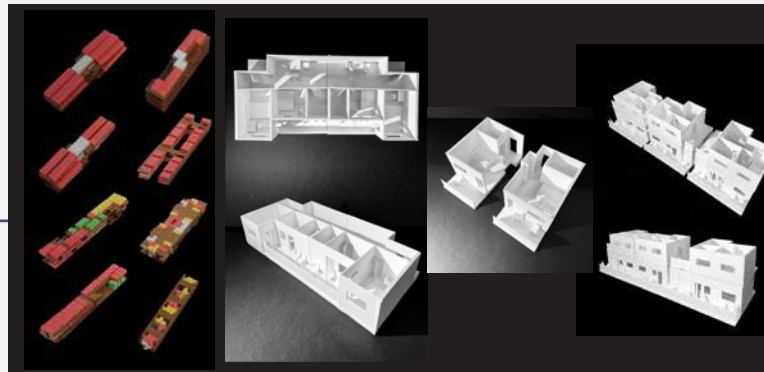
Void for Ground Floor Connection



Programme



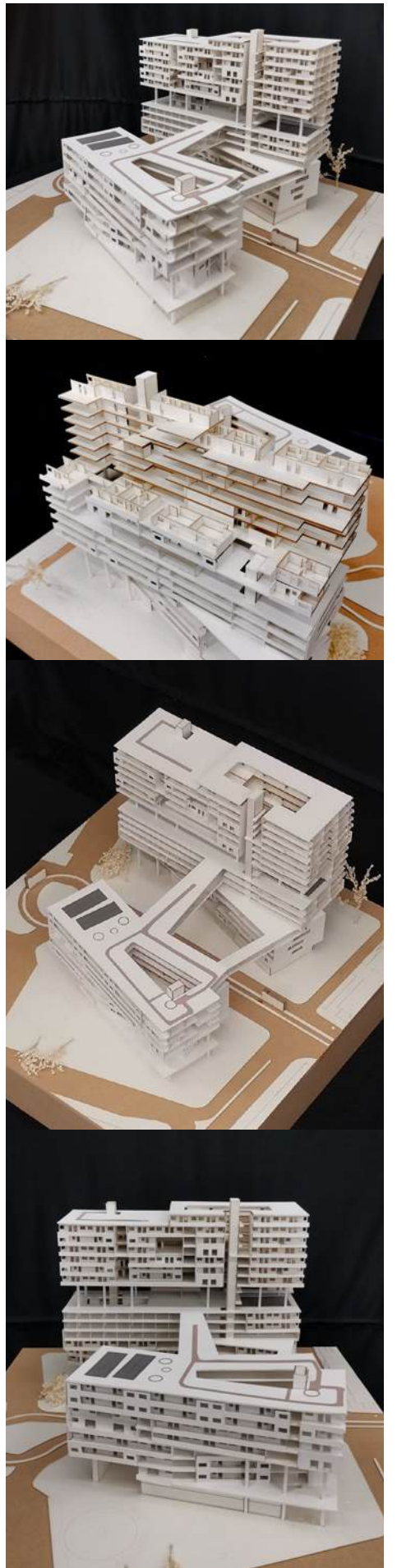
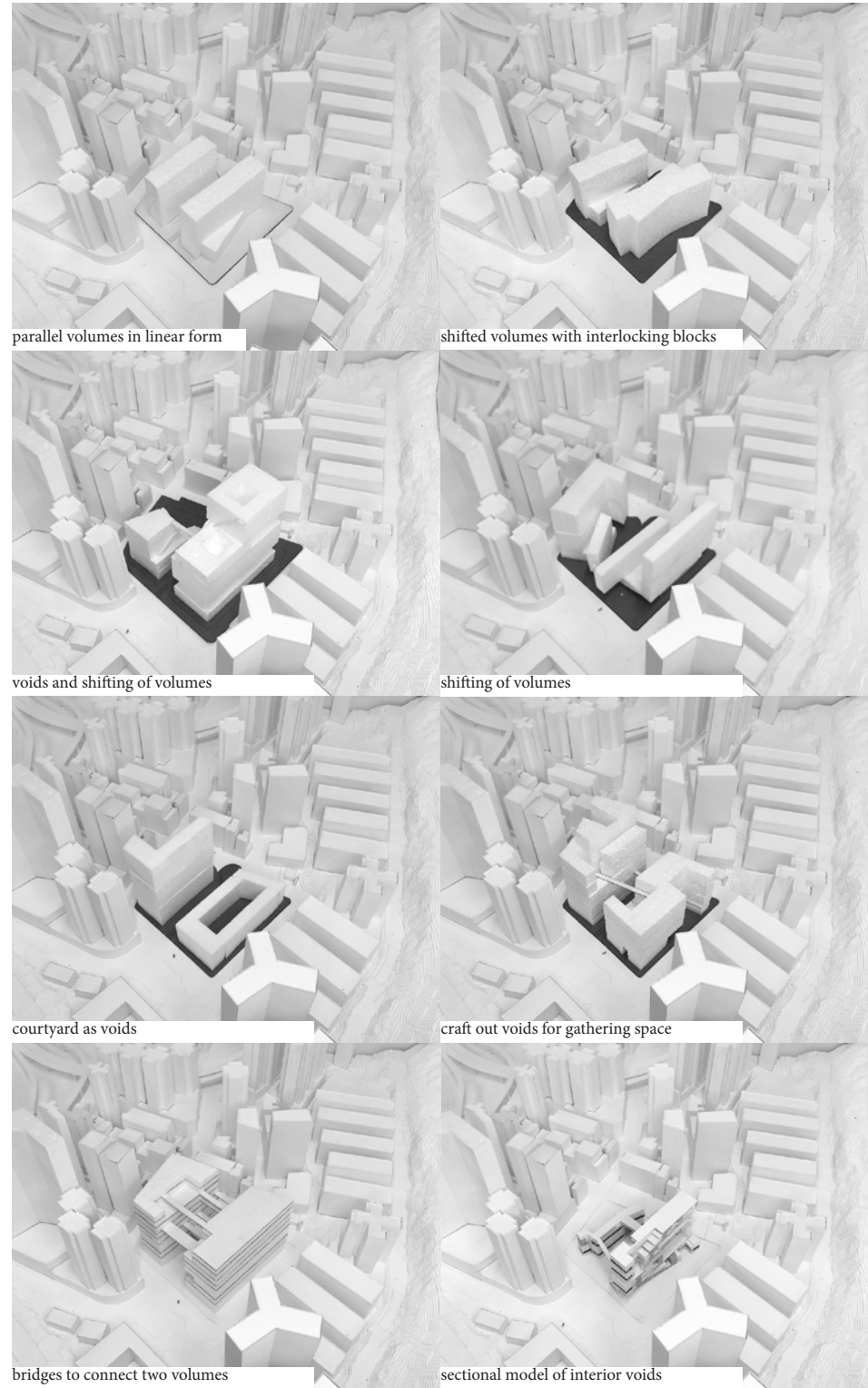
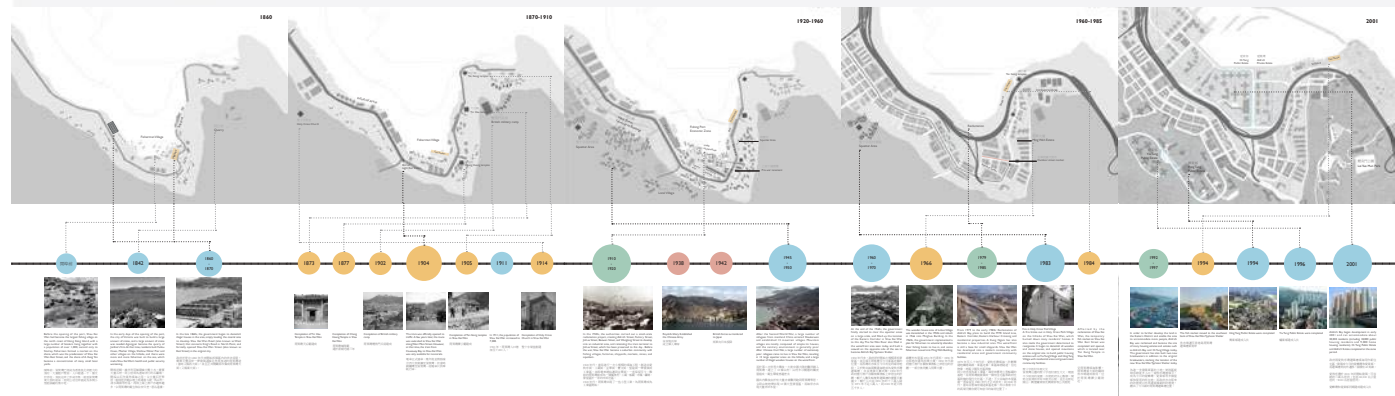
Unit Distribution

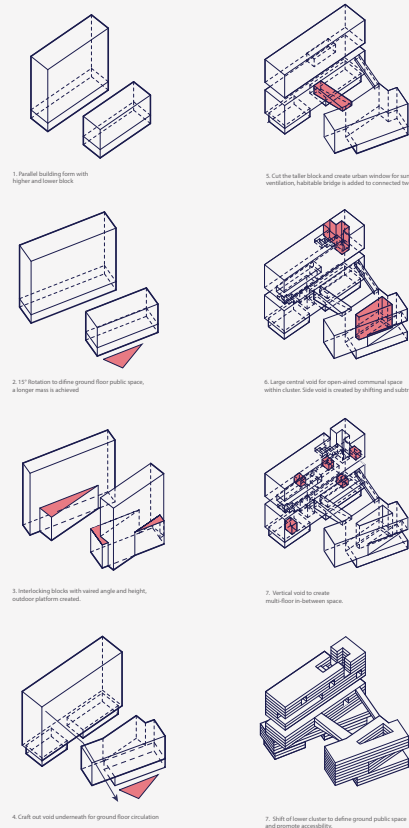




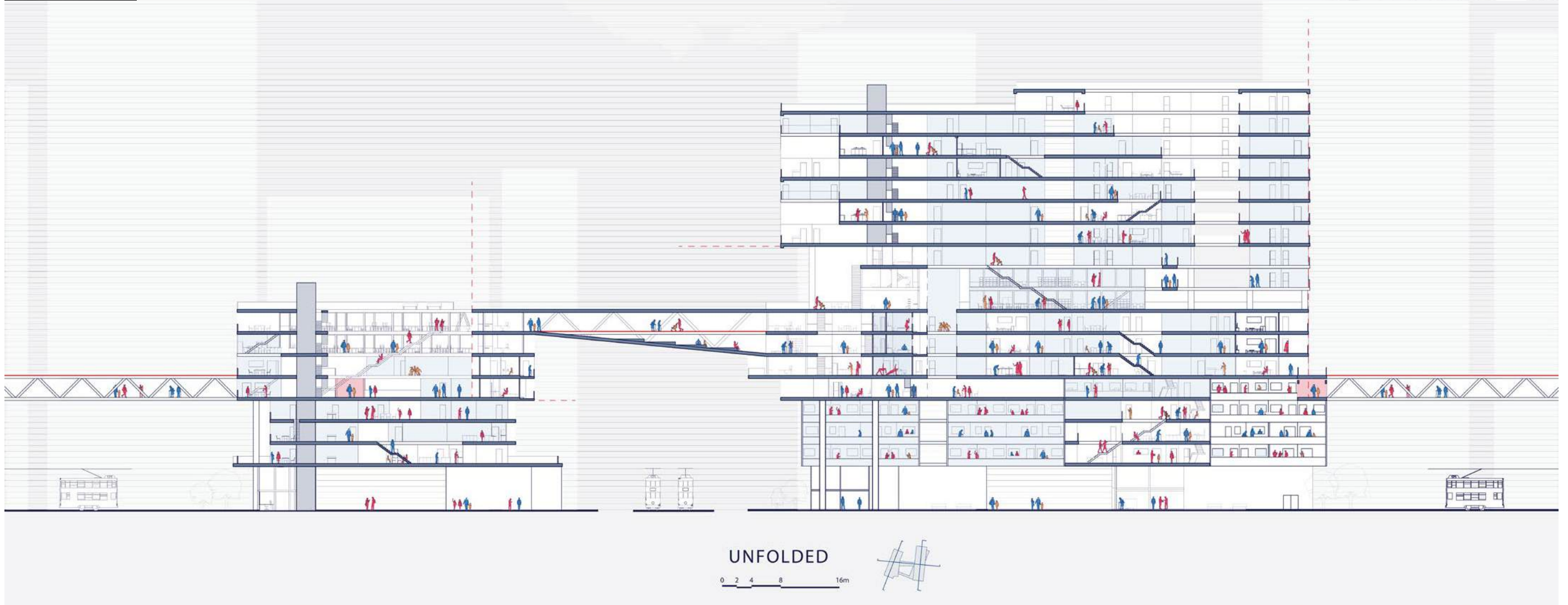
- Food & Market
- Local Restaurant
- Foreign Restaurant
- Fast Food Chain Stores
- Leisure Public Space
- Sitting out Area
- Sport Centre and Library
- Place of Slow
- Temple
- Church
- Main Street East

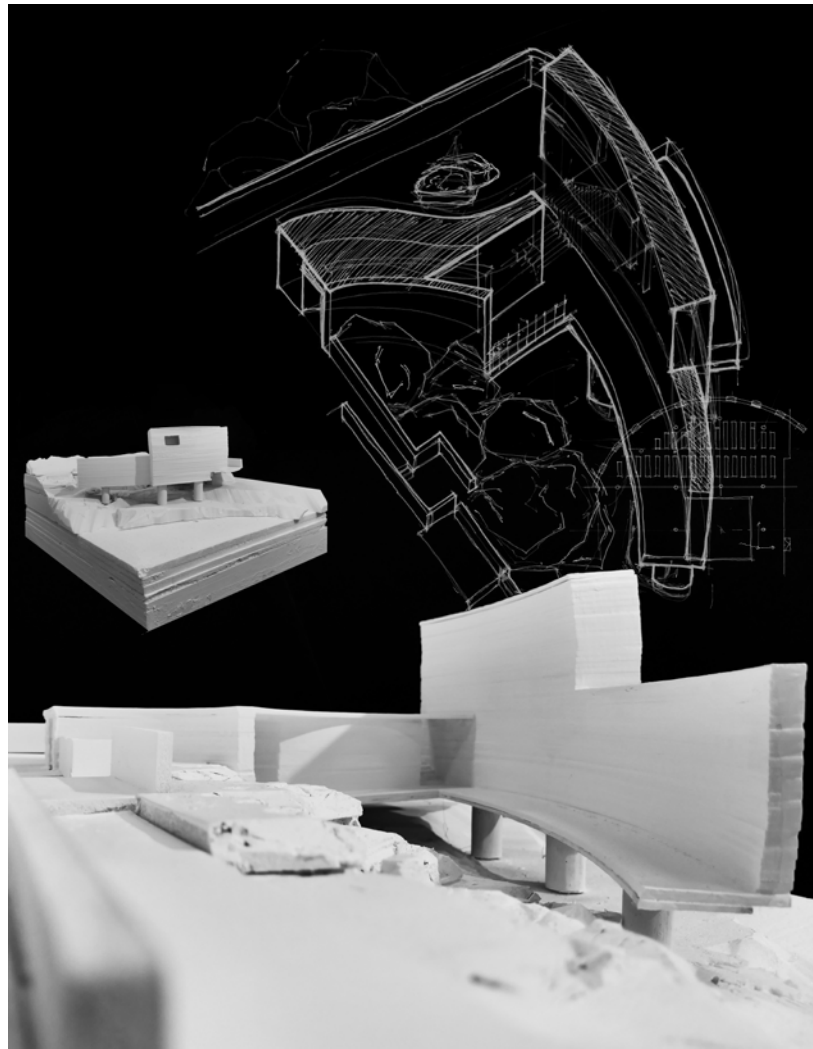
Moment of Slow





Design Development



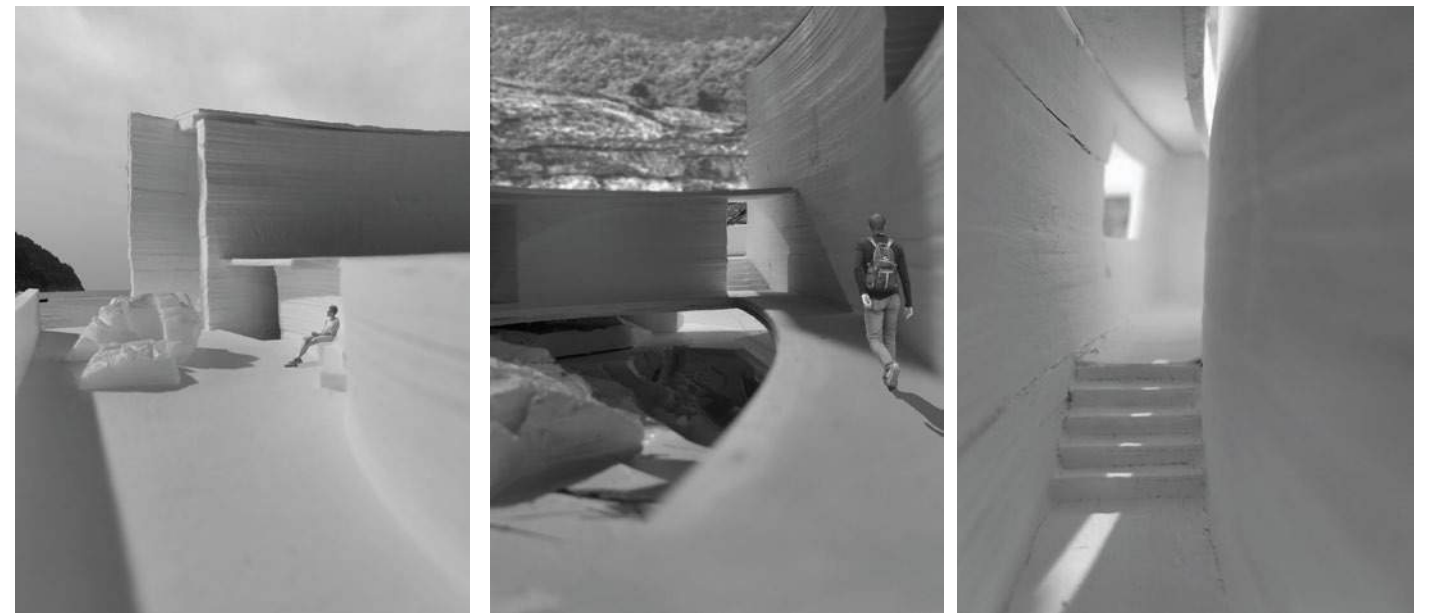
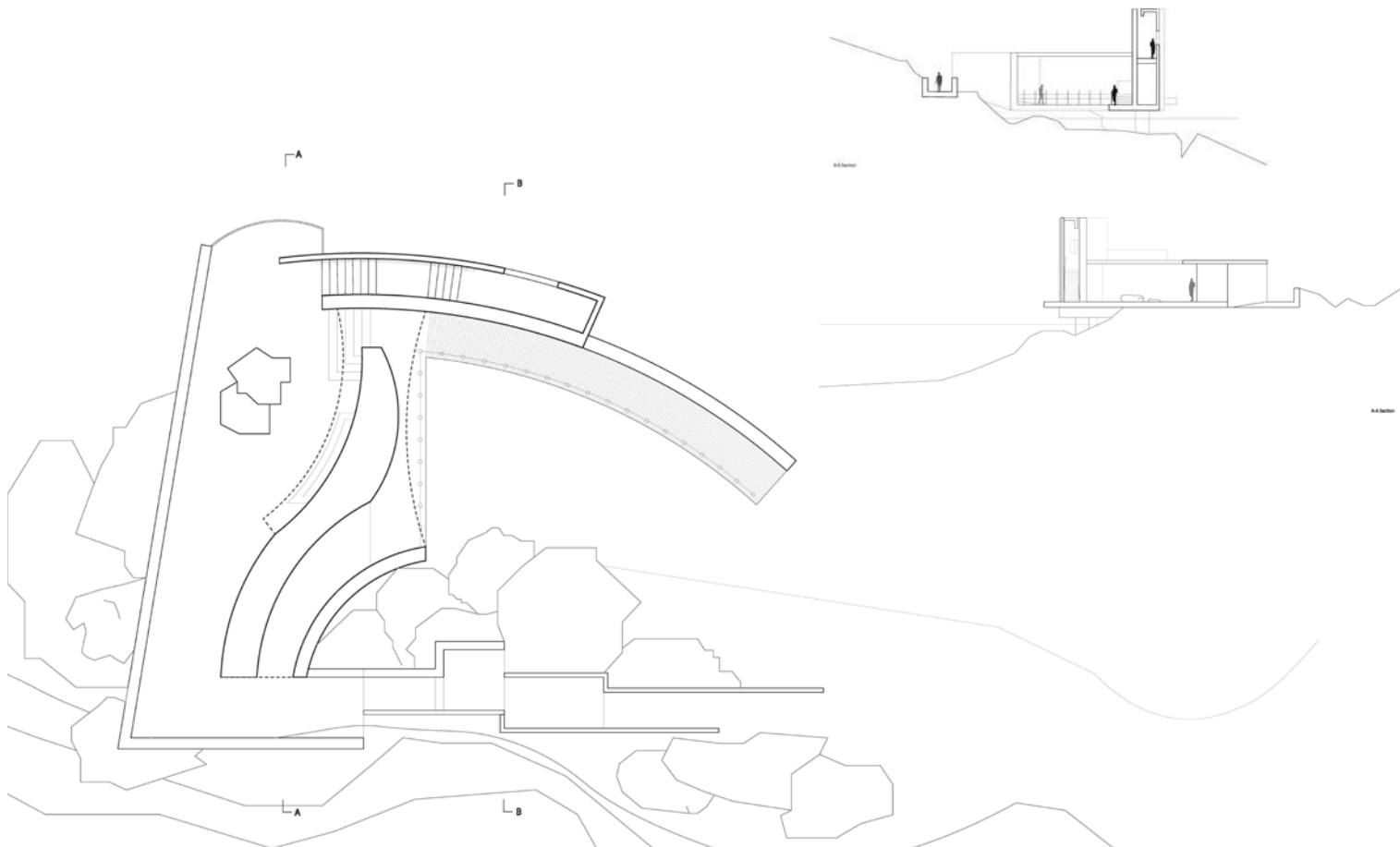


BOTH-AND OBSERVATORY

An Observatory of Sea and Rock
Located Along Seashore

Instructed by Prof. Patrick Hwang

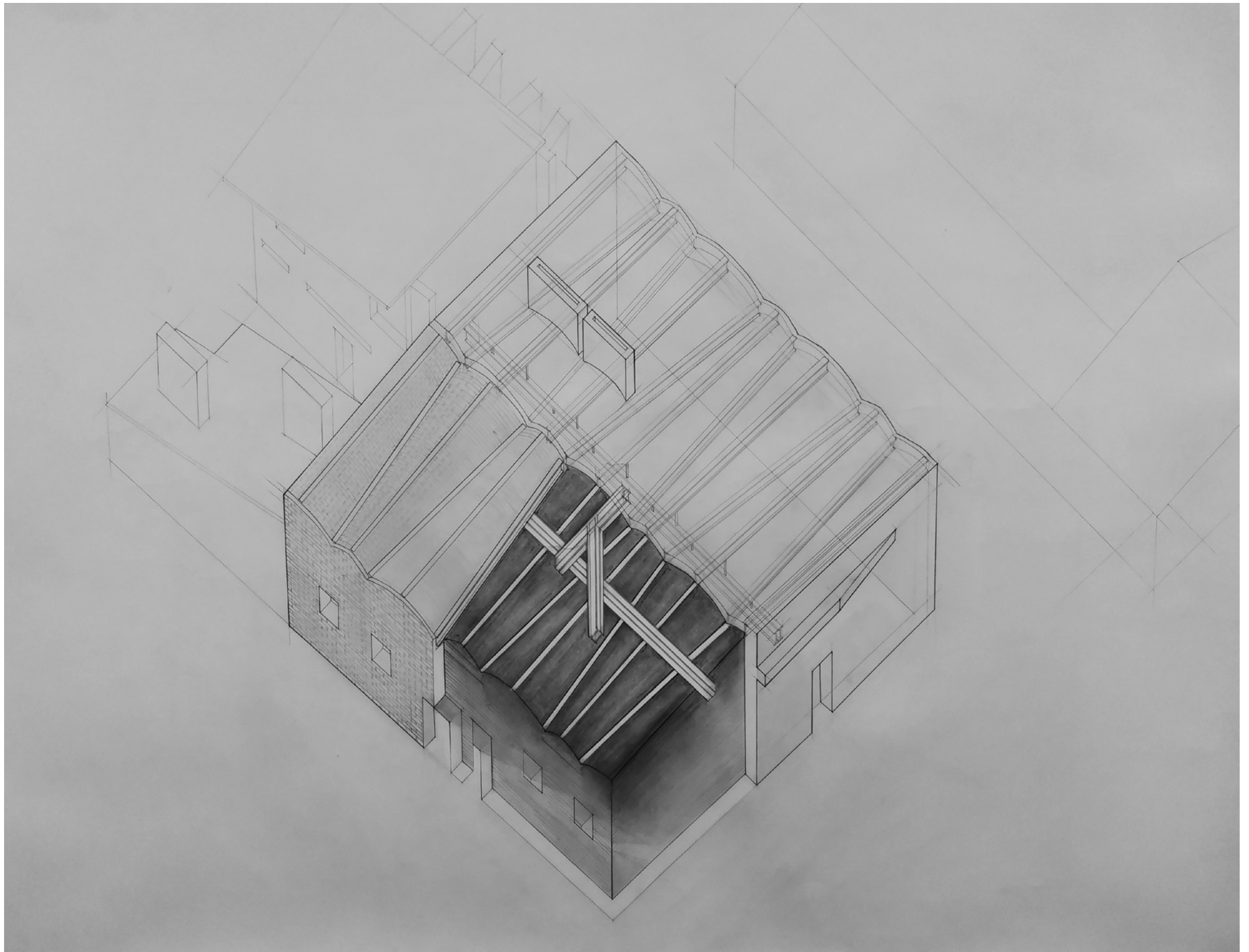
Clear Water Bay, Hong Kong
September 2020- December 2020



Situated along Clear Water Bay coast and embedded into rocky shores, the project acts as an open envelope of rocks that extends out to the sea, on which visitors can view back to the seashore, observing the echoes of wave and rocks with changes of tides level. The enclosure of the observatory is organised based on the natural jaggedness in the massive rock outcrops underneath. With limited interventions on this irregular landscape, rammed earth and reinforced concrete are used as main structural materials.

ANALYSIS OF RENOWNED BUILDING

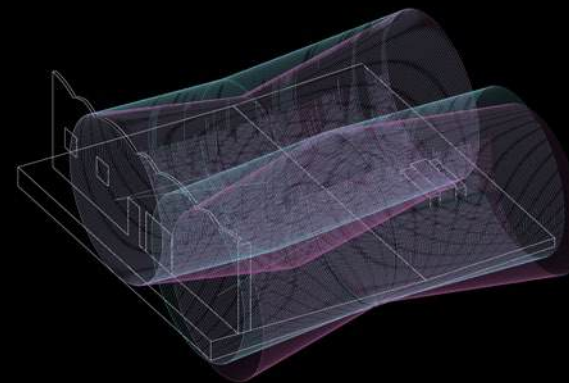
The tectonics and the geometry of the roof in Saint Peter's Church is studied in this project. The architect used I-beams to support a curved brick-and-mortar roof. With a sloping floor and a noncentric T-beam, the balance of weight as well as a gradual change in light quality is well designed in this church. The interior of the church is a large volume which gives a sense of darkness, enclosure and heaviness. However, the lightness of the I-beams contrasts the impression of massiveness from the exterior. The T-beam at the middle served as a symbolic ritual as a church. The design in irregular shape of skylight is also interesting and worth studying in the future. I started to get to know how Swedish Architecture designs atmospherically with structure and materials.



Case Study on Church of St. Peter, by Sigurd Leweren

The Tectonics and Geometry of Roof is studied in this project. The architect used I-beams to support a curved brick-and-mortar roof. With a sloping floor and a non-centric T-beam, the balance of weight as well as a gradual change in light quality is well designed in this church. The interior of the church is a large volume which give a sense of darkness, enclosure and heavy. However, the lightness of the I-beams has contrast the impression of massive from the exterior. The T-beam at the middle served as symbolic ritual as a church. The design in irregular shape of skylight is also be interesting and worth studying in the future. I started get known to how Swedish Architecture design atmospherically with structure and materials.

The tectonics and the geometry of the roof in Saint Peter's Church is studied in this project. The architect used I-beams to support a curved



A TRACE OF WIND

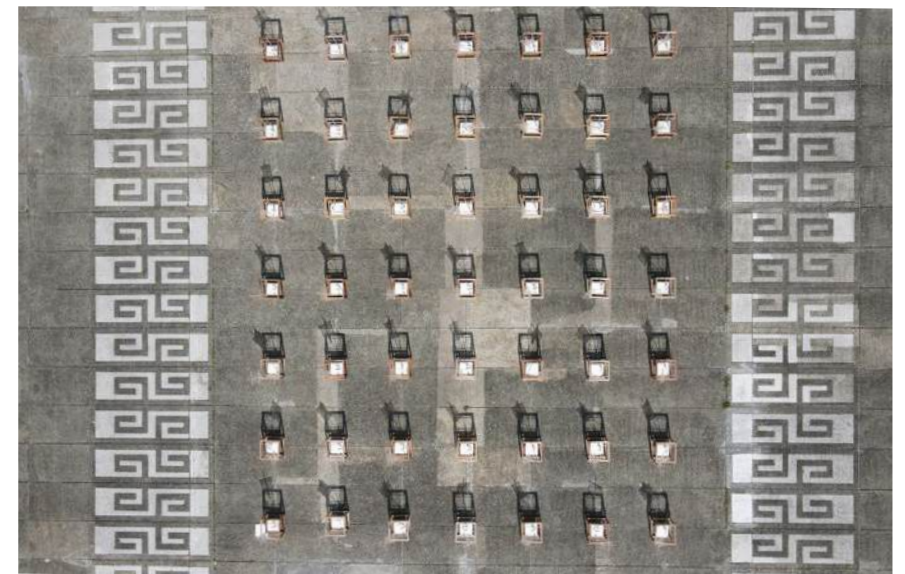
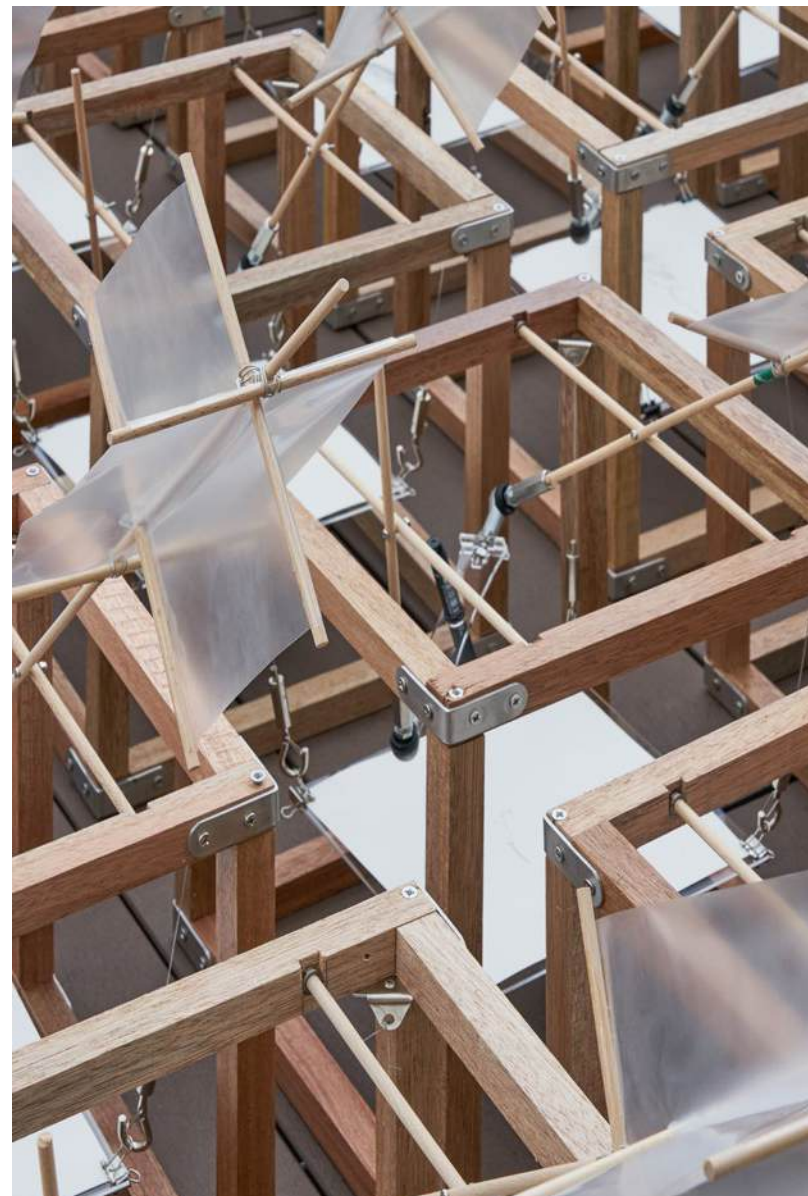
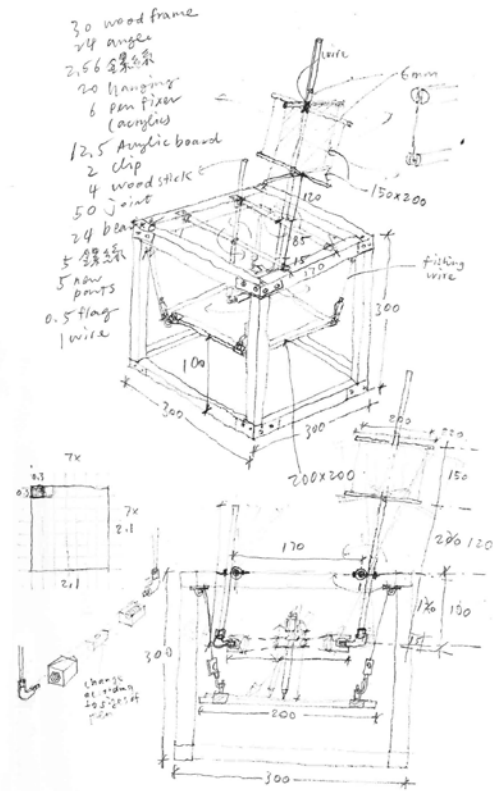
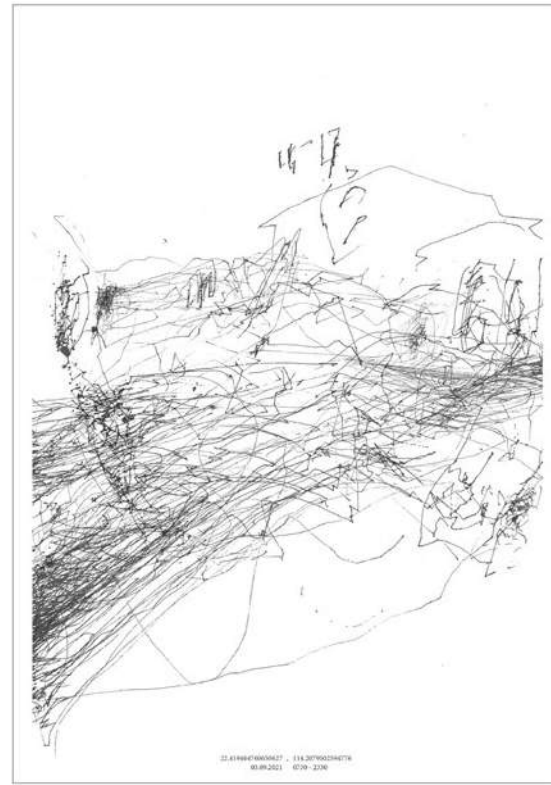
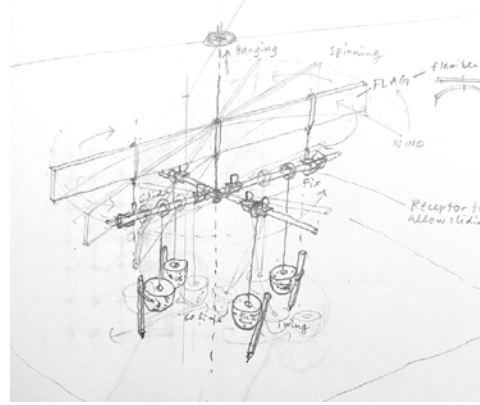
A Research Project on Rural Villages and Designed with a Prototypal Tracing Machine

Awarded in DEZEEN AWARD, and longlisted in TRINITY BUOY WHARF DRAWING PRICE

The project calls into question the process of drawing and memory, instead seeking to highlight elements that can be experienced but not captured such as wind and rain. The 50 machines are made and installed in different locations to capture individual moments and translate them into an abstract form of ink drawings on paper. Designed to be portable and work as individual modules or groups, their structure sees wind flow through, making flags move and pens draw. They come up with drawings that trace moments when and where the machines are placed, capturing an ephemeral memory where the human hand is replaced by the hand of nature.

Instructed by Prof. Peter FERRETTO
Completed with CONDITION LAB

Yim Tin Tsz, Hong Kong
August 2021



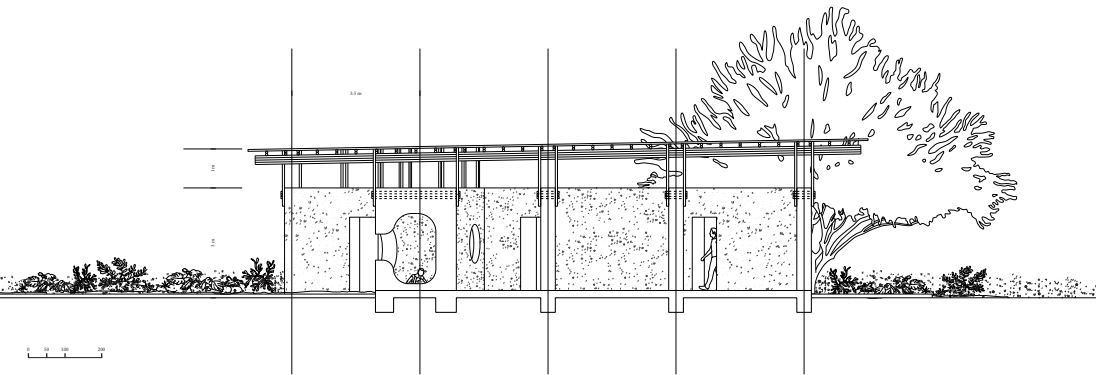
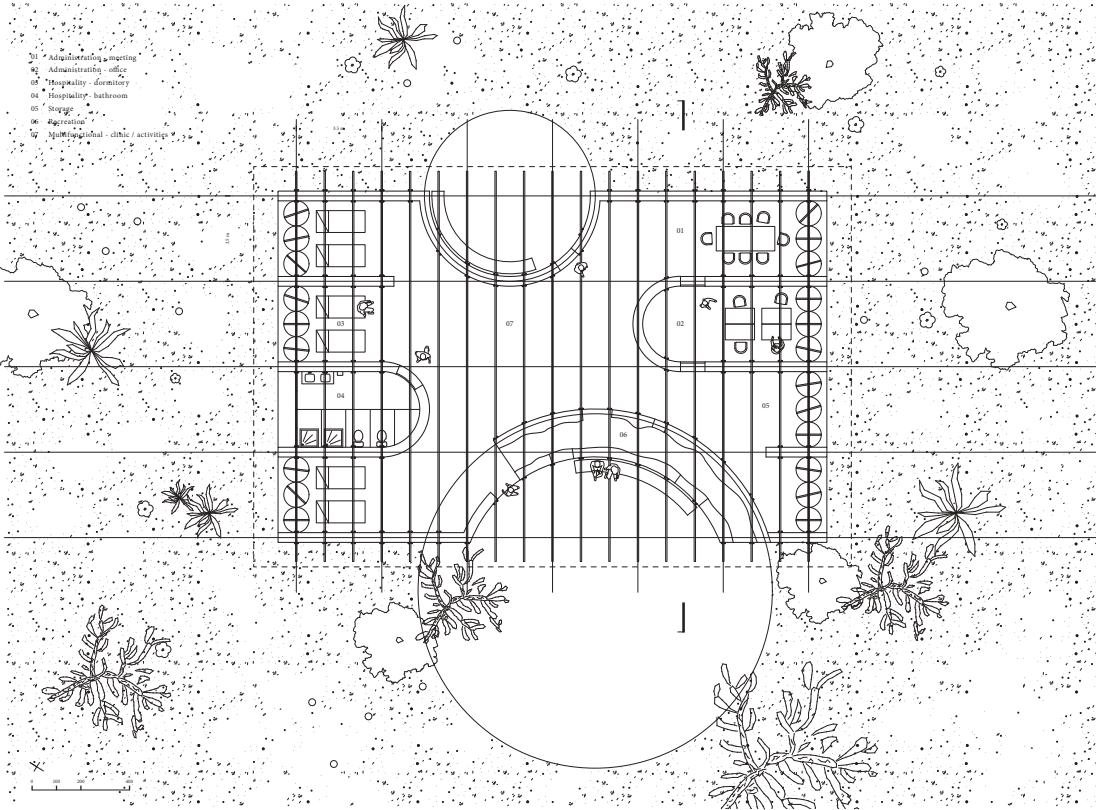
CHILDREN’S HOUSE

Place for Children, Parents and Community
Architecture of Traditional Cultural Heritage

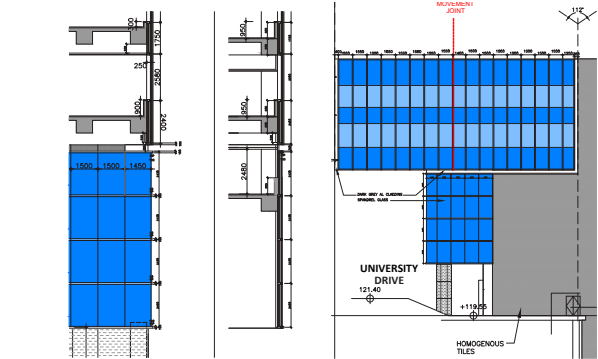


The Children's House is a place patients come for healthcare appointments, nurses come for training, villagers come for different talks and activities, and most importantly, children come to play while they are being monitored or taking treatments. It is also a role model of architecture adopting locally sourced materials, local human resources, at the end helping the locals in a circular ecosystem, while concrete the modern material is slowly invading in Senegal where the village of project site is in. The design continues local building traditions, and together with the Women's House lies in the same village, it is intended to act as an example of modern earthen architecture in Baghere, Senegal.

Not only as a functional space, the Children's House is also a welcoming place for children. The design concept arises from such vision. The grid is stiff and regular yet the rigidity is broken by insertion of several dynamic circular spaces. On top of the grid highlighted by five thick cob walls on the orientation of SE-NW, there inserted four circular spaces, two as relatively more private function (administration office, bathrooms) and two as covered open space which could become community space for the villagers, each sitting on one of the four elevations. Caves carved out of the massive front wall near the entrance provide hiding places where children can play or read or enter the House in a playful way, softening the image of the House as a clinic or medical place.

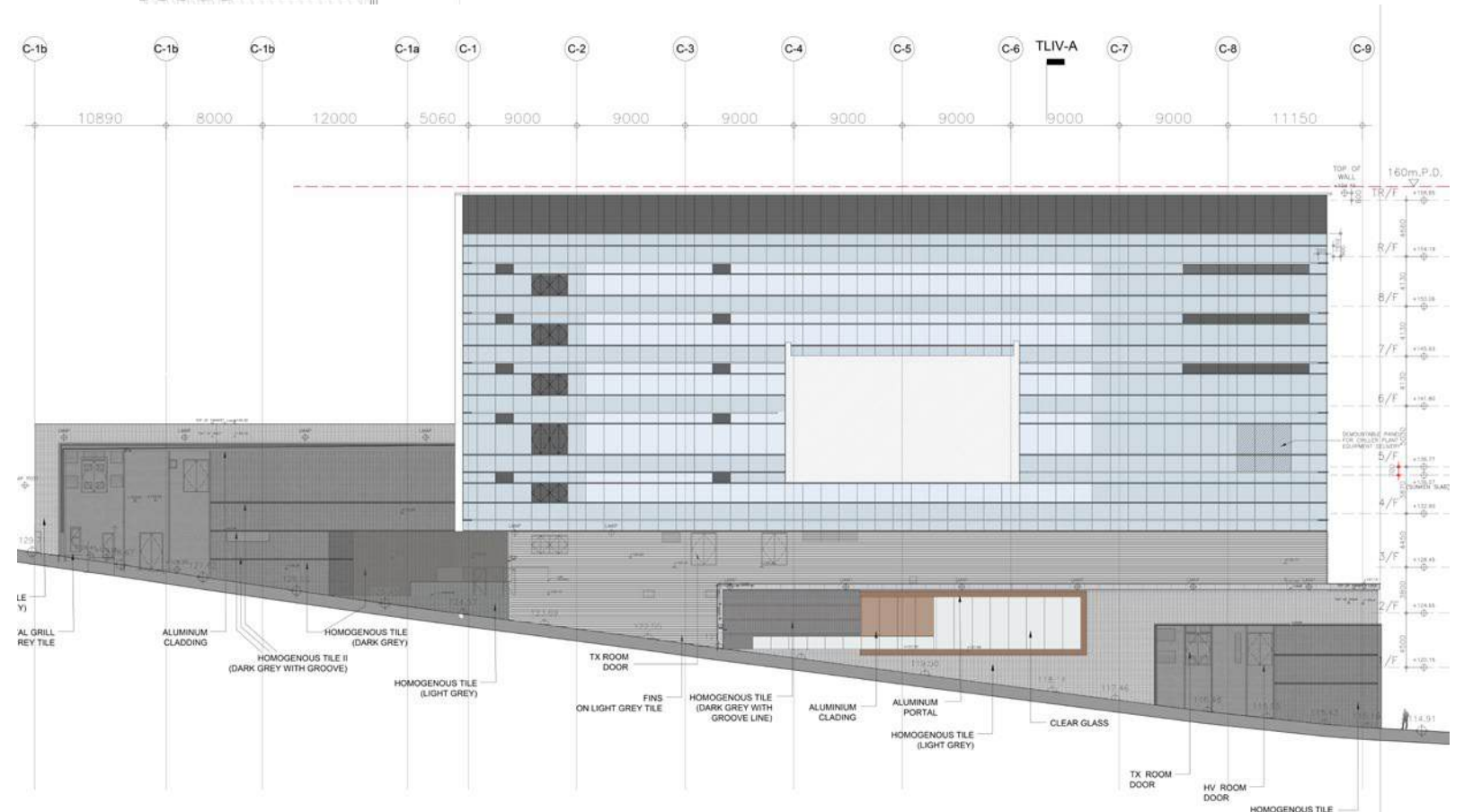
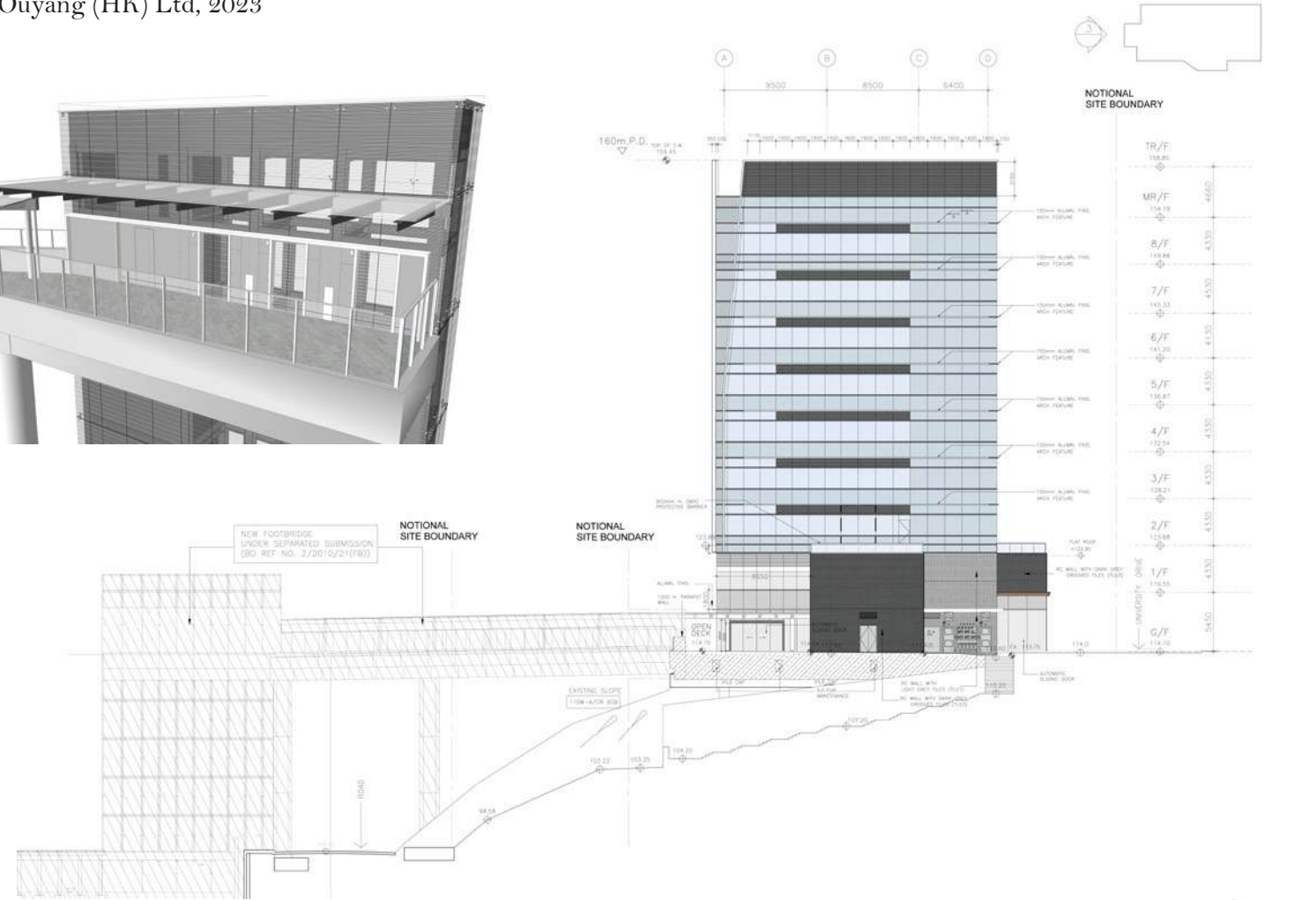


Project Awarded TOP 30 in Kaira Loro 2022 Summer



TECH LANDMARK, HONG KONG UNIVERSITY

Detail Design and Construction Stage, worked in
Wong & Ouyang (HK) Ltd, 2023



BISHOP HILL COMMUNITY HEALTH CENTRE

Tender Bid worked in Wong & Ouyang (HK) Ltd, 2023

